Deriving the Particle Zoo from Observer Consistency

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Abstract

Observer Patch Holography (OPH) reads one screen cell as an outside area and as the electromagnetic observation scale seen from within. Matching the two readings defines a local pixel fixed point. This paper carries the Standard Model structure of the companion compact paper into particle calculations: charge quantization, conditional gauge and gravitational carrier modes, the electroweak bridge, the hierarchy relation, and the Higgs/top quantitative branch. The source-audit weak-boson calculation gives $(m_W, m_Z) = (80.330, 91.119)$ GeV, while the stated criticality branch gives $(m_H, m_t) = (125.77, 172.63)$ GeV.

The same framework provides sharp failure tests. A reciprocal quark-shape candidate misses its held-out Yukawa coordinates by 21.56%, and the weighted-cycle neutrino candidate fails the stated oscillation-data test. For charged leptons, a connected six-dimensional response register and its tracial GNS representation derive the Koide balance $Q = 2/3$. Its attachment to a physical chiral mass response is work in progress; the 2/9 phase and individual mass ratios are comparison inputs. Absolute charged-lepton masses and hadron masses are reported only on their declared empirical-closure surfaces.

What This Paper Contributes

The structural particle content is inherited from the compact SM/GR paper: receipt-conditional compact gauge reconstruction, the realized Standard Model quotient, exact hypercharge, three colors, three generations, and the connection/metric carrier roles. The compact paper gives classical massless carrier modes only after explicit action/background/phase receipts and keeps quantum-particle poles behind a stronger gate. This paper asks what that structure does after the local pixel fixed point is turned on.

The OPH addition is the forward particle readout. A single screen-cell coordinate $P_\star$ feeds the electromagnetic branch, the electroweak transport surface, and the mass rows that can be certified on the declared branch. The hierarchy result is the cleanest example: the 12-port screen sieve and the 24-slot oriented repair register give the capacity-electroweak bridge, and the selected exact branch emits the dimensionless hierarchy $v/E_\star$, with Higgs naturality defect $\epsilon_H = 0$. A physical weak scale in GeV requires an independently source-closed $E_\star$. The same paper keeps weaker entries visible without upgrading them. Charged lepton absolute masses and first-principles hadron masses are outside the emitted theorem, benchmark checks are named as checks, and measured endpoint data are isolated when a transport bridge uses them.

1 Introduction

Observer-Patch Holography asks a concrete follow-up question after the gauge structure is fixed. Can one local screen constant organize the particle spectrum, or are the masses and mixings a collection of unrelated inputs?

The structural companion papers start from observer consistency on a finite screen. Physical data are organized in local patch algebras, nearby observers must agree on overlaps, realized states obey a local maximum-entropy and refinement principle, generalized entropy is recoverable, and the realized low-energy branch is fixed by Minimal Admissible Realization (MAR). On the controlled BW/geometric branch, that basis yields the Lorentzian and Einstein branches; on the realized compact-gauge branch it yields the Standard Model gauge structure

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6},\qquad N_c=3,\qquad N_g=3.$$

The Lorentzian branch, Einstein recovery, and the realized Standard Model gauge structure written above are imported from [5] with their stated branch conditions. That same paper also derives the hypercharge lattice and the realized color triplet $N_c = 3$ and generation count $N_g = 3$. This paper uses those results as input and summarizes only the downstream particle-spectrum continuation. If that derivation is correct, it should also constrain the particle spectrum. The substrate-neutral normal-form framework of Ref. [1] isolates the same-source confluence, cross-source boundary-identification, liveness, and refinement criteria used upstream. It supplies no particle-sector selector or numerical particle output; every such claim below remains conditional on the compact-paper branch and the particle-specific receipts.

The particle branch studied here is a one-fixed-point forward reconstruction problem with an explicit closure matrix. Instead of treating masses, mixings, and Yukawa parameters as independent inputs, it tries to propagate one dimensionless pixel ratio P through the spectrum. The pipeline does not emit photon, gluon, or graviton particle masses from symmetry labels. It records their conditional classical carrier modes and open quantum-particle gates, while the Higgs candidate remains on its declared quantitative surface. The hierarchy/naturality bridge is closed on its selected branch. The restricted quark source-spread non-identifiability theorem, the common-scale rejection of its reciprocal-ray candidate, and the weighted-cycle neutrino lane remain visible as obstruction or comparison surfaces. Target-anchored quark mass textures, the mixed-convention formula diagnostics, and compare-only absolute neutrino attachments are withheld from public prediction tables. The P-closure root, the electroweak W/Z rows, charged leptons, and hadrons have the sector-specific boundary statuses recorded below. This paper develops the derivation chain from the axioms and candidate P trunk to the reported particle outputs, with the closure boundary stated by sector.

1.1 Why this particle content is selected

On the declared branch, particles are not inserted by hand as a list of elementary ingredients. They are the stable observer-visible excitations and carrier modes that are left once three layers of structure have been fixed:

1. overlap consistency on the observer-patch network;
2. compact gauge reconstruction from the theorem-produced transportability criterion and fixed-cutoff bosonic sector category generated under one common stagewise strict representative, together with the compact paper's explicit refinement receipt for the refinement/fiber ladder and realized cofinal MAR-admissible witness data;

3. minimal admissible realization of the low-energy branch.

The first layer says what kinds of local data can be compared consistently across neighboring patches. The second assembles persistent zero-obstruction sector data into a compact gauge structure through the compact paper's transportability, category, and refinement/fiber theorems. The third selects which of the admissible low-energy branches is physically realized. At that point one is not choosing an arbitrary particle catalog. One is reading off the carrier and matter content of the realized branch.

Recovering Relativity and the Standard Model from Observer Overlap Consistency [5] is therefore central to this draft. That paper does the structural work that fixes the realized low-energy package

$$\frac{\mathrm{SU}(3)\times \mathrm{SU}(2)\times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3,$$

together with one admissible Higgs doublet and the realized chiral matter content. The construction does not begin by guessing photons, gluons, quarks, leptons, and weak bosons and then asking how to fit them. It first derives the realized gauge-and-matter branch, and that branch determines which elementary carriers and matter families are available at low energy.

On that selected one-Higgs branch the scalar carrier has a canonical local screen-chart model. Let $C_{\mathrm{EW}} \cong \mathbb{CP}^1$ be the support-visible electroweak chart and fix the positive Hopf line-bundle convention by the neutral component condition $Q(\phi^0) = 0$. Borel–Weil gives

$$H_{\mathrm{OPH}} = H^0(C_{\mathrm{EW}}, \mathcal{O}(1)) \cong \mathbb{C}^2,$$

the first nontrivial holomorphic section space. With OPH's hypercharge and $\mathbb{Z}_6$ normalization this is exactly the $(1,2)_{1/2}$ Higgs carrier. Projectivization classifies a nonzero section direction as a point of $\mathbb{P}(H_{\mathrm{OPH}}) \cong \mathbb{CP}^1$, but it forgets the scalar hypercharge phase. For the lower-component vacuum vector

$$\phi_0 = \frac{v}{\sqrt{2}}\begin{pmatrix}0\\1\end{pmatrix}, \qquad v \neq 0,$$

one has

$$e^{i\alpha T_3} e^{i\beta Y}\phi_0 = e^{i(\beta-\alpha)/2}\phi_0.$$

Consequently $[\phi_0]$ has the projective two-torus stabilizer $\mathrm{U}(1)_{T_3}\times \mathrm{U}(1)_Y$, modulo the inherited finite center, whereas the vector ϕ_0 is fixed only when $\beta = \alpha$, locally. Its connected stabilizer is therefore the electromagnetic diagonal $\mathrm{U}(1)_Q$, generated by $Q = T_3 + Y$. Thus the projective geometry explains the carrier-ray classification, while the chosen nonzero vector supplies the symmetry-breaking statement. This explains the representation, charge, and symmetry-breaking geometry of the one-Higgs slot. It does not derive m_H, the quartic, v, or Coleman–Weinberg dynamics; the weak scale, Higgs/top quantitative surface, and $\epsilon_H = 0$ hierarchy/naturality closure remain the OPH quantitative branch.

This also explains the phrase "and no others." The structural OPH claim concerns the realized low-energy branch selected by the stated admissibility conditions. Once one imposes those conditions, the realized low-energy branch is the Standard-Model branch. Additional connected gauge generators, additional light Higgs multiplets, extra light chiral families, or low-energy supersymmetric partners do not appear on that realized branch because they would enlarge the admissible package that MAR selects against. In that sense the observed light particle zoo is not an arbitrary menu; the representation/content package is the minimal realized one. Propagating gauge-particle existence requires the action, phase, and quantum-pole gates below.

The individual families then emerge for different reasons. The realized electromagnetic, color, and dynamical-metric branches identify connection and metric carrier roles. Explicit Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic actions then yield classical transverse or TT massless modes on their stated backgrounds and phases. Quantum photon, gluon, or graviton states require the separate physical-Hilbert-space, pole-residue, and asymptotic/phase receipt. The weak bosons arise when the electroweak gauge sector is propagated through its quantitative closure branch. Quarks and leptons arise from the realized chiral matter package together with the three-generation structure $N_g = 3$. Their family splittings are then read from the deeper overlap-transport and excitation machinery, not from a second arbitrary postulate that says "copy the family three times and assign masses by hand." Hadrons come one layer later: once color is realized and confined, stable hadrons are composite readout channels of the quark/gluon sector.

The role of the pixel constant P is also important to state clearly. Here, P does not decide whether an electromagnetic particle pole exists, whether there are three colors, or whether quarks come in three generations. The electromagnetic representation role and the color/generation facts belong to the gauge-and-matter branch fixed before the quantitative closure step. What P does is set the shared quantitative scale on which receipt-certified content is read out numerically. So the logic of the paper is:

$$\begin{aligned}\text{observer consistency} &\to \text{realized gauge/matter branch}\\ &\to \text{representation roles} \to \text{conditional action-level carrier modes}\\ &\to \text{common scale } P \to \text{receipt-gated quantitative spectrum.}\end{aligned}$$

That is how the derivation explains why a universe with the realized branch and the shared pixel scale exhibits this particle zoo.

1.2 Logical basis and reading rule

The particle derivation uses the same OPH basis as the SM/GR derivation in Ref. [5]. Its five axioms are the screen net, overlap consistency, local MaxEnt with refinement stability, recoverable generalized entropy, and Minimal Admissible Realization (MAR). On that basis, the SM/GR derivation in Ref. [5] supplies the structural chain used here: receipt-conditional compact gauge reconstruction as a classification step, MAR selection of the realized Standard Model quotient, the hypercharge lattice, the realized color triplet $N_c = 3$, and the generation count $N_g = 3$. Refs. [3, 4] supply the patch-net, repair, measurement, and regulated screen language used later when flavor transport and observer-facing readout enter.

The quantitative particle side adds one local closure variable. The common pixel ratio P, fixed on the synthesis paper's outer/inner closure branch [2], feeds the forward electroweak map and its descendants. The public empirical comparison branch displays

$$\alpha^{-1}(0) = 137.035999177(21), \qquad \alpha(0) \simeq 0.00729735256433, \qquad P_C \simeq 1.6309682094.$$

The comparison pixel P_C is defined from the measured endpoint; it is not a source-root coordinate. The computation has a fixed source order: golden-ratio entropy balance gives φ, boundary Gaussian normalization supplies the $\sqrt{\pi}$ width, a trial P feeds the source map through unification, running, and electroweak anchoring, and Ward-projected electromagnetic transport gives the Thomson endpoint used by the outer/inner pixel fixed point. The value is forced on that declared branch because each ingredient in the closure equation is fixed by the screen balance, normalization, source map, and Ward-projected endpoint; any different electromagnetic readout gives a different P and fails the outer/inner equality for the same cell. The first-principles diagnostic trunk records

$$
\begin{aligned}
P &= \texttt{1.6309720956943290181796789256119188427016}9, \\
\alpha_{\mathrm{root}}^{-1} &= \texttt{136.9948351646216494579499945857871932620}29.
\end{aligned}
$$

The approximately 0.041 inverse-alpha difference to the public Thomson endpoint identifies the declared QCD/hadronic electromagnetic transport boundary; it is not a source-emitted correction. The QCD-free hierarchy witness is the cleaner first-principles stress test. Combining the source/root value with the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel, $\alpha_U(P_C)$, gives the mixed-provenance no-hadron diagnostic $A^{\mathrm{fp}}_{\alpha_U} = 137.0359595008\ldots$, below the Thomson endpoint. It is comparison bookkeeping rather than a source-only fine-structure prediction. The detailed endpoint table appears in Section 4. The separate cosmic record-closure target $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$, with count representation $N_\star = \mathrm{MAR}\arg\max_N[\log|\Omega_N^{\mathrm{sc}}| - N]$, belongs to the cosmological-capacity branch and only to legacy capacity-level neutrino side estimates. The Newton coupling uses the separate selected no-G scale certificate $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, so the particle branch does not derive G by back-solving the pixel ratio. The pair $(P_\star, N_{\mathrm{CRC}})$ determines dimensionless curvature products such as $\Lambda_\star a_{\mathrm{cell}} = 3\pi P_\star/N_{\mathrm{CRC}}$, not the SI scale product $\Lambda_\star N_{\mathrm{CRC}}$ by itself. The rounded $N_{\mathrm{CRC}} \simeq 3.31 \times 10^{122}$ display is a capacity-scale label, not the high-precision input for G. Structural carriers, quantitative outputs, exact sidecars, and continuation lanes are therefore downstream branches of the declared theorem checklist, with particle-facing quantitative burden carried by the local fixed point $P_\star$. The public fine-structure display row is the declared Ward-projected Thomson endpoint of the OPH fixed-point equation. The first-principles diagnostic row and empirical hadron closure rows are separate support records. Section 4 records that boundary in full.

2 P-Closure and the Reverse-Engineering Strategy

Before the claim-tier table, the reverse-engineering claim should be stated plainly. In ordinary phenomenological use, the Standard Model does not derive the observed particle masses and mixings from one common microscopic constant. It is usually presented with 19 free parameters in the minimal massless-neutrino theory, and with at least 26 once neutrino masses and mixing are included, depending on neutrino-sector conventions. The OPH particle derivation compresses that situation into one universal pixel ratio P from the outer/inner closure relation described in the synthesis paper. The same P must drive all downstream bosonic, quark, lepton, neutrino, and hadronic branches.

One clarification matters. The broader zero-input formulation targets the global screen capacity by

$$N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}}),$$

where $F(N)$ is the active cosmic record capacity read back by observers inside the universe supplied with capacity N. The count-density representation is

$$N_\star = \mathrm{MAR}\arg\max_N \left[\log|\Omega_N^{\mathrm{sc}}| - N\right],$$

equivalently, in differentiable form, as the unique stable fixed point of $T_\eta(N) = N + \eta\, d(\log|\Omega_N^{\mathrm{sc}}| - N)/dN$ under the stated strict-concavity certificate. Informally, it is the unique global capacity where the universe reads back its own boundary without deficit or slack, parallel to the local $P_\star$

fixed point where outer screen detuning equals the inner Thomson observation scale. The observed branch reads this same capacity as $N_{\rm scr} \simeq 3.31 \times 10^{122}$. The particle-spectrum derivation studied here uses the local pixel fixed point $P_\star$ in place of a long particle-by-particle parameter list.

Observation is allowed to supply a branch hint. That is a consequence of the OPH picture, where the universe is a closed fixed structure and internal observers can read approximate coordinates from experiments. The proof obligation is stricter: the exact $P_\star$ used by the particle branch must be computed by the self-referential pixel equation, and the endpoint must be unique on the declared interval. A fitted particle mass can identify a neighborhood. It cannot replace the closure solve.

2.1 Why a single P matters

The structural theorems determine *what kind* of particle world is realized, but not *what absolute scale in GeV* that world occupies. The overlap, modular, gauge, anomaly, and admissibility arguments recover the realized Standard Model branch, the exact hypercharge pattern, three generations, and three colors. The carrier-mode theorem is an additional action-level result and does not emit zero-GeV quantum particles. These structural results do not by themselves tell us the numerical values of the W boson mass, the Higgs boson mass, or the up-quark mass.

The role of the pixel closure is to supply one common quantitative scale variable for the entire downstream spectrum. In the synthesis paper, the fine-structure lane asks for the nonzero detuning of a holographic screen cell such that the cell's outer geometric displacement from perfect self-similar equilibrium equals the electromagnetic observation scale emitted by the universe living on that same screen. The outer side of the closure is

$$P = \varphi + \alpha_{\rm in}(P)\sqrt{\pi}.$$

The first-principles computation is a five-step source chain. First, the golden-ratio entropy balance of the local screen cell supplies $\varphi = (1+\sqrt{5})/2$. Second, boundary maximum-entropy normalization fixes the width of the leading electromagnetic detuning to $\sqrt{\pi}$. Third, a trial P is sent through the source map

$$M_U(P) = E_P e^{-2\pi} P^{1/6}, \qquad E_{\rm cell}(P) = \frac{E_P}{\sqrt{P}},$$

followed by the heat-kernel closure

$$\bar{\ell}_{\rm SU(2)}(t_2(P)) + \bar{\ell}_{\rm SU(3)}(t_3(P)) = \frac{P}{4},$$

which selects $\alpha_U(P)$ and the running family $\alpha_i(m_Z; P)$. Fourth, electroweak mixing gives the source anchor

$$a_0(P) = \alpha_2^{-1}(m_Z; P) + \frac{5}{3}\alpha_1^{-1}(m_Z; P).$$

Fifth, Ward-projected $\mathrm{U}(1)_Q$ transport gives the Thomson endpoint

$$A_T(P) = T_Q(a_0(P), F_{\rm src}(P)) = \alpha_{\rm em}^{-1}(0; P),$$

and the cell closes when

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}.$$

The numerical solve is therefore a root problem for

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}.$$

The branch interval is localized from the observer-facing data, the solver evaluates $A_T(P)$ from the declared source map at each candidate point, and the accepted value is the unique zero of H on that interval. The measured endpoint can locate the interval. It cannot replace the root condition. Thus α is the observer-supporting electromagnetic width of the local pixel, and its value is fixed by the unique solution of this outer/inner consistency equation. The OPH fixed-point readout $\alpha^{-1}(0) = 137.035999177(21)$ gives

$$P \simeq 1.6309682094.$$

The same equation is the local ruler for the downstream particle rows. The five-layer OPH diagnostic trunk emits the source-side diagnostic point

$$\begin{aligned} P &= \texttt{1.630972095694329018179678925611918842 70169} \qquad , \\ \alpha_{\mathrm{root}}^{-1} &= \texttt{136.994835164621649457949994585787193262029} \qquad . \end{aligned}$$

The detailed table records how the diagnostic trunk, endpoint residual, and source-spectral payload fit together. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. Once the fixed point is set, the particle question becomes whether all downstream readouts can be written as

$$X_j = G_j(P)$$

with no new sector-specific constants inserted by hand. This paper studies exactly that downstream map.

2.2 How the electroweak branch works

In the implementation used here, the quantitative electroweak branch is the first major readout from P. Its algebraic output family is

$$\{M_W^{(10)},\ M_Z^{(10)},\ \alpha_{\mathrm{em}}^{-1}(q^2),\ \sin^2\theta_W(q^2),\ v\},$$

evaluated from one declared input P on the printed running/matching/threshold/scheme package. The superscript (10) denotes a D10 mass-chart coordinate, not a certified complex pole. The construction is straightforward in spirit: start from P, build the electroweak running family, select the physical carrier point on that family, and read off the W boson and Z boson pair from that selected point. The Higgs/top critical stage inherits that same electroweak core and does not introduce a new free-input sector. On the declared Ward-projected transport branch, the low-energy electromagnetic row is represented by the Thomson endpoint

$$\alpha_{\mathrm{Th}}^{-1}(P) = \lim_{q^2\to 0} \alpha_{\mathrm{em}}^{-1}(q^2; P)$$

of that same electromagnetic transport family. The forward logical order on this branch is:

$$P \longmapsto (M_U(P), E_{\mathrm{cell}}(P)) \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\mathrm{tr}}(P)) \longmapsto (t_2(P), t_3(P), v(P)) \longmapsto \alpha_i(\mu_*; P).$$

The forward transmutation certificate records that the same runtime basis reconstructs the unified diffusion parameter $t_U(P) = 4\pi^2\alpha_U(P)$ and transmutation exponent $t_{\mathrm{tr}}(P) = 2\pi/((N_c+1)\alpha_U(P))$ as the pixel-closure solve itself. The runtime subgraph does not read measured electroweak data until its validation surface. This does not establish a target-free historical dependency graph or select one unique repair law.

The electroweak branch is the place where the named pixel coordinate organizes the weak-sector rows. The target-frozen inverse adapter displays

$$M_W = 80.3625 \text{ GeV}, \qquad M_Z = 91.1879 \text{ GeV}.$$

The selected-carrier chart, the archived target-free-formula candidate, and the freeze-once coherent repair surface are recorded separately in the support table below.

2.3 What the electroweak branch fixes

On the declared electroweak running/matching/threshold/scheme surface, the pixel variable P supplies the source basis $(\alpha_U, \alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v)$ used by the archived chart. The displayed value law is an exact implication of the five-part quotient-path certificate stated below, but the finite carrier does not yet emit that certificate. The candidate law evaluates the mass-side pair (W, Z) together with the running-family anchor (a_0, s_0, v). The electromagnetic row is physically read only after Ward projection to the unbroken $\mathrm{U}(1)_Q$ channel; its low-energy value is the $q^2 \to 0$ endpoint of the same transport family. The derivation also includes two explicit benchmark surfaces beneath the public theorem output: the exact selected-carrier chart and the freeze-once coherent repair surface. The same support logic is used later in the paper because flavor continuations have different support levels, whereas the electroweak quantitative lane has an explicit declared-surface contract. Its source payload, same-scheme remainder, and interval-certificate records are public bookkeeping artifacts for the declared bridge from P to the fine-structure endpoint.

The empirical closure surface for that bridge is populated. A documented piecewise $e^+e^- \to$ hadrons compilation, built from resonance parameters and perturbative QCD, evaluates the subtracted dispersion integral to $\Delta\alpha^{(5)}_{\text{had}}(m_Z) = 0.0267591200153 \pm 0.0007524955365$. Inserting it into the endpoint map with the frozen source anchor and lepton transport packet gives $\alpha^{-1} = 136.2636589431$ on $[136.1583832834, 136.3690975240]$, at $P = 1.6310415204$, on the empirical-closure row class. The comparison to the measured endpoint sits in an explicitly compare-only block: the measured value lies outside the interval, and the certified same-scheme anchor gap is $[0.6485541111, 0.8547920666]$ inverse-alpha units. The anchor $a_0(P) = \alpha^{-1}_{\text{em}}(m_Z^2; P)$ is the one-loop renormalization-group value run from the OPH unification scale to m_Z; the physical five-flavor on-shell value is 128.939, and the difference from the OPH anchor 128.308 is 0.631, at the lower edge of the certified gap. The gap records transport missing from the one-loop anchor; it is not evidence by itself that a source chain closes. A source-only bridge that closes this gap reduces to the OPH hadronic spectral measure, which needs a working hadron construction; the empirical-closure endpoint is the working surface in the interim. On the paper surface, the transmutation factor is $\beta_{\text{EW}} = N_c + 1$; overloaded β-ratios appear only on benchmark readouts and are not part of the theorem contract.

3 Results at a Glance

The particle derivation carries the local pixel scale into the directly comparable rows below. Detailed scope statements are collected in the support sections that follow.

Two points are worth stating explicitly. First, the theorem rows and the exact-hit rows are not the same object. The theorem table carries a two-modulus quark non-identifiability result and a corpus-limited charged no-go boundary, while the exact-hit surface above it contains target-anchored same-family witnesses and diagnostic sidecars. Second, "the bosons are finished" is too blunt. The electromagnetic, color, and metric carrier roles have conditional classical-mode receipts but no promoted quantum mass rows; the W boson and Z boson are benchmark frozen-adapter values; and the Higgs boson sits on the declared Higgs/top critical surface.

This support boundary governs the rest of the paper. Whenever a later chapter gives more detail about a continuation family, it should be read through this table. The table records the constructive chain and keeps its support status explicit.

Non-hadron output surface. The paper-facing non-hadron bundle splits by lane as follows:

Concretely, the public values and benchmark checks shown on this surface are

$$\alpha^{-1}(0) = 137.035999177(21), \qquad P \simeq 1.6309682094,$$

$$N_{\mathrm{CRC}}^{\mathrm{EW}} = 3.5323546226929906511\ldots \times 10^{122}, \qquad \mathcal{B}_{\mathrm{EW}} = 0, \qquad \epsilon_H = 0,$$

No nonzero particle mass appears in this prediction ledger. The selected-carrier, value-law, inverse-adapter, and conditional Higgs coordinates are retained only in the technical audit sections below. Target-anchored charged-lepton and quark witness values, including the mixed-convention quark target packet, are retained only in the exact-fit audit artifacts. Compare-only absolute neutrino attachments are likewise withheld from public prediction tables. The neutrino comparison surface keeps representative splitting checks such as

$$\Delta m_{21}^2 = 7.488059465106851 \times 10^{-5}\,\mathrm{eV}^2,$$
$$\Delta m_{31}^2 = 2.5123118727618473 \times 10^{-3}\,\mathrm{eV}^2,$$
$$\Delta m_{32}^2 = 2.4374312781107786 \times 10^{-3}\,\mathrm{eV}^2.$$

The table is lane-based. It states the derivation chain that emits each exact-hit surface and the caveat that prevents a stronger theorem claim.

How to read mismatches and exact hits. Exact numerical agreement is not by itself a proof of a blind prediction. The provenance record classifies the W/Z row as target-used frozen-reference reproduction, the charged-lepton triple as an empirically anchored current-family witness, and the mixed-convention quark packet as a target-anchored selected-frame audit rather than a public source-only prediction. The same record classifies the hierarchy/naturality formula as a zero-residual identity on its declared selected surface, with $\epsilon_H = 0$; the strict source-root and physical-scale certificates remain separate open gates. Where a row does not match, or where it matches only on a constrained surface, the explanations are as follows. The P-closure and $\alpha(0)$ displacement is localized to the Ward-projected Thomson residual. At the implemented fixed point the residual is

$$0.041164012378350542050005414212806738$$

inverse-alpha units. Combining the undressed source/root inverse coupling with the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel gives the mixed-provenance no-hadron diagnostic

$$A_{\alpha_U}^{\mathrm{fp}} = 136.99483516462165\ldots + 0.041124336195630495 = 137.03595950081728\ldots,$$

The sum uses $\alpha_U(P_C)$, while the stored source-audit candidate is paired with P_{cand}. It lands close to the Thomson endpoint and does not reach it, but its mixed provenance prevents source-only promotion. OPH source-only closure requires the Ward-projected hadronic spectral transport. The remaining inverse-alpha shortfall is

$$137.035999177 - A_{\alpha_U}^{\mathrm{fp}} = 0.000039676182720047050005414212807\ldots.$$

Equivalently, the calibrated fixed-point residual can be written

$$\Delta^{\rm fp}_{\rm H,cal} = \alpha_U(P_C)\, C_{24,Q}, \qquad C_{24,Q} = 1.000964785973232625384951114070 2475\ldots.$$

The $C_{24,Q}$ factor is endpoint accounting rather than a source-emitted theorem. At the public endpoint pixel value

$$P \simeq 1.6309682094$$

the endpoint residual package requires

$$\Delta_{\rm source}(P) = 0.041465861005223389053448715357314044\ldots.$$

That scalar belongs to the source-side hadronic spectral transport and scheme-remainder map. The declared empirical route would use a separately labeled $e^+e^- \to$ hadrons spectral input. No same-scheme integrated endpoint payload is emitted here. The first-principles transport row excludes an inserted comparison endpoint. The electroweak support record names the residual map and the RG/matching/threshold/scheme packet. An observer inside the branch measures the dressed Thomson coupling, not the undressed source diagnostic $1/136.994835\ldots$, because a zero-momentum electromagnetic measurement sees the Ward-projected $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme finite endpoint matching have been included. The electroweak W/Z row is therefore a benchmark check. Charged leptons carry a corpus-limited no-go because the determinant trace-lift attachment from the D10 descendants of P to physical charged data is absent. The auxiliary direct-top PDG row differs from the theorem coordinate by 0.20673301656674425 GeV, or 0.28458848947515303 combined standard deviations, because Q007TP and Q007TP4 are distinct extraction codomains; the direct-top response kernel has a corpus-limited no-go boundary. Neutrino absolute masses are not directly measured, and PMNS-angle residuals remain visible comparison tension outside the theorem branch. First-principles hadron masses have no emitted prediction because the required Ward-projected hadronic spectral measure must come from a working OPH hadron construction. Empirical hadron closure rows carry a separate support class.

Local unification surface. The bosonic rows carry one cross-lane statement. The same pixel input P, fixed on the synthesis-paper outer/inner closure relation, fixes the D10/D11 bosonic trunk

$$P \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\rm tr}(P)) \longmapsto v(P) \longmapsto (M_W, M_Z)\,.$$

$$\sigma_{D11,\rm HT} = \alpha_U(P)\cos(2\theta_{W0})/\sqrt{\pi} \longmapsto (m_H, m_t)\,.$$

Here $\alpha_U(P)$ is the branch value selected by the same forward D10 pixel-closure solve, so the bosonic trunk contains no inverse electroweak readback of the internal transmutation data. On the companion gravity side, the same pixel law packages

$$\bar\ell_{\rm SU(2)}(t_{2,\rm run}) \;+\; \bar\ell_{\rm SU(3)}(t_{3,\rm run}) = P/4.$$

The gravity normalization itself is emitted by the selected scale certificate $\gamma_\star = \ell_\star \nu_{\rm Cs}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. With $a_{\rm cell} = P\ell_\star^2$, the Newton area law gives $G_{\rm geom} = \ell_\star^2$, so the pixel cancels. The local unification surface separates one explicit familiar-unit readout package from three support surfaces: the W/Z benchmark sidecar, the conditional D11 declared-surface Higgs/top coordinate, and the gravity-side release with its strict classical-regime clause. On the gravity side, the stated local extension surface uses the lifted product presentation of the realized quotient branch and identifies

$$\bar\ell_{\rm shared} = \bar\ell_{\rm SU(2)}(t_{2,\rm run}) \;+\; \bar\ell_{\rm SU(3)}(t_{3,\rm run}).$$

On that same surface the D10 pixel law fixes $\bar{\ell}_{\text{shared}} = P/4$, and the local SI readout is

$$G_{\text{SI}} = \frac{c^3 \ell_\star^2}{\hbar}$$

from the selected scale certificate. The χ_ν protected-reserve branch uses the same public endpoint convention as

$$P_\chi = P_{\text{C}} = 1.630968209403959\ldots, \qquad \epsilon_{\text{res}} = \frac{P_\chi}{24},$$

after the same-collar shared-edge budget and one-class $\mathbb{Z}_6$ trace receipts pass. This does not make $P/4$ a primitive Hilbert-space dimension: $P/4$ is the local screen-cell entropy budget in the particle and hierarchy lanes, while $P_\chi/24$ is the shared scalar-reserve density used by the χ_ν collar branch. On that declared extension surface the same familiar-unit package reads

$$L_{\text{loc}} = \sqrt{a_{\text{cell}}}\,\widehat{L}(P), \qquad t_{\text{loc}} = \frac{\sqrt{a_{\text{cell}}}}{c}\,\widehat{T}(P),$$

$$E_{\text{loc}} = \frac{\hbar c}{\sqrt{a_{\text{cell}}}}\,\widehat{E}(P), \qquad \Theta_{\text{loc}} = \frac{\hbar c}{k_B\sqrt{a_{\text{cell}}}}\,\widehat{\Theta}(P),$$

with dimensionless $\widehat{L}, \widehat{T}, \widehat{E}, \widehat{\Theta}$. Thus, at fixed P, the local ruler is $\sqrt{a_{\text{cell}}}$; seconds are that ruler divided by the structural Lorentz output c; and GeV and Kelvin are downstream familiar-unit displays of the inverse local ruler through $\hbar$ and k_B. On that declared extension surface the local release bundle is

$$c = 299792458\,\text{m/s},$$
$$G = 6.674299995910528 \times 10^{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2},$$

No nonzero particle-mass coordinate belongs to this release ledger. The W/Z inverse adapter and the conditional Higgs/top coordinates are audit-only quantities discussed in their technical sections. The structural Lorentz output is the common invariant null cone and speed $c_\star$. The decimal 299792458 m/s is exact by the SI definition of the metre, not a predicted magnitude. The paper keeps the G row as an exact scale-readback value, with no literal zero-difference identity against the rounded benchmark 6.6743×10^{-11}.

The detailed inventory below records the public theorem/continuation rows sector by sector. Whenever an exact sidecar or same-family witness is stronger than the public theorem row, the caveat is the one stated in the exact-hit table above.

3.1 Detailed particle inventory

Family	Particle	Support status	OPH value or strongest derived benchmark	Remaining step
Conditional carrier modes	electromagnetic	action/phase receipt	two transverse classical $k^2 = 0$ modes; $\mu^2_{Q,\text{quad}} = 0$ only in the displayed Maxwell action	construct the physical Hilbert space, positive-residue pole, and stable asymptotic photon receipt
Conditional carrier modes	color	perturbative/pre-confinement action receipt	$2\dim G$ transverse classical $k^2 = 0$ modes; $\mu^2_{\text{YM,quad}} = 0$ only in the displayed pure-Yang–Mills expansion	construct a physical positive-residue pole and deconfined asymptotic colored sector; no free gluon is claimed on the confining branch
Conditional carrier modes	tensor	action/background receipt	two Einstein transverse-traceless classical $k^2 = 0$ modes; $\mu^2_{\text{EH,TT}} = 0$ only on the pure-Einstein branch	construct metric quantization, a positive physical Hilbert space, a positive-residue pole, and an asymptotic/EFT particle interpretation
Electroweak branch	W boson	no source-only physical mass	withheld from the prediction ledger	selected-carrier, conditional value-law, and inverse-calibration coordinates remain audit-only; no pole receipt

Family	Particle	Support status	OPH value or strongest derived benchmark	Remaining step
Electroweak branch	Z boson	no source-only physical mass	withheld from the prediction ledger	same boundary as the W row
Higgs/top critical stage	Higgs boson	no source-only physical mass	withheld from the prediction ledger	close the independent source root and physical scale, QT1–QT5, RG/scheme and threshold transport, target-independent D11 rigidity, complex-pole, uncertainty, and no-target dependency-DAG gates
Quark family	top quark	separate target-audit coordinate	withheld from public prediction table	cross-section pole-mass extraction coordinate; it is not a sixth entry in one common running-mass chart and is not selected by the source spread laws
Quark family	up quark	reciprocal-ray candidate rejected	withheld from public prediction table	common-scale dimensionless Yukawa test fails; a source-derived flavor-orbit selector is absent
Quark family	down quark	reciprocal-ray candidate rejected	withheld from public prediction table	same common-scale rejection and six-scalar interface boundary
Quark family	strange quark	held-out reciprocal-ray failure	withheld from public prediction table	remains part of the 19.9% to 21.6% held-out failure across tested scales
Quark family	charm quark	held-out reciprocal-ray failure	withheld from public prediction table	stored GeV-valued matrices are mixed-convention mass textures, not physical Yukawas
Quark family	bottom quark	reciprocal-ray candidate rejected	withheld from public prediction table	common-scale source transport, Higgs normalization, and flavor-orbit selection remain absent
Charged leptons	electron	continuation gap	n/a; exact same-carrier centered readback exists once a charged source pair is emitted, but the absolute scale is blocked by the closed common-shift no-go; benchmark target $g_e^\star = 0.0457789$, equivalently $\Delta_e^{\mathrm{abs},\star} = 3.00398633$	close branch-generator splitting, then emit the lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$; within that lift the exact smaller forcing object is the physical identity-mode equalizer, after which $\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\mathrm{phys}}(Y_e)\,\mathbf{1}$, $A_{\mathrm{ch}}(Y_e) = \mu_{\mathrm{phys}}(Y_e)$, and the readouts g_e, Δ_e^{abs} follow
Charged leptons	muon	continuation gap	n/a; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\widehat{C}_e^{\mathrm{cand}} \to$ branch-generator splitting $\to$ lift $\to \mu_{\mathrm{phys}}(Y_e) \to A_{\mathrm{ch}} \to g_e$ closure chain, with the physical identity-mode equalizer beneath the descended scalar
Charged leptons	tau lepton	continuation gap	n/a; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\widehat{C}_e^{\mathrm{cand}} \to$ branch-generator splitting $\to \mu_{\mathrm{phys}}(Y_e) \to A_{\mathrm{ch}} \to g_e$ closure chain, with the physical identity-mode equalizer beneath the descended scalar
Neutrino comparison	declared f-basis weighted-cycle matrix	rejected target-informed candidate	no flavor-particle mass row; the frozen declared-basis coordinate gives $\theta_{12} = 34.2259^\circ$, $\theta_{23} = 49.7228^\circ$, $\theta_{13} = 8.68636^\circ$, $\delta = 305.581^\circ$, $J = -0.02753$, and, under its declared normal-ordering labels, $\Delta m_{21}^2/\Delta m_{32}^2 = 0.03072111$	the upstream flavor kernel is a hand-written template; the physical charged basis and mass-label rule are open; target-ranked selector development prevents historical-blindness status; the NuFIT 6.1 correlated profile rejects the stored candidate; Majorana and absolute-attachment values remain compare-only
Hadrons	proton	strong-binding construction absent; empirical closure policy emitted	no first-principles emitted prediction	first-principles prediction requires a working OPH-QCD hadron backend, such as GLORB/Echosahedron, plus Ward-projected two-current, higher-point, and transition spectral exports, same-scheme remainder, and production systematics; empirical closure rows use separate $e^+e^- \to$ hadrons input
Hadrons	neutron	strong-binding construction absent; empirical closure policy emitted	no first-principles emitted prediction	same OPH construction gate, with isospin-resolved hadron systematics in the construction support class
Hadrons	neutral pion proxy	strong-binding construction absent; empirical closure policy emitted	no first-principles emitted prediction	same OPH construction gate; local stable-channel surrogate output is not a paper prediction
Hadrons	$\rho(770)^0$ proxy	strong-binding construction absent; empirical closure policy emitted	no first-principles emitted prediction	same OPH construction gate, plus finite-volume resonance extraction in the construction support class

4 Detailed Closure Values, Screen Architecture, and Theorem Packages

This section records the detailed closure-value and theorem checklist used by the particle derivation: the five axioms, the screen-capacity closure, the local pixel closure, the regulated screen realization, and the imported OPH theorem packages. The order is the logical order used later in the paper: axioms and screen language first, closure values next, and theorem packages after that.

4.1 Canonical OPH basis

The particle-spectrum derivation uses the same five axioms as the compact reconstruction paper [5]. They are restated here for local readability because every later particle-family chapter depends on them either directly or through the structural OPH chain imported from that paper.

Axiom 4.1 (Screen Net)**.** *Physical data is organized on a horizon screen S^2 carrying a net of local algebras*

$$P \longmapsto \mathcal{A}(P)$$

for connected patches $P \subset S^2$, with isotony

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

Axiom 4.2 (Overlap Consistency)**.** *For overlapping patches $P_1 \cap P_2 \neq \varnothing$, the local states induced on the shared algebra agree:*

$$\omega_{P_1}|_{\mathcal{A}(P_1 \cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1 \cap P_2)}.$$

Axiom 4.3 (Local MaxEnt and Refinement Stability)**.** *At the regulator scale ℓ_{UV}, the realized branch is selected by maximizing entropy subject to the finitely many homogeneous global-sum constraints*

$$\Big\langle \sum_x O_a(x) \Big\rangle = C_a, \qquad a = 1, \ldots, N_{\mathrm{con}},$$

built from gauge-invariant local densities of support radius $O(\ell_{\mathrm{UV}})$; there is one constraint and one Lagrange multiplier per density label a, not per regulator cell, so the multiplier count is cutoff-independent by construction. The axiom additionally asserts a refinement-closure clause: under refinement, the coarse-grained realized state again lies in the exponential family generated by the same finite density list at the coarser scale. Closure is a substantive renormalization condition, since coarse-graining a finite-range Gibbs family generically generates interactions outside any fixed finite list. The closure defect, the moment-matching I-projection realizing the induced refinement map on the multiplier space, and the Pinsker trace-norm residual bound controlling a nonzero defect are constructed in the companion SM/GR derivation and synthesis papers [5, 2]; the clause is exactly the statement that this defect vanishes along the realized branch, and that vanishing is assumed there rather than proved. Granting it, the realized states belong to one common finite-dimensional MaxEnt family instead of unrelated maximizers.

Axiom 4.4 (Recoverable Generalized Entropy)**.** *A generalized entropy functional exists on caps,*

$$S_{\mathrm{gen}}(C) = S_{\mathrm{bulk}}(C) + \langle L_C \rangle,$$

where L_C is a positive edge-center entropy functional and the semiclassical branch identifies its leading coarse-grained contribution with $A(\partial C)/(4G)$. The functional obeys the recoverability and focusing structure required by the collar and null-modular arguments.

Axiom 4.5 (Minimal Admissible Realization). *Among admissible realized low-energy sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the EFT branch, its admissible light chiral matter content, and one Higgs doublet, the realized package is lexicographically minimal under*

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}}, N_{\mathrm{nonab}}, N_c, N_g),$$

subject to loop coherence, anomaly freedom, refinement-stable light chiral matter, single-Higgs Yukawa completable structure with one connected abelian charge factor, intrinsic CP capability, and weak-sector UV completeness.

For particle physics, the first four axioms provide the observer-centric kinematic and entropic/modular background. The fifth axiom, MAR, is the selector that turns "some compact gauge group reconstructed from the transportable edge-sector category on a cofinal tail carrying the compact paper's explicit refinement receipt, coherent pullback ladder, symmetry, and forgetful-fiber conditions" into the realized Standard Model branch. This paper therefore uses MAR constantly, but only after the compact-gauge reconstruction step has been separated from the later realized-branch selection.

4.2 Regulated screen architecture and patch language

The SM/GR derivation in Ref. [5] states the axioms in algebraic language. Ref. [4] supplies one concrete regulated realization of that language: a federation of finite observer patches with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap packets, record algebras, and local repair instruments. A spherical screen is an observer-visible regulator chart, not a literal microscopic computer. This regulated architecture is important for the particle derivation for three reasons.

First, it makes the screen picture operational. A patch is a finite local algebra in a bounded carrier, and an overlap is a boundary-visible algebra with declared shared observables. Second, it makes the edge-sector language concrete. The particle derivation uses edge sectors, transport, refinement, fusion, and overlap data to build the gauge and flavor branches; Ref. [4] shows how those objects can be realized at fixed cutoff by finite gauge-aware patch carriers. Third, it clarifies the measurement interface. The observer-facing measurement package is a theorem-bearing fixed-cutoff statement about a central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting algebra.

This regulated architecture should be read as a reference implementation, not as the unique OPH UV completion. Ref. [4] is explicit on that boundary: it selects a federated patch-carrier architecture because it is local, finite-dimensional, gauge-aware, naturally compatible with collars, overlap observables, and repair maps, and does not ask the reader to identify the spherical regulator chart with the literal substrate. Hardware evidence is imported only when it appears in a public OPH evidence bundle with stable hashes and verifier receipts. This paper inherits that architecture as a concrete screen-side realization of the OPH language, while keeping microscopic uniqueness separate from the support-visible continuum theorem surfaces supplied by the compact paper.

4.3 Quantitative inputs and where they enter the particle derivation

The quantitative derivation developed here uses two closure coordinates together with the selected no-G scale certificate:

$$P \equiv a_{\text{cell}}/\ell_\star^2, \tag{1}$$

$$N_{\text{CRC}} \equiv \log \dim \mathcal{H}_{\text{tot}}, \tag{2}$$

$$\gamma_\star \equiv \frac{\ell_\star \nu_{\text{Cs}}}{c}, \qquad B_\star \equiv \frac{3\pi}{\ell_\star^2}. \tag{3}$$

The particle-spectrum draft needs to keep these closure values visible because several later sections depend on them in different ways.

Pixel area P. The pixel area is the declared local UV-area ratio. In this derivation it feeds the heat-kernel / edge-law input stage and, through that route, the forward D10 electroweak surface. It is the common upstream numerical variable for the electroweak, Higgs/top, flavor, quark, and hadron-facing branches. The particle paper therefore treats $P = P_\star$ as the inherited outer/inner fixed point

$$P_\star = \varphi + \frac{\sqrt{\pi}}{A_T(P_\star)}$$

from the synthesis paper [2], not as a quantity derived again inside the particle-spectrum argument itself.

Screen capacity N_{CRC}. The screen capacity is not part of the local recovered-core gravity/gauge derivation. It enters the separate screen-capacity branch through the cosmic record-closure target

$$N_{\text{CRC}} = F(N_{\text{CRC}}), \qquad \Lambda_{\text{CRC}}\ell_\star^2 = \frac{3\pi}{N_{\text{CRC}}}, \qquad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

The last equality is the scale-certified display. It does not determine an SI curvature scale from $P_\star$ and N_{CRC} alone. The count-density representation is

$$N_\star = \text{MAR} \arg\max_N \left[\log |\Omega_N^{\text{sc}}| - N\right].$$

In differentiable form, with $\ell(N) = \log |\Omega_N^{\text{sc}}| - N$, this same branch is the unique stable fixed point of $T_\eta(N) = N + \eta\ell'(N)$ when the OPH normal-form count gives a strictly concave normalized density. Thus the capacity is the unique OPH-derived global readback balance point, not a separately fitted cosmological number. In the particle context, this matters mainly for a legacy capacity-level neutrino side estimate and for the cosmological-capacity discussion that frames why local null data do not determine the cosmological constant. The implemented branch uses the static-patch normalization

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P}\right)^2 \simeq 1.05 \times 10^{122}, \qquad N_{\text{scr}} = \pi N_{\text{patch}} \simeq 3.31 \times 10^{122}.$$

This paper should therefore keep the neutrino chapter direct: legacy capacity-level order-of-magnitude bookkeeping and the weighted-cycle neutrino derivation are not the same thing.

Selected scale certificate. The gravity scale is a separate observation-located certificate. On the declared branch,

$$\gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star = \frac{3\pi}{\ell_\star^2}, \qquad G_{\mathrm{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

The local pixel $P_\star$ then identifies $a_{\mathrm{cell}} = P_\star \ell_\star^2$ and $\bar{\ell}_{\mathrm{shared}} = P_\star/4$. In the Newton area law $P_\star$ cancels, so this paper does not treat G as a particle-sector consequence of $P_\star$. $\bar{\ell}_{\mathrm{shared}}$ is a shared-cut density, not the logarithm of an autonomous cell Hilbert factor. The particle branch uses it for cell/edge consistency; it does not select a fixed primitive observer capacity.

Why these are the declared continuous closure/readback values. One of the discipline constraints of the OPH derivation is that particle-sector freedom should not be hidden in a large uncontrolled parameter set. This paper exposes $P_\star$, the N_{CRC} readback fixed point with $N_\star$ as its count representation, and the selected scale certificate, then asks the overlap, modular, gauge, and admissibility machinery to do the rest. Whether later continuation branches stay faithful to that discipline is one of the central questions this paper is meant to make explicit. For the particle-spectrum branch specifically, the nontrivial common quantitative burden sits on $P_\star$: N_{scr} enters the separate capacity/cosmology side and $\gamma_\star$, equivalently $B_\star$, enters the separate gravity-scale side, whereas the reverse-engineering claim for masses and couplings is that one shared pixel fixed point should drive the spectrum instead of a long sector-by-sector input list.

4.4 Technical theorem data carried into the particle-spectrum derivation

This paper uses the same theorem checklist as *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [5], together with the regulator-language clarification integrated into the consensus and microphysics appendices. The ledger matters only insofar as it separates branch-internal statements from declared inputs.

The third axiom internalizes several pieces of infrastructure that often appear as separate assumptions: the local Gibbs form of the regulator-scale state, the quasi-local propagation bound, the endpoint-control estimate for bounded intervals, and the meaning of refinement stability itself. In regulator language, one works with finite local Hilbert spaces and a boundary-fixed overlap action on cut data, often summarized as the R0/R1 presentation. Beyond that regulator package, the compact paper supplies the support-visible BW scaling theorem for the Lorentz/null-modular/Einstein branch. Transportability and the fixed-cutoff bosonic sector category are theorem-produced in the compact paper; the refinement/fiber ladder and cofinal MAR-admissible compact-gauge witness additionally require its explicit compact-gauge refinement receipt. On the central branch the matching gluing obstruction is the combined zero-obstruction transport criterion: vanishing triangle defect plus trivial represented holonomy of the strictified edge 1-cocycle. It introduces no second theorem-side input. Any local-Gibbs or collar-mixing language on that branch is fixed-cutoff recoverability/support control; it is not the source of the compact-gauge witness. For the BW/geometric side, the target is the support-visible extracted prime geometric subnet. The compact paper proves the needed scaling theorem by combining regularized support-visible modular transport, weak-$*$/GNS extraction of the cap pair, support-readable modular covariance, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric 2π-KMS convergence with wrong-scale controls. The unregularized full-algebra common-floor route is deliberately not claimed, because off-support directions may collapse without affecting observer-facing matrix elements. Bare finite consensus does not by itself produce the finite cap-normal BW certificate used by that theorem.

For the particle-spectrum derivation, this has a practical consequence. Structural statements such as compact-gauge reconstruction, the realized Standard Model quotient, the exact hypercharge lattice, the color count $N_c = 3$, and the generation count $N_g = 3$ are not floating independently of the declared input and theorem checklist. The additional classical carrier modes live on explicit quadratic action/background/phase receipts, and particle poles require the stronger quantum receipt. These statements live on a controlled chain whose scope conditions need to be visible in this paper, especially once later chapters turn from structural branches to quantitative closure, continuation, and nonperturbative sectors.

4.5 Core theorem packages used later

This paper will repeatedly use earlier OPH results without re-proving them in full. The theorem packages imported into the present draft are the following.

1. **Patch-net and overlap language.** Ref. [3] organizes overlap repair, fixed-point language, cycle holonomy, and gauge-as-gluing in a form that is especially useful for flavor transport and observer-centric interpretation.

2. **Screen-side regulated architecture.** Ref. [4] gives the federated echosahedral patch-carrier model, the regulated patch-net embedding theorem, and the fixed-cutoff edge heat-kernel / Casimir theorem.

3. **Measurement interface.** Ref. [4], together with the integrated measurement appendices and the synthesis paper *Observers Are All You Need* [2], supplies the fixed-cutoff central-record, Born-rule, and Lüders-conditioning package together with the fixed-cutoff Bell/CHSH theorem stack used when later chapters discuss how measurements pick out definite observer-accessible outcomes and how two-wing comparison laws stay on the quantum side of the classical Bell bound.

4. **Compact gauge reconstruction and Standard Model structure.** *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [5] and the dedicated gauge-group fragment together provide the path from a receipt-certified cofinal tail of refinement-stable edge sectors to a compact group, then from MAR to

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3.$$

5. **Continuation-level topic notes.** The integrated lane appendices carry useful material on supersymmetry, heuristic baryogenesis continuations, proton stability, generation structure, and Yukawa hierarchy, but those sections have to be read with their stated claim boundaries intact. They are sources for later discussion and continuation chapters, not automatic upgrades to recovered-core theorems.

This reading rule governs the whole paper. Structural gauge and carrier results can be used as structural results. The electroweak branch is a forward-emitted Phase-II quantitative branch. The Higgs/top critical stage is a conditional downstream calculation on its declared running, matching, and threshold surface. The quark, charged-lepton, neutrino, and hadron chapters retain their stated continuation or comparison labels unless the underlying derivation chains close more strongly.

5 Gauge Architecture, Standard Model Structure, and Structural Carriers

The particle-spectrum derivation does not start from a list of particles. It starts from the compact-gauge branch. On a cofinal tail carrying the explicit refinement receipt, OPH reconstructs a compact gauge group from the tensor-generated zero-obstruction edge-sector category and forgetful fiber produced by the compact paper's transportability, fixed-cutoff category, and refinement/fiber theorems. The overlap/transport obstruction calculus is the classification stage, not the selection stage. The receipt-conditional compact-gauge witness theorem supplies nonempty realized MAR-admissible branch data, and MAR then selects the realized low-energy package. That logical order matters for this paper. The gauge branch fixes the structural carrier roles before any mass readout. It does not fix their quantum-particle masses; the action-level and quantum spectral gates below decide whether a classical mode or particle row exists.

On the realized branch, the gauge/content result used throughout this paper is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3.$$

Recovering Relativity and the Standard Model from Observer Overlap Consistency [5], the longer main derivation surface, and the dedicated gauge-group proof fragment all agree on the logical split: compact-gauge reconstruction first, realized Standard Model selection second. The first stage gives a compact internal symmetry group under the stated zero-obstruction bosonic sector conditions and compact-gauge refinement receipt. The second stage uses MAR and anomaly cancellation to fix the realized quotient and hypercharge lattice; the minimal coupled carrier fixes $N_c = 3$; intrinsic CP capability together with weak-sector UV completeness bounds N_g; MAR then fixes $N_g = 3$; and Witten parity remains a consistency check on the realized triplet-doublet package.

For the particle zoo this structural branch fixes three carrier roles, not three particle masses. The compact paper's carrier-mode theorem then supplies the following conditional action-level statements. The Maxwell action on the ordinary unbroken electromagnetic vacuum has two transverse classical modes. The pure Yang–Mills action about a flat connection has $2 \dim G$ perturbative transverse modes, but the confined color phase has no asserted gauge-invariant asymptotic gluon. The pure two-derivative Einstein–Hilbert action about Minkowski space has two classical transverse-traceless modes. In every case a quantum particle additionally requires a positive- energy physical Hilbert space, a positive-residue physical two-point pole, and the appropriate asymptotic or deconfinement receipt.

These conditional classical modes should not be confused with quantum-particle mass rows. The group and gravity branches identify the carrier roles; the displayed actions and phases supply their classical propagation; and only a separate quantum spectral receipt can promote them to photon, gluon, or graviton particles. The later boson sections address the W, Z, and Higgs rows on their own electroweak and Higgs/top quantitative stages.

6 Electroweak D10 Branch and the Higgs/Top Critical Stage

The reported bosonic sector distinguishes sharply between emitted numerical rows and scoped non-emitted rows. The active bosonic stages are the electroweak D10 prediction stage and the Higgs/top critical stage, while charged-lepton rows carry their no-go boundary, the neutrino lane contains a rejected weighted-cycle comparison candidate, and hadron rows are backend-gated. The bosonic numerical sector is therefore concentrated in two places: the D10 electroweak branch for the W boson and Z boson, and the Higgs/top critical stage for the Higgs boson together with its conditional D11 companion top coordinate.

6.1 Electroweak chart on the D10 quantitative branch

The electroweak derivation consists of a single-P running family followed by a reduced two-scalar carrier, a selector on that carrier, an exact carrier mass chart, and a conditional quotient-transport theorem beyond the selected carrier. In practical terms, the construction starts from the declared pixel input P, builds the running electroweak family, reduces it to the selected carrier, reads the W boson/Z boson pair from the selected point on that carrier, and then evaluates the mass pair together with the Ward-projected electromagnetic transport family. On the D10 branch, a live forward prediction would have to respect exactly that ordering: first certify the shared pixel input P on the declared D10 running/matching/threshold/scheme surface, thereby fixing the source basis

$$(\alpha_U, \alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v),$$

with $\alpha_U(P)$, equivalently $t_U(P)$ and $t_{\text{tr}}(P)$, fixed by the same forward pixel-closure solve, and only then emit the electroweak transport family forward from that source basis. The completion pipeline classifies that full chain outside the promoted theorem tier. Its W/Z rows are the frozen validation adapter values, and their exact numerical agreement is recorded as a comparison label instead of as a closed D10 Phase-II prediction theorem.

The same quantitative lane has one exact golden-ratio benchmark. If one writes

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)}$$

for the total/bulk/edge hierarchy and imposes exact self-similar balance

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle},$$

then x obeys $x^2 - x - 1 = 0$, so the unique positive equilibrium point is

$$x = \varphi := \frac{1+\sqrt{5}}{2}.$$

Equivalently the order parameter $A_\varphi(x) = x - 1 - \frac{1}{x}$ vanishes there. The synthesis paper fixes the exact numerical value of P through the outer/inner closure relation. The point of the equilibrium theorem here is that the proximity to φ has a structural reason. Exact balance is too symmetric to support durable records, structure, and dynamics, so the closure solve fixes a small equilibrium-breaking detuning away from that balance point.

The compare-only W/Z adapter rows in the final bundle are

$$m_W = 80.3625 \text{ GeV}, \qquad m_Z = 91.1879 \text{ GeV}.$$

These numbers sit on the frozen validation surface, not on a target-free prediction theorem. The exact selected-carrier chart is explicit on disk and emits the local pair

$$m_W^{\text{carrier}} = 80.38629169244275 \text{ GeV}, \qquad m_Z^{\text{carrier}} = 91.18290444674243 \text{ GeV},$$

so the mathematics distinguishes the selected-carrier chart, the archived value-law candidate $(80.3770000154, 91.1879780779)$ GeV, the conditional QT1–QT5 implication, and the target-frozen inverse repair surface. The older pair $(80.377, 91.1879780919)$ GeV is a rounded legacy alias of the candidate, not the inverse adapter. None is a physical pole theorem.

Theorem 6.1 (Freeze-once coherent electroweak repair law)**.** *Let the emitted running/core electroweak basis be*

$$Q_{\rm run}(P) = \big(\alpha_{Y,m_Z}(P), \alpha_{2,m_Z}(P), v(P), \eta_{\rm source}(P)\big),$$

and write

$$\tau_Y(\tau_2) = -\frac{\tau_2 + 2\eta_{\rm source}}{1 + 4\tau_2^2}, \qquad n_{\rm fiber}(\tau_2) = 1 + \frac{\alpha_{Y,m_Z}\tau_Y(\tau_2) + \alpha_{2,m_Z}\tau_2}{\alpha_{Y,m_Z} + \alpha_{2,m_Z}}.$$

Freeze one authoritative target pair

$$\mathcal{T}^\dagger = (M_W^\dagger, M_Z^\dagger), \qquad 0 < M_W^\dagger < M_Z^\dagger,$$

and define the charged anchor

$$\tau_{2,W}^\dagger = \frac{M_W^{\dagger\,2}}{\pi v^2 \alpha_{2,m_Z}} - 1, \qquad \delta\alpha_2^\dagger = \frac{M_W^{\dagger\,2}}{\pi v^2} - \alpha_{2,m_Z}.$$

Then the fiber-parallel hypercharge leg is

$$\tau_Y^\dagger = \tau_Y(\tau_{2,W}^\dagger), \qquad n_{\rm fiber}^\dagger = n_{\rm fiber}(\tau_{2,W}^\dagger),$$

$$M_{Z,\rm fiber}^\dagger = v\sqrt{\pi(\alpha_{Y,m_Z} + \alpha_{2,m_Z})\, n_{\rm fiber}^\dagger},$$

and the orthogonal neutral-shear scalar is

$$\delta M_Z^\perp = M_Z^\dagger - M_{Z,\rm fiber}^\dagger,$$

equivalently

$$\delta n^\dagger = \frac{(M_Z^\dagger + M_{Z,\rm fiber}^\dagger)\,\delta M_Z^\perp}{\pi v^2(\alpha_{Y,m_Z} + \alpha_{2,m_Z})}, \qquad \delta\alpha_Y^\perp = \frac{(M_Z^\dagger + M_{Z,\rm fiber}^\dagger)\,\delta M_Z^\perp}{\pi v^2}.$$

The fiber-parallel hypercharge motion is

$$\delta\alpha_Y^\parallel = \alpha_{Y,m_Z}\,\frac{8\eta_{\rm source}(\tau_{2,W}^\dagger)^2 - \tau_{2,W}^\dagger}{1 + 4(\tau_{2,W}^\dagger)^2}.$$

Therefore the unique coherent repair package is

$$\Sigma_{EW}^\dagger = (\delta\alpha_2^\dagger, \delta\alpha_Y^\parallel, \delta\alpha_Y^\perp),$$

equivalently $(\tau_{2,W}^\dagger, \delta n^\dagger)$*. If the selected carrier anchor is the* $\tau_2 = 0$ *fiber point, so that*

$$\alpha_{2,*} = \alpha_{2,m_Z}, \qquad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\rm source}),$$

then the coherent validation couplings are

$$\alpha_{2,\dagger} = \alpha_{2,*} + \delta\alpha_2^\dagger, \qquad \alpha_{Y,\dagger} = \alpha_{Y,*} + \delta\alpha_Y^\parallel + \delta\alpha_Y^\perp,$$

and they emit one coherent quintet

$$M_W^\dagger = v\sqrt{\pi\alpha_{2,\dagger}}, \qquad M_Z^\dagger = v\sqrt{\pi(\alpha_{Y,\dagger} + \alpha_{2,\dagger})},$$

$$\alpha_{\rm em,\dagger}^{-1} = \frac{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}{\alpha_{Y,\dagger}\alpha_{2,\dagger}}, \qquad \sin^2\theta_{W,\dagger} = \frac{\alpha_{Y,\dagger}}{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}.$$

The frozen-target electroweak validation law is one unique coherent value-emission law on top of the closed fiber map.

For the target-frozen inverse-adapter audit surface,

$$\begin{aligned}
\alpha_{2,\star} &= 0.03377843630219015, \\
\alpha_{Y,\star} &= 0.009682831911900495, \\
\eta_{\text{source}} &= 0.022147000871961295, v \quad = 246.76711732749683 \text{ GeV}, \\
M_W^\dagger &= 80.3625 \text{ GeV}, \\
M_Z^\dagger &= 91.1879 \text{ GeV},
\end{aligned}$$

and the law emits

$$\begin{aligned}
\tau_{2,W}^\dagger &= -0.0005918464744071317, \\
\delta\alpha_2^\dagger &= -1.9991648436440412 \times 10^{-5}, \\
M_{Z,\text{fiber}}^\dagger &= 91.16822266259587 \text{ GeV}, \\
\delta M_Z^\perp &= 19.6773374041328 \text{ MeV}, \\
\delta\alpha_Y^\parallel &= 5.9969727526906094 \times 10^{-6}, \\
\delta\alpha_Y^\perp &= 1.8756950557734103 \times 10^{-5}, \\
\delta n^\dagger &= 4.2716772021678714 \times 10^{-4}.
\end{aligned}$$

So the coherent validation couplings are

$$\begin{aligned}
\alpha_{2,\dagger} &= 0.03375844465375371, \\
\alpha_{Y,\dagger} &= 0.00970758583521092,
\end{aligned}$$

and the same coherent branch emits

$$M_W^\dagger = 80.3625 \text{ GeV}, \qquad M_Z^\dagger = 91.1879 \text{ GeV},$$

$$\begin{aligned}
\alpha_{\text{em},\dagger}^{-1} &= 132.6344411210187, \\
\sin^2\theta_{W,\dagger} &= 0.22333729871365943.
\end{aligned}$$

These two carrier-side scalars are bookkeeping data on the frozen validation surface. They are not the public electromagnetic readout on the Ward-projected D10 lane. That freeze-once law provides an exact coherent inverse-calibration surface. On that target-frozen surface the reference W/Z pair is hit exactly, but only as compare-only validation. It is excluded from every prediction ledger. The archived forward value law has a stronger conditional theorem: it follows uniquely from a finite quotient-path certificate. The certificate is not emitted by the five axioms or by the existing carrier artifacts.

Definition 6.2 (D10 quotient-path certificate)**.** *After quotienting gauge representatives, port labels, scheduler data, and other hidden presentation coordinates, require:*

QT1. *the real, CP-even, color-singlet, charge-preserving response through two returns is exactly* $\mathbb{R}q_2 \oplus \mathbb{R}q_n$*, with no third scalar or charged-neutral off-block term;*

QT2. *the primitive response is bilinear in* $(\eta_{\text{source}}, \alpha_U)$*, one primitive event has fixed unit response normalization, and the quotient probability measure gives each of the four transmutation slots weight* $1/4$*, yielding* $\lambda_{\text{EW}} = \eta_{\text{source}}\alpha_U/4$*;*

QT3. *explicit carrier path lists, carrying one factor of* η_{source} *per return and uniform color-orbit measure* $1/3$*, have primitive, one-return, and two-return incidences* $(1, 2/3, 1)$ *for* q_2 *and* $(1, 4/3, 2)$ *for* q_n*, while the diagonal* $\mathbb{Z}_6$ *projector in the declared response representation has normalized trace* $1/6$*, source amplitude* ρ_{EW}*, and subtracts* $(\rho_{\text{EW}}/6)\eta_{\text{source}}^2$ *from both degree-two characters;*

QT4. *mismatch descent fixes the charged sign as contracting and the neutral sign as uplifting, and the parallel hypercharge fibre is the least-norm minimizer with Gram factor* $1+4\tau_2^2$ *and residual pairing* $\tau_2 + 2\eta_{\text{source}}$*;*

QT5. *the listed paths exhaust the admissible class: deeper paths, alternative central weights, and mixed terms are absent, quotient-trivial, or separated by a positive target-independent MAR gap.*

Theorem 6.3 (Conditional D10 quotient-transport value law)**.** *Let the D10 source-only emitted basis be*

$$Q_{\text{src}}(P) = (\alpha_U(P), \alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\text{source}}(P), v(P)),$$

and define

$$\rho_{\text{EW}} := \frac{\alpha_{2,m_Z} - \alpha_{Y,m_Z}}{\alpha_{2,m_Z} + \alpha_{Y,m_Z}}, \lambda_{\text{EW}} := \frac{\eta_{\text{source}}\alpha_U}{4}.$$

Assume that every entry is emitted on the same source branch, that $\eta_{\text{source}} = \rho_{\text{EW}}\alpha_U$*, and that no measured* W/Z*, measured* v*, fitted electroweak parameter, or calibrated proxy is an ancestor of the tuple. Hence* $\lambda_{\text{EW}} = \eta_{\text{source}}^2/(4\rho_{\text{EW}})$*. Assume also the finite certificate of Definition 6.2 and the physical-domain inequalities* $\alpha_2' > 0$*,* $\alpha_Y' > 0$*. Then the beyond-selected-carrier transport map is uniquely*

$$\tau_2^{\text{exact}} = -\lambda_{\text{EW}}\left(1 + \frac{2}{3}\eta_{\text{source}} + \left(1 - \frac{\rho_{\text{EW}}}{6}\right)\eta_{\text{source}}^2\right),$$

$$\delta n^{\text{exact}} = \lambda_{\text{EW}}\left(1 + \frac{4}{3}\eta_{\text{source}} + \left(2 - \frac{\rho_{\text{EW}}}{6}\right)\eta_{\text{source}}^2\right),$$

with fiber hypercharge transport

$$\tau_Y^{\text{fiber}} = -\frac{\tau_2^{\text{exact}} + 2\eta_{\text{source}}}{1 + 4(\tau_2^{\text{exact}})^2}.$$

The coherent transport shifts are

$$\delta\alpha_2 = \alpha_{2,m_Z}\tau_2^{\text{exact}}, \qquad \delta\alpha_Y^{\parallel} = \alpha_{Y,m_Z}\frac{8\eta_{\text{source}}(\tau_2^{\text{exact}})^2 - \tau_2^{\text{exact}}}{1 + 4(\tau_2^{\text{exact}})^2},$$

$$\delta\alpha_Y^{\perp} = (\alpha_{2,m_Z} + \alpha_{Y,m_Z})\,\delta n^{\text{exact}}.$$

With

$$\alpha_{2,*} = \alpha_{2,m_Z}, \qquad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\text{source}}),$$

the coherent transport couplings are

$$\alpha_2' = \alpha_{2,*} + \delta\alpha_2, \qquad \alpha_Y' = \alpha_{Y,*} + \delta\alpha_Y^{\parallel} + \delta\alpha_Y^{\perp},$$

and they emit one coherent D10 mass chart

$$M_W^{(10)} = v\sqrt{\pi\alpha_2'}, \qquad M_Z^{(10)} = v\sqrt{\pi(\alpha_2' + \alpha_Y')}, \qquad v = v(P).$$

$$a_0(P) := \frac{\alpha_{2,m_Z}(P) + \alpha_{Y,m_Z}(P)}{\alpha_{2,m_Z}(P)\alpha_{Y,m_Z}(P)}, \qquad s_0(P) := \frac{\alpha_{Y,m_Z}(P)}{\alpha_{2,m_Z}(P) + \alpha_{Y,m_Z}(P)}.$$

Thus the certificate fixes the candidate mass pair together with the running-family anchor (a_0, s_0, v) from the D10 basis. The physical electromagnetic row is read from the Ward-projected unbroken $\mathrm{U}(1)_Q$ channel in Theorem 6.5.

Proof. QT1 reduces every admissible response to the two displayed coordinates. QT2 fixes their common primitive activity. QT3 gives the charged and neutral path characters, including the common $\mathbb{Z}_6$ subtraction, and QT4 fixes their signs. The fibre functional

$$\mathcal{J}_Y(t) = \frac{1}{2}(1 + 4(\tau_2^{\mathrm{exact}})^2)t^2 + (\tau_2^{\mathrm{exact}} + 2\eta_{\mathrm{source}})t$$

is strictly convex, so its unique minimizer is $-(\tau_2^{\mathrm{exact}} + 2\eta_{\mathrm{source}})/(1 + 4(\tau_2^{\mathrm{exact}})^2)$. Substitution gives the repaired couplings and the one-Higgs mass chart. QT5 excludes every output-changing admissible deformation, which proves uniqueness on the certified class. $\square$

Using the same frozen electroweak validation basis

$$\begin{aligned}
\alpha_{2,m_Z} &= 0.03377843630219015,\\
\alpha_{Y,m_Z} &= 0.010131601067241624,\\
\alpha_U &= 0.04112498041477454,\\
\eta_{\mathrm{source}} &= 0.022147000871961295,\\
v &= 246.76711732749683\ \mathrm{GeV},
\end{aligned}$$

the conditional quotient-transport law evaluates to

$$M_W^{(10)} = 80.37700001539531\ \mathrm{GeV}, \qquad M_Z^{(10)} = 91.18797807794321\ \mathrm{GeV},$$

$$\begin{aligned}
a_0 &= 128.30576920234813,\\
s_0 &= 0.23073542347506173.
\end{aligned}$$

This specialization verifies the archived value-law algebra. It does not derive QT1–QT5, establish that this calibration tuple belongs to the strict source-audit pixel branch, or identify W, Z as complex propagator poles. A machine-checkable promotion receipt must provide the finite path lists, quotient canonicalizer, exact incidence and central-trace checks, fibre Gram calculation, deformation enumeration or positive MAR gap, and a same-branch no-target dependency DAG.

Proposition 6.4 (Local nondegeneracy of the two-output D10 mass chart)**.** *Define*

$$n_{\mathrm{fib}}(\tau_2) := 1 + \frac{\alpha_2\tau_2 + \alpha_Y\tau_Y(\tau_2)}{\alpha_2 + \alpha_Y}.$$

On the physical domain $1 + \tau_2 > 0$ and $n_{\mathrm{fib}}(\tau_2) + \delta n > 0$,

$$\det \frac{\partial(M_W^{(10)}, M_Z^{(10)})}{\partial(\tau_2, \delta n)} = \frac{M_W^{(10)} M_Z^{(10)}}{4(1+\tau_2)\,[n_{\mathrm{fib}}(\tau_2) + \delta n]} > 0.$$

Holding $(\alpha_2, \alpha_Y, v, \eta)$ fixed, the two-output chart is locally nondegenerate. This determinant neither proves QT1 nor excludes an additional physical response channel. Within the displayed two-coordinate chart, the missing content is a source selector rather than another fit coordinate.

Proof. $M_W^{(10)}$ is independent of δn, while

$$\frac{\partial M_W^{(10)}}{\partial \tau_2} = \frac{M_W^{(10)}}{2(1+\tau_2)}, \qquad \frac{\partial M_Z^{(10)}}{\partial \delta n} = \frac{M_Z^{(10)}}{2[n_{\mathrm{fib}}(\tau_2)+\delta n]}.$$

The determinant is the product of these diagonal entries. $\square$

Theorem 6.5 (Ward-projected $\mathrm{U}(1)_Q$ transport law and Thomson-limit electromagnetic readout)**.** *Let the D10 source-only running family be locked by the forward pixel solve on the declared running/matching/threshold/scheme surface, with source basis*

$$Q_{\mathrm{src}}(P) = (\alpha_U(P), \alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\mathrm{source}}(P), v(P)).$$

Assume the realized D9 branch, so that

$$Q = T_3 + Y$$

on the realized low-energy branch and color-singlet physical states carry integer Q-charge. Assume further that the post-selector electroweak transport object is the populated quotient-local kernel

$$K_{D10}^{\mathrm{EW}}(q^2;P) = \{\Pi_{AA}(q^2;P), \Pi_{AZ}(q^2;P), \Pi_{ZZ}(q^2;P), \Pi_{WW}(q^2;P)\}$$

on the declared physical observable algebra, together with a Ward projector

$$\mathcal{W}_Q : K_{D10}^{\mathrm{EW}}(q^2;P) \longrightarrow \Pi_Q(q^2;P)$$

onto the unbroken electromagnetic channel satisfying at $q^2 = 0$

$$\mathcal{W}_Q[\Pi_{AZ}(0;P)] = 0, \qquad \mathcal{W}_Q[\Pi_{AA}(0;P)] \neq 0.$$

Assume the abelian projected edge-sector probabilities obey the U(1) *heat-kernel law*

$$p_n(q^2;P) \propto e^{-t_Q(q^2;P)\lambda_n}, \qquad \lambda_n = n^2,$$

equivalently

$$g_Q^2(q^2;P) = \frac{t_Q(q^2;P)}{2\pi},$$

and that the scalar readout package satisfies the exact provenance lock

$$\begin{aligned} \textit{family\_source\_id} &= \texttt{d10\_running\_tree}, \\ \textit{scheme\_id} &= \texttt{freeze\_once}, \\ \textit{origin\_kernel\_id} &= \texttt{EWTransportKernel\_D10}. \end{aligned}$$

Then the unique electromagnetic coupling readout on the Ward-projected D10 lane is

$$\alpha_{\mathrm{em}}^{-1}(q^2;P) = \frac{8\pi^2}{t_Q(q^2;P)}.$$

If

$$a_0(P) := \alpha_{\mathrm{em}}^{-1}(m_Z^2;P),$$

then the unique zero-normalized affine scalar on the same source-locked family is

$$\delta_\alpha(q^2; m_Z^2; P) := \frac{t_Q(m_Z^2; P)}{t_Q(q^2; P)} - 1,$$

so that

$$\alpha_{\rm em}^{-1}(q^2; P) = a_0(P)(1 + \delta_\alpha(q^2; m_Z^2; P)).$$

Hence the Thomson-limit electromagnetic readout on that same Ward-projected electromagnetic kernel:

$$\alpha_{\rm Th}^{-1}(P) := \lim_{q^2 \to 0} \alpha_{\rm em}^{-1}(q^2; P) = a_0(P)\,\frac{t_Q(m_Z^2; P)}{t_Q(0; P)}.$$

Consequently, the low-energy electromagnetic row on this lane is read as the Thomson endpoint of the D10 electromagnetic transport family instead of as a mass-chart byproduct.

Proof. The realized D9 branch fixes the electric charge operator by $Q = T_3 + Y$, and the quotient-protected charge theorem fixes the physical color-singlet Q-lattice to be integral. The abelian edge-sector theorem supplies the heat-kernel extraction rule

$$p_n \propto e^{-t_Q \lambda_n}, \qquad g_Q^2 = \frac{t_Q}{2\pi},$$

so the Ward-projected Q-channel determines a unique transport time $t_Q(q^2; P)$. Because the Ward projector kills A - Z mixing at $q^2 = 0$, the selected channel is the physical unbroken electromagnetic channel. Then

$$\alpha_{\rm em} = \frac{g_Q^2}{4\pi} = \frac{t_Q}{8\pi^2},$$

hence

$$\alpha_{\rm em}^{-1} = \frac{8\pi^2}{t_Q}.$$

Evaluating at $q^2 = m_Z^2$ defines the source-locked anchor $a_0(P)$, and dividing by the same expression at general q^2 yields

$$\delta_\alpha = \frac{t_Q(m_Z^2)}{t_Q(q^2)} - 1.$$

The provenance-equality clause forces one running-family source, one frozen scheme, and one origin kernel for all scalar readouts, so no mixed-family or Z-only surrogate is promotable. The Thomson formula is the $q^2 \to 0$ limit of the same identity. □

Corollary 6.6 (Conditional Ward-projected Maxwell normalization)**.** *On the same Ward-projected* $\mathrm{U}(1)_Q$ *lane, assume in addition the low-energy Maxwell action with the displayed nonzero kinetic normalization. Then the electromagnetic field strength and current obey*

$$F_Q = dA_Q, \qquad dF_Q = 0, \qquad d{*}F_Q = g_Q^2(q^2; P){*}J_Q,$$

with

$$g_Q^2(q^2; P) = \frac{t_Q(q^2; P)}{2\pi}, \qquad \alpha_{\rm em}^{-1}(q^2; P) = \frac{8\pi^2}{t_Q(q^2; P)}.$$

At the Thomson endpoint, this is the Maxwell normalization used by the public fine-structure row.

Proof. The compact reconstruction supplies the unbroken abelian electromagnetic connection direction $\mathrm{U}(1)_Q$ with charge generator $Q = T_3 + Y$, but does not supply the Maxwell kinetic action. On an abelian factor the compact-gauge curvature restricts to $F_Q = dA_Q$, so $d^2 = 0$ gives $dF_Q = 0$. Varying the quadratic electromagnetic action in the OPH coupling convention gives $d{*}F_Q = g_Q^2 {*}J_Q$. The theorem above identifies the same branch coupling as $g_Q^2 = t_Q/(2\pi)$, hence $\alpha_{\mathrm{em}}^{-1} = 8\pi^2/t_Q$. $\square$

Corollary 6.7 (Source-only release criterion on the Ward-projected D10 lane)**.** *The Ward-projected D10 lane supports a source-only fine-structure release row only when the populated Ward-projected transport kernel satisfies*

$$\lim_{q^2\to 0} \alpha_{\mathrm{em}}^{-1}(q^2; P) = \alpha_{\mathrm{Th}}^{-1}(P),$$

on the same source-locked family and without any separate endpoint value feeding that lane. Concretely, the release packet must contain `WARD_Q_SPECTRAL_MEASURE_RECEIPT`, `J24Q_SOURCE_JACOBI_RECEIPT`, `OMEGAQ_NORMALIZATION_RECEIPT`, `XI_SAME_SCHEME_REMAINDER_RECEIPT`, `NO_THOMSON_TARGET_LEAK_DAG`, `FULL_PIXEL_CONTRACTION_INTERVAL`, *and* `ALPHA_PREDICTION_BEATS_DIRECT_ME` *under the direct-measurement interval adopted for that release. Until those objects are emitted, the public endpoint remains an empirical hadron-closure row.*

This release criterion is the two-current marginal of the larger OPH-QCD hadronic precision backend. It is adequate for the Thomson endpoint and HVP branch only if the same source law also records the QCD quotient ensemble, source parameter map, Ward current normalization, systematics ledger, and no-target-leak DAG. Full hadronic precision claims require the higher-point and transition spectral exports from that same backend rather than reusing the scalar fine-structure residual.

The endpoint table records the source-locked anchor from the runtime candidate,

$$a_0(P) = 128.307965473286248209961108741 7567\ldots,$$

and the OPH-plus-empirical hadron-closure value

$$\alpha_{\mathrm{emp}}^{-1}(0) = 136.2636589431201, \qquad \alpha_{\mathrm{emp}}^{-1}(0) \in [136.1583832834267, 136.3690975239657].$$

The repository's `code/P_derivation` witness path records the source-side audit trunk without a built-in reference inverse-α value. The measured endpoint 137.035999177(21) is excluded as a source-only closure-solve input and remains a separate comparison coordinate. The source-side audit trunk emits

$$\alpha_{\mathrm{root}}^{-1} = \texttt{136.9948351646216494579499945857871932620 29} \qquad ,$$

so the difference from the endpoint readout is recorded as an endpoint-and-matching audit packet.

Remark 6.8. *The empirical closure row is a Thomson-limit inverse fine-structure evaluation using external hadron data. The source anchor* $a_0(P) = 128.3079654732862482099611087417567\ldots$ *is the electroweak-scale running-family value at* m_Z^2*, while the measured row is separate. The 2022 CODATA/NIST reference value is* $\alpha^{-1} = 137.035999177(21)$ *[6].*

The target-frozen inverse repair surface provides compare-only validation and is excluded from the prediction ledger. The companion carrier-side pair

$$\alpha_{\mathrm{em},\dagger}^{-1} = 132.6344411210187,$$
$$\sin^2\theta_{W,\dagger} = 0.22333729871365943$$

is mass-chart bookkeeping on that frozen validation surface. It is not the public electromagnetic readout on the Ward-projected $U(1)_Q$ lane.

The electroweak quantitative-closure branch also contains several subordinate theorem objects: an underdetermination theorem for the quadratic repair family, the smallest theorem route through $\mathrm{ColorBalancedQuadraticRepairDescent}_{EW}$, the candidate mass-emitter contract beneath the quarantined W/Z audit rows, and the Ward-projected electromagnetic transport theorem above the source-locked running-family anchor. The scalar-package consistency clause is the shared provenance lock

$$(\texttt{d10\_running\_tree}, \texttt{freeze\_once}, \texttt{EWTransportKernel\_D10}).$$

A further forward transmutation certificate sits directly beneath the candidate readout and records that the runtime source-side basis reconstructs the same $\alpha_U(P)$, $t_U(P)$, and $t_{\mathrm{tr}}(P)$ as the pixel-closure solve without reading them back from measured couplings. This runtime separation does not prove target-free historical ancestry. These objects locate the archived repair formula inside the quantitative branch; they do not supply QT1–QT5 or a physical pole receipt.

6.2 Hierarchy interpretation on the local transmutation branch

The selected branch certifies a dimensionless electroweak hierarchy/naturality relation. The large cosmic horizon ratio and the electroweak hierarchy belong to different OPH branches. The global capacity branch N_{CRC} accounts for the cosmic area and horizon-length hierarchy. The local pixel/transmutation readout is the ratio

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P_\star)}\right].$$

The hierarchy certificate records the interval

$$I_U = [0.041123336195630494,\ 0.041125336195630496],$$

the Krawczyk inclusion

$$K(I_U) \subset [0.0411243357185544983,\ 0.0411243366727064662] \subset \mathrm{int}(I_U),$$

and a strictly negative derivative enclosure $[-10.995768, -10.985284]$. Thus the local source equation has a unique zero in that enclosure. On the public endpoint branch,

$$\begin{aligned}\alpha_U(P_\star) &= 0.041124336195630495,\\ \frac{v}{E_\star} &= 2.0199803239725553 \times 10^{-17}.\end{aligned}$$

The source-audit branch gives

$$\begin{aligned}P_{\mathrm{cand}} &= \texttt{1.630972095694329018179678925611918842701669},\\ \frac{v(P_{\mathrm{cand}})}{E_\star} &= 2.0198114150099223 \times 10^{-17},\end{aligned}$$

with the public Thomson endpoint excluded upstream. These two named branches must not be merged. The dimensionless ratio is generated by the ordinary exponential transmutation factor controlled by the unified diffusion coupling; a weak scale in GeV additionally requires an independently source-closed $E_\star$. The twenty-four-round capacity contraction belongs to the global horizon-length ratio.

This is also the OPH reading of Higgs naturalness on the declared D10/D11 surface. The observer-visible scalar mass is the normal-form readout

$$m_H = H_{\mathrm{OPH}}(P_\star)$$

on the selected branch. The split into a bare Higgs mass plus a cutoff correction is a regulator coordinate split outside the observer-facing physical variable. A scalar relevant deformation absent from the retained branch constraints is off-branch; a scalar deformation present on the selected branch has its value fixed by the branch equations. The claim inherits the D10/D11 quantitative surface and the $\mathcal{R}_U$ precision policy. The selected exact source-to-Higgs coarse-graining square has $\epsilon_H = 0$, with $\epsilon_H \in [0,0]$, as stated below.

6.3 Local/global resonance continuation for the hierarchy

The local/global hierarchy resonance compares the D10 transmutation exponent with the global capacity branch at the same joint pair $(P_\star, N_{\mathrm{CRC}})$. The target relation is

$$t_{\mathrm{tr}}(P_\star) = \frac{P_\star}{12}\log\left(\frac{N_{\mathrm{CRC}}}{\pi}\right),$$

or equivalently

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N_{\mathrm{CRC}}}{\pi}\right)^{-P_\star/12}.$$

Writing

$$|g'_\star| = \left(\frac{N_{\mathrm{CRC}}}{\pi}\right)^{-1/48}$$

turns the same expression into

$$\frac{v}{E_{\mathrm{cell}}} = |g'_\star|^{4P_\star}.$$

This packages the electroweak hierarchy as a resonance between the local pixel fixed point and the global screen-capacity fixed point. The screen branch supplies twelve irreducible curvature ports. Including reversible write/check orientation gives a 24-slot oriented repair register, and the hierarchy exponent uses this 24-fold repair normalization through $m_{\mathrm{rep}} = 24$. The local weak hierarchy reads the global screen depth through a $4P_\star$ electroweak projection.

The repair count is fixed by the observer-visible product-adjoint spectrum. On the realized product branch,

$$m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8+3+1) = 24.$$

The factor 2 is the reversible write/verify orientation of each repair channel. The $\mathfrak{su}(5)$ adjoint has the same single-orientation integer for a different support: its X/Y mixed gauge channels are absent from the OPH product branch.

Lemma 6.9 (Global repair-tick lemma)**.** *Let $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$ be the cosmic record-capacity fixed point with observed-branch readout $N_{\mathrm{CRC}} = S_{\mathrm{dS}}$ and D6 normalization*

$$N_{\mathrm{CRC}} = \pi\left(\frac{r_{\mathrm{CRC}}}{\ell_\star}\right)^2.$$

Work in the multiplicative screen-radius coordinate $\rho = \ell/r_{\mathrm{CRC}}$, so the horizon sits at $\rho = 1$ and the local cell at $\rho_\star = \ell_\star/r_{\mathrm{CRC}}$. Define the global repair cycle G_N at the fixed point as the scale-free *readback transport of the horizon record coordinate, with fixed-point closure (readback without deficit or slack) meaning* $G_N(1) = \rho_\star$ *exactly. The representation-to-spectrum theorem gives* $m_{\text{rep}} = 24$*, so the selected resonance branch decomposes as* $G_N = g_N^{24}$ *with* g_N *the positive root. Then*

$$G_N(\rho) = \left(\frac{N_{\text{CRC}}}{\pi}\right)^{-1/2}\rho, \qquad g_N(\rho) = \left(\frac{N_{\text{CRC}}}{\pi}\right)^{-1/48}\rho, \qquad |g_\star'| = \left(\frac{N_{\text{CRC}}}{\pi}\right)^{-1/48}.$$

More generally an m*-tick decomposition gives* $|g_\star'| = (N_{\text{CRC}}/\pi)^{-1/(2m)}$*;* $m_{\text{rep}} = 24$ *gives the stated exponent.*

Proof. The D6 normalization gives $\rho_\star = \ell_\star/r_{\text{CRC}} = (N_{\text{CRC}}/\pi)^{-1/2}$. The closure condition follows at the readback interface. The Observers synthesis defines the readback map as $F(N) = \text{Cap}_{\text{read}}(\text{Obs}(\text{nf}_{r,N}(\mathfrak{U}_{r,N})))$ in the refinement limit, the capacity reconstructed by the interior observer sector from the horizon record. Modeling Cap_{read} by the D6 area law at the effective delivery resolution ρ_{read}, the canonical scale-covariant extension of $N = \pi(r/\ell)^2$, gives the D6-consistent counting identification. The corpus defines Cap_{read} as reconstructed capacity; this certificate adds the D6-consistent counting rule. The readback reconstructs $F(N) = \pi/\rho_{\text{read}}^2$, so the fixed-point equation $N_{\text{CRC}} = F(N_{\text{CRC}}) = \pi/\rho_\star^2$ forces $\rho_{\text{read}} = \rho_\star$ for positive roots: deficit ($\rho_{\text{read}} > \rho_\star$) gives $F(N) < N$, slack ($\rho_{\text{read}} < \rho_\star$) gives $F(N) > N$. This is exactly $G_N(1) = \rho_\star$. Homogeneity forces $G_N(\rho) = Q\rho$ with $Q > 0$, hence $Q = (N_{\text{CRC}}/\pi)^{-1/2}$. The homogeneous one-tick map with $g_N^{24} = G_N$ is the positive 24th root, so its multiplier, and hence its absolute derivative in this coordinate, is $(N_{\text{CRC}}/\pi)^{-1/(2\cdot 24)} = (N_{\text{CRC}}/\pi)^{-1/48}$. □

The emitted particle rows continue to use the $\mathcal{R}_U$ certificate above. Lemma 6.9 closes the global tick side with the derived representation-to-spectrum count $m_{\text{rep}} = 24$. The proof derives the closure transport at the interface level of F under the D6 area-law counting model of Cap_{read}.

Theorem 6.10 (Joint product fixed-point branch)**.** *Let* $x = \log N$*, let* $X = I_P \times I_x$*, and equip* X *with*

$$d((P,x),(Q,y)) = \max\{|P-Q|,\ |x-y|\}.$$

On the product-separated source branch the joint map is

$$\mathcal{J}(P,x) = (\Gamma(P), \widehat{C}(x)), \qquad \widehat{C}(x) = \log F(e^x),$$

where Γ *is the local pixel closure map and* F *is the global capacity readback map. If*

$$|\Gamma(P) - \Gamma(Q)| \le q_P|P-Q|, \qquad |\widehat{C}(x) - \widehat{C}(y)| \le q_N|x-y|, \qquad q_P, q_N < 1,$$

then $\mathcal{J}$ *has a unique fixed point* $(P_\star, \log N_{\text{CRC}})$*, and every iteration in* X *converges to it with contraction constant* $q = \max\{q_P, q_N\}$*.*

Proof. For any two points in X,

$$d(\mathcal{J}(P,x), \mathcal{J}(Q,y)) \le \max\{q_P|P-Q|,\ q_N|x-y|\} \le \max\{q_P, q_N\} d((P,x),(Q,y)).$$

Banach's theorem gives existence, uniqueness, and stability. A genuinely coupled source map $\mathcal{J}(P,x) = (\Gamma(P,x), \widehat{C}(P,x))$ requires derivative bounds a, b, c, d and a weight $r > 0$ with

$$\max\{a + b/r,\ d + rc\} < 1$$

in the weighted sup metric. Without that bound, the coupled branch has residual freedom and no uniqueness theorem. □

Theorem 6.11 (Electroweak tick-projection bridge)**.** *On the selected D10 source branch,*

$$\frac{v}{E_{\text{cell}}} = \exp\left[-\frac{2\pi}{4\alpha_U(P)}\right].$$

Together with the global repair tick

$$|g'_\star| = \left(\frac{N}{\pi}\right)^{-1/48},$$

define the electroweak projection exponent by

$$\Pi_{\text{EW}}(P,N) := \frac{-\log(v/E_{\text{cell}})}{-\log|g'_\star|} = \frac{24\pi}{\alpha_U(P)\log(N/\pi)}.$$

Since $0 < |g'_\star| < 1$, this exponent is unique. The target projection

$$\Pi_{\text{EW}}(P_\star, N_{\text{CRC}}) = 4P_\star$$

is equivalent to the bridge residual

$$\mathcal{B}_{\text{EW}}(P,N) := \alpha_U(P)\log(N/\pi) - \frac{6\pi}{P} = 0.$$

Equivalently, the exact bridge target is

$$N_{\text{EW}}(P) = \pi\exp\left[\frac{6\pi}{P\alpha_U(P)}\right].$$

For the public endpoint branch,

$$N_{\text{EW}}(P_\star) = 3.5323546226929906511187512962330547600462 \times 10^{122}.$$

For the bridge-refined capacity certificate

$$N_{\text{CRC}}^{\text{EW}} = \pi\exp\left[\frac{6\pi}{P_\star\alpha_U(P_\star)}\right],$$

one has $\mathcal{B}_{\text{EW}}(P_\star, N_{\text{CRC}}^{\text{EW}}) = 0$, and hence

$$\frac{v}{E_{\text{cell}}} = \left(\frac{N_{\text{CRC}}^{\text{EW}}}{\pi}\right)^{-P_\star/12} = |g'_\star|^{4P_\star}.$$

The factor 4 is the electroweak transmutation multiplicity $\beta_{\text{EW}} = N_c + 1 = 4$. The factor 48 is $2 \cdot 24$ from the global repair tick, hence $12 = 48/4$. The rounded $N_{\text{CRC}} \simeq 3.31 \times 10^{122}$ display is a capacity-scale label and fails the exact bridge residual.

Proof. The D10 relation gives

$$-\log(v/E_{\text{cell}}) = \frac{2\pi}{4\alpha_U(P)} = \frac{\pi}{2\alpha_U(P)}.$$

The global tick relation gives

$$-\log|g'_\star| = \frac{1}{48}\log(N/\pi).$$

Taking the ratio gives the displayed Π_{EW}. Setting $\Pi_{\text{EW}} = 4P$ and rearranging gives $\alpha_U(P)\log(N/\pi) = 6\pi/P$, exactly $\mathcal{B}_{\text{EW}}(P,N) = 0$. Substitution gives the displayed $N_{\text{EW}}(P)$ and the hierarchy identity. □

Theorem 6.12 (Icosahedral screen sieve and capacity projection). *On the OPH triangulated S^2 screen branch with a locally six-valent MaxEnt quotient-ensemble background, convex defect cost, edge-center collar readout, and no-marked-point finite isotropy, the screen has exactly twelve irreducible curvature ports. The ports form the* 12*-vertex icosahedral orbit, and an invariant screen load X is read locally as $X/12$. For $X = \log(N/\pi)$, local cell entropy $P/4$, and electroweak multiplicity $\beta_{\mathrm{EW}} = 4$,*

$$\Gamma_{\mathrm{EW}} = \beta_{\mathrm{EW}} \frac{P}{4} \frac{1}{12} \log(N/\pi) = \frac{P}{12} \log(N/\pi).$$

Consequently

$$\log(E_{\mathrm{cell}}/v) = \frac{P}{12} \log(N/\pi), \qquad \alpha_U(P)^{-1} = \frac{P}{6\pi} \log(N/\pi),$$

which is equivalent to the electroweak bridge residual $\mathcal{B}_{\mathrm{EW}}(P, N) = 0$.

Proof. For a triangulated sphere with V vertices, E edges, and F triangular faces, $V - E + F = 2$, $3F = 2E$, and hence $E = 3V - 6$. With defect charge $q_v = 6 - \deg(v)$,

$$\sum_v q_v = 6V - 2E = 12.$$

The local vacuum is six-valent, so the total spherical curvature appears as positive defects. Convex defect cost spreads the fixed charge 12 into twelve unit $q_v = 1$ defects. Tetrahedral and octahedral alternatives carry the same total charge while concentrating it into four $q = 3$ defects or six $q = 2$ defects, placing them outside the MaxEnt/refinement-stable screen branch. Edge-center collars expose each unit fivefold defect as one central port, and no-marked-point finite isotropy places the twelve ports on the maximal finite rotation orbit of S^2. For $I \simeq A_5$, orbit-stabilizer gives $|I|/|C_5| = 60/5 = 12$. An invariant scalar contraction on that transitive orbit is split equally, giving the $1/12$ sieve factor. Multiplying the sieved screen load by $P/4$ and $\beta_{\mathrm{EW}} = 4$ gives $\Gamma_{\mathrm{EW}} = (P/12) \log(N/\pi)$. Exponentiating gives the displayed hierarchy relation, and matching it to the source transmutation exponent gives the displayed α_U equation and $\mathcal{B}_{\mathrm{EW}}(P, N) = 0$. □

Theorem 6.13 (EW-refined exact capacity fixed point). *Define the log-capacity map*

$$\mathcal{C}_{\mathrm{EW}}(P, x) = (1 - \lambda)x + \lambda \frac{6\pi}{P\alpha_U(P)}, \qquad 0 < \lambda \leq 1.$$

For fixed $P = P_\star$ and $\lambda = 1/2$, this is a contraction with Lipschitz constant $1/2$. Its unique fixed point is

$$x_\star = \frac{6\pi}{P_\star \alpha_U(P_\star)}, \qquad N_{\mathrm{CRC}}^{\mathrm{EW}} = \pi e^{x_\star}.$$

Therefore

$$\mathcal{B}_{\mathrm{EW}}(P_\star, N_{\mathrm{CRC}}^{\mathrm{EW}}) = 0.$$

The rounded 3.31×10^{122} capacity display remains diagnostic for this bridge.

Proof. The map has derivative $1 - \lambda$ in x, hence is a contraction for $0 < \lambda \leq 1$. Solving $x = \mathcal{C}_{\mathrm{EW}}(P_\star, x)$ gives the displayed $x_\star$. Substitution into $\mathcal{B}_{\mathrm{EW}}$ gives zero. □

Theorem 6.14 (Conditional RG/Higgs naturality identity). *On a declared hierarchy branch satisfying the selected-surface premises, let Q_s be the source hierarchy quotient and Q_H the Higgs/electroweak quotient. Define*

$$\rho_{sH}([x]_s) = [P(x), N(x), \Theta(P(x), N(x)), \Pi_{HT} F_{D11} F_{D10}(P(x), N(x)), 0]_H,$$

with

$$\Theta(P,N) = \left(\frac{N}{\pi}\right)^{-P/12} = |g'_\star|^{4P}.$$

For the corresponding normal-form maps and obstruction maps,

$$\rho_{sH} n_s = n_H \rho_{sH}, \qquad \chi_{sH} h_s = h_H \rho_{sH}.$$

Consequently,

$$\varepsilon_H = \max\{\varepsilon^n_{sH}, \varepsilon^h_{sH}\} = 0, \qquad \varepsilon_H \in [0,0].$$

Proof. The two routes through the source and Higgs quotients preserve the same tuple

$$(P, N, \Theta(P,N), \Pi_{HT} F_{D11} F_{D10}(P,N)).$$

The equality with $|g'_\star|^{4P}$ follows from $|g'_\star| = (N/\pi)^{-1/48}$. A hidden scalar relevant deformation is not an independent physical input on the selected branch: if absent from the retained finite constraint family, it is off-branch by refinement stability; if retained, its value is fixed by the branch equations and belongs to the displayed Higgs/top coordinate. Both controlled squares commute exactly, so the product defect is zero. □

The conditional RG/Higgs naturality map does not itself use the measured weak scale. Measured W/Z, Higgs/top, G, Planck-area, and Λ data are excluded by its declared input contract. The local/global package records the closed global repair tick, joint product branch, electroweak projection bridge, EW-refined exact capacity fixed point, finite readback-resolution certificate, and RG naturality square. Conditional on the declared branch, this certifies the dimensionless electroweak hierarchy/naturality identities; the strict source-root certificate remains open, and the result does not independently attach $E_\star$ to a physical GeV scale. The finite readback certificate defines

$$F_r(N) = \mathrm{Cap}_{\mathrm{read}}\left(\mathrm{Obs}(\mathrm{nf}_{r,N}(\mathfrak{U}_{r,N}))\right), \qquad \rho_{\mathrm{read}}(r,N) = \sqrt{\pi/F_r(N)},$$

and proves one selected effective delivery resolution on the finite branch, with $\rho_{\mathrm{read}}(r, N_{\mathrm{CRC}}) \to (N_{\mathrm{CRC}}/\pi)^{-1/2}$ in the positive-root refinement limit. The representation-to-spectrum theorem gives $m_{\mathrm{rep}} = 2(8+3+1) = 24$, the 24-slot oriented repair register on the selected branch, so the local/global hierarchy-resonance package closes on the selected branch. The promoted hierarchy row does not use the local/global resonance as an input.

6.4 Higgs/top critical stage

The Higgs/top critical stage is a conditional quantitative stage built on top of the electroweak gauge core. It is described by a downstream split law together with a Jacobian readout map on the declared D10/D11 running, matching, and threshold surface. Write the coupling asymmetry as ρ_{EW}, distinct from $b_{\mathrm{tr}} = N_c + 1 = 4$. The candidate D10 tuple

$$(\eta_{\mathrm{source}}, \rho_{\mathrm{EW}}, \lambda_{EW}, \tau^{\mathrm{exact}}_{2,\mathrm{tree}}, \delta n^{\mathrm{exact}}_{\mathrm{tree}})$$

emits the shared scalar

$$\rho_{HT} = \log\!\left(1 + \tau^{\mathrm{exact}}_{2,\mathrm{tree}}\right)$$

and the declared residual formulas

$$R_T = -\tau^{\mathrm{exact}}_{2,\mathrm{tree}} \eta^2_{\mathrm{source}} + \left(1 + \frac{\rho_{\mathrm{EW}}}{28}\right)\eta^6_{\mathrm{source}} + \frac{\eta^8_{\mathrm{source}}}{14} + \frac{\eta^9_{\mathrm{source}}}{27},$$

$$R_H = \eta_{\rm source}^5 - \frac{3}{25}\eta_{\rm source}^6 + \frac{\lambda_{EW}\eta_{\rm source}^6}{18} + \frac{\eta_{\rm source}^8}{2\rho_{\rm EW}}.$$

The split coordinates are

$$\pi_y = \frac{\eta_{\rm source} + \left(\frac{3}{2} + \frac{\rho_{\rm EW}}{4}\right)\rho_{HT} + R_T}{\sqrt{\pi}}, \qquad \pi_\lambda = \frac{\eta_{\rm source} - \left(\frac{4}{3} - \frac{\rho_{\rm EW}}{54}\right)\rho_{HT} + R_H}{\sqrt{\pi}},$$

so the declared D11 Jacobian reads out

$$\delta y_t(\mu_t) = \pi_y\, y_t^{\rm core}(\mu_t), \qquad \delta\lambda(\mu_t) = -\frac{16}{9}\pi_\lambda\, \lambda^{\rm core}(\mu_t).$$

These equations are promoted from a declared formula surface only if a finite D11 split-character certificate passes DS1–DS5: one strict source branch and frozen pre-split carrier; an exact two-coordinate quotient response; explicit characters deriving $\rho_{HT}, R_T, R_H, \pi_y, \pi_\lambda$ and $-16/9$; oriented mismatch descent with canonical normalization and positivity; and an exhaustive deformation quotient proving rigidity or a target-free positive-gap selector. The repository evaluates the displayed formulas without emitting that certificate. This gives

$$m_H = 125.1995304097179\ \text{GeV}, \qquad m_t^{D11} = 172.3523553288312\ \text{GeV}.$$

This is a coordinate on the declared D10/D11 Jacobian surface, obtained by back-solving from the measured pair through the synchronization-scale scan, so it is a target-anchored fit and not a certified Higgs or top complex pole; the target-free headline is the double-criticality family evaluated below. The same surface emits a companion top coordinate. The selected-class quark support witness carries a target-anchored running-top audit row using the PDG 2025 cross-section entry; it is not a separate public source-only top prediction. The bridge to the auxiliary direct-top PDG row is closed as a corpus-limited codomain no-go; the auxiliary row is compare-only. The one-scalar companion seed

$$\sigma_{D11,\rm HT} = \frac{\alpha_U \cos(2\theta_{W0})}{\sqrt{\pi}}$$

is on disk only as the fixed-ray companion branch with $\pi_y = \pi_\lambda = \sigma_{D11,\rm HT}$. Separately, the same D11 Jacobian admits a compare-only inverse slice that hits the canonical Higgs/top reference pair exactly. That exact sidecar is useful for bookkeeping and does not define the forward branch used for the public rows. The Higgs boson belongs naturally in the boson discussion, while the top quark will be revisited in the quark-family chapter as the third-generation up-type quark. The top coordinate reported here is the conditional D11 split coordinate on this surface, not a separate public top-mass prediction row. The selected-class quark support wrapper carries only a target-anchored audit witness under the strict public-output policy.

The local implication from the candidate D10 tuple through the displayed D11 formulas is executable and single-valued. Strict Higgs promotion remains blocked by the source-root, independent-scale, QT1–QT5, RG/scheme, split-rigidity, top/threshold, complex-pole, uncertainty, and prospective provenance gates. The exact inverse slice is a compare-only validation surface.

6.5 Source-closure and physical-pole envelope

The preceding formulas end at running or declared-surface coordinates. A physical $W/Z/H$ theorem requires the following additional composition.

Proposition 6.15 (Conditional physical clock attachment)**.** *If the same strict source branch emits the dimensionless energy $\varepsilon_{\mathrm{clk}} > 0$ of a declared clock transition whose physical frequency ν_{clk} defines the operational unit, then*

$$E_\star = \frac{h\nu_{\mathrm{clk}}}{\varepsilon_{\mathrm{clk}}}.$$

Equivalently, $\gamma_\star = \ell_\star \nu_{\mathrm{clk}}/c$ gives $E_\star = \hbar\nu_{\mathrm{clk}}/\gamma_\star$. An operator-norm perturbation of at most ϵ changes an isolated selected clock gap by at most 2ϵ. This is a conversion and stability theorem; the required source clock packet is absent from the source.

The atomic component of that packet has an exact scalar reduction. Let $P = P_3 + P_4$ project onto the isolated zero-field $6S_{1/2}$, $I = 7/2$ cesium ground manifold, $\dim V_3 = 7$, $\dim V_4 = 9$, and let $Q = 1 - P$. On every real interval where $Q(\widehat{H} - z)Q$ is invertible, define the Feshbach–Schur operator $\mathcal{F}_P(z) = P(\widehat{H} - z)P - P\widehat{H}Q\,[Q(\widehat{H} - z)Q]^{-1}Q\widehat{H}P$.

Proposition 6.16 (Cesium channel scalarization and monotone clock roots)**.** *$z \in \operatorname{spec}\widehat{H}$ exactly when $0 \in \operatorname{spec}\mathcal{F}_P(z)$, with matching multiplicities. Rotational invariance and the multiplicity-free decomposition $V_3 \oplus V_4$ force $\mathcal{F}_P(z) = f_3(z)P_3 + f_4(z)P_4$, and $\mathrm{d}\mathcal{F}_P/\mathrm{d}z \preceq -P$ gives $f_F'(z) \leq -1$. One sign bracket per channel therefore isolates exactly one level, and the clock normal form is $H^{\mathrm{eff}}_{\mathrm{clock}} = E_0 P + \widehat{a}_{\mathrm{Cs}}\,\mathbf{I}\cdot\mathbf{J}$ with $E_0 = (9E_4 + 7E_3)/16$ and $\varepsilon_{\mathrm{Cs}} = E_4 - E_3 = 4\widehat{a}_{\mathrm{Cs}}$.*

The public atomic packet may therefore export two monotone scalar interval evaluators with resolvent certificates and sign brackets in place of a correlated 55-electron matrix. The open clock parents are the atomic-scheme electromagnetic convention, the absolute electron ratio $m_e c^2/E_\star$, the cesium nuclear mass/spin/current/Compton packet, the two scalar evaluators, summable refinement tails, and the no-target provenance freeze. A non-entailment theorem fixes the direction of effort: two source extensions that agree on every named object and differ in one unfixed parent produce different unique gaps, so no unique $\varepsilon_{\mathrm{Cs}}$ is a theorem emitted by the source, and further evaluation of the existing formulas cannot close the clock lane.

Proposition 6.17 (Operational scale metrology)**.** *In SI units $E/\mathrm{GeV} = [E/(h\nu_{\mathrm{Cs}})]\,[h\nu_{\mathrm{Cs}}/\mathrm{GeV}]$, and the second factor is an exact unit conversion. A source-only numerical GeV mass is therefore equivalent to a source-only dimensionless operational clock ratio. Replacing cesium by another clock species adds the burden of predicting that clock's frequency ratio to the defining cesium transition.*

The stored clock-gap candidate reproduces the displayed gravitational coupling to a relative agreement of order 10^{-49} through the clock–gravity identity, so the audit lane classifies that decimal as a calibration checksum outside every prediction ancestry (`clock_source_energy_closure_audit.json`). The selection mechanism for source laws themselves is closed separately by a source-action rigidity theorem: on a finite quotient with a faithful reference measure, an affinely independent feature basis, and a source-emitted moment vector, the maximum-entropy law is unique, every feasible competitor pays a positive relative-entropy gap with the Pinsker lower bound, and the surviving action freedom consists of additive constants, exact feature redundancies, and BRST-exact terms (`qme_source_action_rigidity_mechanism.json`). The physical Standard-Model operator basis and its moment vector $c_r(P_\star, N_\star)$ are the open inputs of that mechanism.

Proposition 6.18 (Unique frozen RG and threshold transport)**.** *Let the canonically normalized coefficient vector $X(\mu)$ obey a locally Lipschitz beta system on each interval of a frozen finite threshold list. Assume deterministic matching maps, fixed loop order, field content, renormalization scheme, threshold locations, and decoupling order. Then one source initial condition determines one low-energy coefficient vector. If L_j bounds the beta-system Lipschitz constant and K_j bounds* *threshold map j, perturbations obey the corresponding product of $e^{L_j \Delta t_j}$ and K_j, plus the declared truncation and matching remainders.*

Proof. Picard–Lindelöf gives existence and uniqueness between thresholds. Grönwall bounds perturbations on each interval; applying each deterministic Lipschitz threshold map and inducting through the ordered list gives the stated composition. □

The repository's printed running/matching packet is a declared-convention contract, not this completed source theorem: its beta provenance, threshold origins, matching interval composition, and truncation enclosure remain promotion gates.

Theorem 6.19 (Conditional complex-pole promotion)**.** *For $B = W, Z, H$, let $\Gamma_B^{\text{phys}}(s, \xi)$ be the BRST-complete physical inverse two-point block, including every declared mixing field after the Ward/Slavnov–Taylor projection, analytically continued to a declared Riemann sheet, and set*

$$D_B(s, \xi) = \det \Gamma_B^{\text{phys}}(s, \xi).$$

Assume a reference determinant has exactly one simple physical zero inside a certified contour, no zero lies on the contour, and the full determinant obeys $|D_B - D_{B,0}| < |D_{B,0}|$ there. In the neutral sector also assume the Ward-protected photon line has been separated from the massive eigenvalue. Then the enclosed pole is unique and stable. With the exact convention

$$s_B = \left(M_B - \frac{i}{2}\Gamma_B\right)^2, \qquad M_B > 0, \quad \Gamma_B \geq 0,$$

it determines one mass and width. If a Nielsen identity has the form $\partial_\xi D_B = C_B^{(\xi)} D_B$, the simple pole is gauge-parameter independent.

Proof. Rouché's theorem preserves the number of enclosed zeros under the certified perturbation. Simplicity gives the implicit-function stability bound and a nonzero residue. Differentiating $D_B(s_B(\xi), \xi) = 0$ and using the Nielsen identity gives $ds_B/d\xi = 0$. An invertible analytic field redefinition multiplies D_B by a nonzero analytic factor and therefore preserves the pole set. □

No source-emitted $W/Z/H$ self-energy kernels, analytic-sheet receipt, contour enclosures, widths, residues, or pole-convention uncertainty map are present on the displayed mass-chart surface. Historical code fields named `MW_pole`, `MZ_pole`, or `mt_pole` retain legacy schema names; those names do not supply this theorem.

Theorem 6.20 (Rigidity, dependency, and W/Z/H composition criterion)**.** *Let $\mathcal{F}_{\text{EW}}$ be the fully enumerated class of source maps that preserve the declared locality, symmetry, dimensional, refinement, and provenance constraints, modulo gauge and trivial reparameterization. Suppose either the pole readout is constant on this class or a target-independent selector has one minimizer with a positive winner gap. Suppose further that a hash-bound acyclic dependency manifest, committed before evaluation, records the source artifacts, code, selectors, conventions, uncertainty model, and outputs, and that measured $W/Z/H$ data and calibrated proxies are absent from every predicted ancestor set.*

If, in addition, a unique source root and independently source-closed physical $E_\star$ emit the canonical D10/D11 coefficients, the hypotheses of Definition 6.2, DS1–DS5, the frozen RG proposition, and the complex-pole theorem hold, then OPH determines one source-separated pole triple (s_W, s_Z, s_H) with a certified uncertainty enclosure.

Proof. Rigidity or the positive-gap selector removes alternative source maps. A deterministic leaf depends only on its ancestors, so the committed DAG gives formal source separation. The unique root and scale emit one coefficient packet, frozen RG and matching transport it uniquely, and the pole theorem gives one stable pole per particle. Composition proves the claim. □

This criterion is not satisfied by the displayed numbers. QT1–QT5 remain certificate assumptions, the public and source-audit pixel lanes differ, the physical scale and concrete RG receipt are open, DS1–DS5 are not emitted, and the complex-pole and prospective-manifest gates are absent.

The missing object is a source-emitted certificate. The structural theory admits target-free analytic counterfamilies $\tau_2 = -c\eta^2$, $\delta n = d(1-\rho_{\rm EW})\eta^2$ and $(\pi_y, \pi_\lambda) \mapsto (\pi_y + a\eta^N, \pi_\lambda + b\eta^N)$, all on the same open physical domain but with different mass-chart outputs. Likewise, a fixed running mass is compatible with both $s - m_R^2$ and $s - m_R^2 - \epsilon$, which have different poles. These countermodels prove that two-channel exhaustion and analyticity do not emit the D10 character, D11 split, or physical kernel. Nor is a formal path realization sufficient: any finite polynomial can be encoded by assigning one weighted path to each monomial. A non-vacuous certificate must independently fix its transitions, path measure, signs, quotient action, exhaustive census, rigidity gap, and no-target DAG.

Once genuine source packets are supplied, no stochastic simulation is needed. Summable clock-Hamiltonian refinements give a unique limiting gap; frozen Lipschitz RG segments compose uniquely; Rouché, Nielsen, and analytic conjugacy isolate gauge-independent poles. If a determinant defect is bounded by ϵ, $a = D'(s_0) \neq 0$, and $|D''| \le K$, every r with $\epsilon < |a|r - Kr^2/2$ encloses the unique displaced pole and propagates to mass/width bounds through the selected square root. The four absent source objects are therefore the factorized clock packet, independently weighted D10 carrier, D11 split-character carrier, and BRST-complete pole-kernel packet. A runtime DAG proves computational separation but not historical blindness; the latter requires disclosure and a prospective frozen claim.

The electroweak chapter therefore presents four mathematically distinct surfaces: the selected-carrier chart, the archived value-law candidate, the conditional QT1–QT5 theorem that would select that candidate, and the freeze-once repair surface used for compare-only W/Z values. A physical pole surface requires the additional source-root, scale, RG/scheme, two-point-kernel, rigidity, uncertainty, and provenance receipts.

6.6 Repair-tuple selection and the color amplitude/loop split

The archived D10 tuple sits inside a two-parameter freedom that the emitted corpus does not close. With $\rho_{\rm EW} = (\alpha_2 - \alpha_Y)/(\alpha_2 + \alpha_Y)$, the coherent quadratic family is

$$\tau_{2,\rm tree}^{\rm exact} = -c\,\eta_{\rm source}^2, \qquad \delta n_{\rm tree}^{\rm exact} = d\,(1-\rho_{\rm EW})\,\eta_{\rm source}^2,$$

and an open neighborhood of distinct (c, d) values preserves the positive mass-chart domain and defines consistent D10 repairs. The charged leg is the $\mathrm{SU}(2)_L$ coupling correction $\delta\alpha_2 = \alpha_2\tau_2$; the neutral leg is the hypercharge screening correction carried by δn.

The compact-gauge reconstruction fixes the color count but does not fix the channel assignment or normalization that would close this freedom. Doplicher–Roberts/Tannaka reconstruction returns the color triplet sector with statistical dimension $d(\rho_3) = N_c = 3$, and its standard conjugate intertwiner satisfies $R^*R = d(\rho_3) = N_c$, so R has norm $\sqrt{N_c}$ while $R/\sqrt{N_c}$ is the isometry; a closed color loop carries the full dimension N_c. The two repair legs sit at different levels. The charged leg is the broken, mass-generating $\mathrm{SU}(2)_L$ channel, driven by the color-singlet condensate amplitude whose large-N_c scaling is the decay-constant scaling $\sqrt{N_c}$ assigned to one standard conjugate intertwiner. The neutral leg would be the unbroken hypercharge screening channel, a vacuum-polarization loop that closes the color line and carries raw trace weight N_c. Under these extra hypotheses the alternative quadratic model has

$$c = \frac{\sqrt{N_c}}{2} = \frac{\sqrt{3}}{2}, \qquad d = \frac{N_c}{2} = \frac{3}{2},$$

where the chart coefficient $d = N_c/2$ represents the raw neutral trace because $(\alpha_2 + \alpha_Y)(1 - \rho_{\mathrm{EW}}) = 2\alpha_Y$.

The resulting comparison is a target-informed, fixed-slice diagnostic of that alternative model. Profiling the charged coefficient against the D10 running-tree W coordinate does not use the D11 Jacobian core, but it does inherit the declared α_2, v, running, and scheme surface. It selects $c = 0.8670$ against the proposed $\sqrt{3}/2 = 0.8660$, agreement to about one part in a thousand. Profiling the neutral coefficient against the Z mass selects $d = 1.461$ against the predicted $3/2$, agreement to about three percent. The joint $W+Z$ one-sigma band on c is $[0.731, 1.065]$; the color amplitude/loop value lies inside, while the loop-symmetric values $N_c/2$ and N_c lie outside on this fixed-d, one-dimensional alternative-model slice. That slice is not a test of the complete archived value law, which changes both coordinates and includes higher path characters; it therefore neither excludes that law nor proves or excludes a separate running-tree companion.

The color-balanced rule does not equal the complete archived value law in Theorem 6.3; it gives a different W coordinate. A proof of its amplitude and trace hypotheses would establish that alternative quadratic model, not QT1–QT5. The complete value-law promotion task is the finite quotient-path campaign: construct the two-channel quotient, enumerate the charged and neutral path lists, verify the $(1,2,1)$ and $(1,4,2)$ incidences with their $1/3$ color measure and $1/6$ central trace, compute the fibre Gram form and residual pairing, enumerate admissible deformations, and prove a positive MAR gap. Measured W/Z agreement cannot replace that certificate. The first implementation step is to freeze the actual pre-repair carrier independently of these desired outputs as an observer-like self-reading patch with bounded local state, typed ports, readback records, and admissible feedback/repair moves. That carrier may contain neither the target masses nor the desired incidence and central-trace coefficients. An exact path/orbit generator can then be treated as untrusted and checked by a smaller independent verifier; two canonicalizers should agree on orbit hashes, and the final formal checker must recompute path legality, exhaustion, exact response rank and nullspace, central action, oriented mismatch first variation, fibre Gram form, and the deformation census. Agreement between two canonicalizers is regression evidence rather than a completeness proof. Moreover, the order of $\mathbb{Z}_6$ supplies the averaging coefficient $1/6$, not automatically a normalized trace $1/6$: the checker must separate the trivial physical matter-kernel action from the declared D10 center-label/transport trace space and derive the trace, source amplitude, and subtraction sign in the latter.

6.7 Selector rigidity and the discrete repair-law boundary

The archived D10 selector closes the continuous (c, d) freedom of the preceding subsection. Writing $x = \tau_{2,\mathrm{tree}}^{\mathrm{exact}}$, $y = \delta n_{\mathrm{tree}}^{\mathrm{exact}}$, and $\kappa = (\alpha_Y + \alpha_2)/\alpha_Y$, the selector

$$J_{10}(x, y) = \frac{x^2\left[1 + (2\eta_{\mathrm{source}} + x)^2 + 4x^2\right]}{1 + 4x^2} + \frac{\kappa^2}{4}(1 + 4x^2)y^2$$

admits the exact factorization

$$J_{10} - \frac{3}{4}x^2 - \frac{\kappa^2}{4}y^2 = \frac{x^2}{4(1 + 4x^2)}\left[8(x + \eta_{\mathrm{source}})^2 + 8\eta_{\mathrm{source}}^2 + 1 + 4\kappa^2(1 + 4x^2)y^2\right],$$

so $J_{10} \geq \frac{3}{4}x^2 + \frac{\kappa^2}{4}y^2$ with equality exactly at the origin, and $x^2 + y^2 \geq r^2$ forces $J_{10} \geq \min\{3/4, \kappa^2/4\}r^2$. The selector picks the zero deformation with a quantitative gap. The archived value-law tuple has $J_{10} \approx 3.12 \times 10^{-7} > 0$, so it is a replacement law for the selector rather than a refinement, and the residual electroweak freedom is one discrete choice between two incompatible laws rather than a continuous chart family.

On the strict source-audit branch the two branch points give the tree/chart coordinates

$$\begin{aligned} \text{zero-selector law:} \quad & M_W/E_\star = 6.579631 \times 10^{-18}, \quad M_Z/E_\star = 7.463335 \times 10^{-18}, \\ \text{carrier value law:} \quad & M_W/E_\star = 6.578870 \times 10^{-18}, \quad M_Z/E_\star = 7.463750 \times 10^{-18}, \end{aligned}$$

with unclosed-clock displays (80.3301, 91.1191) and (80.3208, 91.1242) GeV and discrete ambiguity widths of 9.3 MeV on M_W and 5.1 MeV on M_Z (`wzh_residual_elimination_boundary.json`). Numerical W/Z agreement cannot decide between the laws; the decision object is a source-law selection principle, and the quantitative mechanism of the source-action rigidity theorem applies once the electroweak feature basis and moment vector are emitted.

A prospective finite C10 root/Cartan carrier emits the value-law coefficients exactly: the transitive C_3 color action has invariant measure 1/3, the regular $\mathbb{Z}_6$ register has rank-one trace 1/6, the transitive four-slot register has slot measure 1/4, the frozen independent-product source law gives $\lambda_{\rm EW} = \eta^2_{\rm source}/(4\rho_{\rm EW})$, and the depth-two path census reproduces the archived quadratic response polynomials. A companion C11 SU(3)-Casimir carrier emits the declared D11 normalizations, $\kappa_\lambda = C_F^2 = 16/9$, the response coefficients $3/2 + \rho_{\rm EW}/4$ and $4/3 - \rho_{\rm EW}/54$, and the logarithmic character $\rho_{HT} = \log(1 + \tau_2)$ from multiplicativity with unit derivative. Both carriers are post-exposure candidates; the open promotion object is a prospective selection or census certificate of the QT1–QT5 type.

The declared D11 Jacobian entries follow exactly from the core, $\partial m_t/\partial y_t = m_{t,0}/y_0$ and $\partial m_H/\partial\lambda = m_{H,0}/(2\lambda_0)$, and the exact readout is $m_t = m_{t,0}(1 + r_y)$ and $m_H = m_{H,0}\sqrt{1 - r_\lambda}$ with the exact linearization error $m_{H,0}r^2/(2(1 + \sqrt{1 - r})^2)$. The synchronized D11 core is target-ancestral: its synchronization scale minimizes an objective containing the Higgs and top comparison values. Removing that switch and integrating the literal archived one-loop source equations gives the target-free tree coordinates $m_t^{\overline{\rm MS}}/E_\star = 1.267758 \times 10^{-17}$ and $m_H^{\rm tree}/E_\star = 9.427653 \times 10^{-18}$, with unclosed-clock displays 154.78 GeV and 115.10 GeV and a QCD-converted top display of 164.13 GeV. These are one-loop tree coordinates on the declared surface; they are not physical poles, and the pole gates of §6.5 keep their status.

The archived one-loop coordinates are one branch of a zero-continuous-parameter family. The literal core derives its top boundary from the gauge sector through the double-criticality condition $\lambda(\mu_b) = 0$, $\beta_\lambda(\mu_b) = 0$, which fixes $y_t(\mu_b) = [(2g_2^4 + (g_2^2 + g_Y^2)^2)/16]^{1/4}$. The archived branch imposes this at the gauge-unification scale μ_U, while the electroweak transmutation in the same model anchors v at the pixel energy $E_{\rm cell} = E_\star/\sqrt{P}$, so the archived model carries two different high-scale anchors. Evaluating the same law at the named source scales gives, at one loop, $(m_t, m_H) = (164.1, 115.1)$ GeV at μ_U, (170.4, 127.8) at $E_{\rm cell}$, and (170.7, 128.3) at $E_\star$; a benchmark-validated two-loop upgrade shifts these to (169.4, 119.4), (175.7, 131.8), and (175.9, 132.3). The family brackets the measured pair in both channels at both loop orders, so the archived deficit decomposes into the boundary-scale choice plus loop truncation, with no continuous freedom anywhere. Along the fit-free curve the Higgs coordinate at the measured top mass is 125.7 GeV at two loops, within 0.5 percent of measurement and inside the declared tree-to-pole matching band; the implied boundary scale is 4.8×10^{17} GeV, between μ_U and $E_{\rm cell}$. The open object is a boundary-scale selection theorem (`d11_criticality_boundary_scan.json`, `d11_criticality_comparison.json`); numerical agreement cannot select the scale, and the declared-surface values obtained by back-solving from the measured pair stay classified as target-anchored fits.

The selection question admits a near closure. The only scale-selecting condition available inside the flow itself, the triple-criticality root $\mathrm{d}\beta_\lambda/\mathrm{d}\ln\mu = 0$ at the critical point, has no solution: along the family the flow derivative is dominated by $-24y_t^3\beta_{y_t} > 0$ and stays bounded away from zero across the whole window, so every family point is a clean λ minimum and the boundary scale must be selected by the source structure. That structure supplies a variational selection principle with three premises: (AR1) the criticality boundary is a record that reconciles the model's two anchor records, the gauge-unification record at μ_U and the transmutation record at $E_{\rm cell}$; (AR2) the reconciliation cost is quadratic in renormalization time; and (AR3) the two anchor records carry equal capacity. These premises give a strictly convex cost with a unique minimizer at the log-midpoint $\sqrt{\mu_U E_{\rm cell}} = E_\star e^{-\pi}P^{-1/6}$, hence $(m_t, m_H) = (172.63,\ 125.77)$ GeV at two loops; the implication is proved exactly on rational sample points (`d11_boundary_scale_midpoint_selection_theorem.json`).

Two of the three premises reduce to the axioms. The quadratic-cost premise AR2 is a theorem under the canonical record model: the one-loop chart is inverse-affine, so $1/\alpha(t)$ is exactly linear in renormalization time, and the Kullback–Leibler divergence between fixed-resolution Gaussian records with an affine stored coordinate is exactly $(s^2/2\sigma^2)(t - t_i)^2$, a quadratic cost with no leading-order approximation. The reconciliation placement in AR1 is also a theorem: the mismatch functional is port-additive by construction and the repair move settles a record at the cost minimizer, so a record with two scale-parents settles at the capacity-weighted log-mean. The equal-capacity premise AR3 reduces to equal slope and readback resolution, which same-class registers at equal refinement depth supply. What remains are two finite carrier facts about the D11 carrier: CF1, that the criticality boundary record carries exactly two parent ports, one to the gauge-unification register and one to the transmutation register; and CF2, that those two anchor registers are the same register class at equal refinement depth. Given the canonical record model and CF1 and CF2, the axioms force the boundary scale, and with it $(m_t, m_H) = (172.63,\ 125.77)$ GeV, with no remaining choices. CF1 and CF2 are statements about the same D11 carrier census certificate (of the QT1–QT5 class) required by the W/Z value law; one certificate closes the Higgs-scale selection, the W/Z law, and the D10 two-law choice together. The equal-capacity assumption is measurable: a capacity asymmetry moves the selected scale off the midpoint by a computable amount, about 2.1 GeV in m_H per e-fold, and the registered three-loop implied scale reads it off (`d11_anchor_reconciliation_reduction_theorems.json`). The candidate registry is frozen before any three-loop computation exists, so the three-loop implied scale is the registered discriminating test.

A conditional selection theorem accompanies the registry. If the boundary record reconciles the two anchor records with port-additive quadratic cost in renormalization time and equal anchor capacities, the boundary scale is exactly the log-midpoint. Two of the three premises reduce to the axioms' record and repair structure: the one-loop chart is inverse-affine in $\ln\mu$, so the canonical Gaussian MaxEnt record model gives an exactly quadratic reconciliation cost, and repair minimization of port-additive mismatch places a two-parent record at the capacity-weighted log-mean. The surviving content is two finite carrier facts, the two-parent port structure of the boundary record and the equal-class equal-depth status of the anchor registers; a carrier certificate of the census class closes both (`d11_boundary_scale_midpoint_selection_theorem.json`, `d11_anchor_reconciliation_reduction_theorems.json`).

6.8 Conditional electroweak mass envelope

Evaluating two historical repair candidates at the endpoint pixel and across the empirical-closure interval gives the following legacy comparison envelope. It is not a source interval: its candidate set is not an exhaustive source-derived deformation class, and it mixes the public endpoint lane with the strict source-audit question.

$$\begin{aligned} m_H &\in [125.183,\ 125.232]\ \text{GeV}, &&\text{measured } 125.13 \pm 0.11,\\ m_t &\in [172.278,\ 172.352]\ \text{GeV}, &&\text{measured } 172.1 \pm 0.6,\\ m_W &\in [80.3692,\ 80.3774]\ \text{GeV}, &&\text{measured } 80.3692 \pm 0.0133,\\ m_Z &\in [91.1880,\ 91.1983]\ \text{GeV}, &&\text{measured } 91.1876 \pm 0.0021. \end{aligned}$$

The Higgs, top, and W envelopes lie inside the one-sigma experimental band; the Z envelope overlaps it, with the upper edge an artifact of holding the running couplings fixed while shifting the pixel through the endpoint interval. At the endpoint pixel the two repair selections give m_Z values 91.18798 and 91.18801 GeV, within 1.3 standard deviations. The selection spread between the two displayed laws is below the quoted experimental resolution in every observable, at roughly eight MeV in m_W, thirty keV in m_Z, sixteen MeV in m_H, and thirty MeV in m_t. These figures compare two ansätze; they do not bound the full QT5 deformation class.

6.9 Residual attribution across the electroweak rows

On this legacy comparison envelope, the four electroweak rows sit at 0.31, 0.70, 0.36, and 2.63 standard deviations from their measured values for m_W, m_H, m_t, and m_Z. Each residual carries a named source. The m_W offset is repair-selection freedom: the two candidate repair laws give m_W values that differ by about eight MeV, and the measured value falls between them. The m_H and m_t offsets track the same repair-selection width folded through the declared Higgs and top Jacobian cores.

The m_Z row is the sharpest case. Within the frozen adapter model, its 2.63-standard-deviation excess is a pixel-channel effect. The empirical-closure pixel inherits the frozen electromagnetic anchor deficit through the fine-structure endpoint, and m_Z tracks the pixel with $\mathrm{d}m_Z/\mathrm{d}P \approx 123$ GeV per unit P. Two propagation branches separate the mechanism. The direct branch injects the certified anchor shift into the running couplings and re-evaluates the candidate W/Z chart. That branch shifts m_Z by -66 to -87 MeV on the hypercharge line and by roughly -230 to -302 MeV on the proportional line, overshooting the excess by an order of magnitude, so it is excluded. The pixel branch moves the endpoint to the measured fine-structure value and propagates the induced pixel shift through $\mathrm{d}P/\mathrm{d}A_{\mathrm{Th}}$. That branch moves the empirical pixel onto the calibration pixel to within 4×10^{-7} in P, equivalently -0.05 MeV in m_Z. In this adapter diagnostic the displayed m_Z excess is the frozen-anchor deficit propagating through the endpoint pixel interval. The coincidence of the independently constructed calibration pixel and the endpoint-corrected pixel at the seventh decimal is a consistency check on the pixel architecture itself.

For this comparison surface, a closed fine-structure anchor bridge would collapse the endpoint pixel interval onto the calibration pixel and remove this particular m_Z residual. It would not close QT1–QT5, the independent scale, the RG/scheme certificate, or the complex-pole gate. The anchor bridge is the object developed in the fine-structure companion, and its source branch is the subject of Section 6.10.

6.10 Source branch of the fine-structure anchor bridge

The certified anchor gap [0.649, 0.855] in inverse fine-structure units names the shift the source side supplies at the electroweak anchor for the empirical-closure endpoint to reach the measured fine-structure value. The gap is defined by that measured requirement, so inserting it back is circular.

The non-circular source route continues the declared running convention to the next order and tests whether the induced anchor shift lands inside the certified gap.

That test is executed. The one-loop anchor is reproduced from the declared coefficients and the transmutation-certificate boundary data to a residual of 10^{-11}. The standard two-loop running system, integrated down the same 33.2 e-folds with a top-Yukawa convention scan, shifts the anchor by [+1.62, +2.14] in inverse fine-structure units for every convention. The shift has the correct sign and overshoots the certified gap by a factor of two to three. Restoring the balance requires the threshold and scheme-conversion data, which the matching contract classifies as hidden fit parameters when they are undeclared. The balance requirement is quantitative: the threshold map removes about 2.14 inverse-alpha, equivalent under naive coefficient accounting to a single effective threshold near 458 GeV. The source branch of the anchor bridge is work in progress, and the empirical class carries the endpoint in the interim. The refined missing objects are the declared scheme-lock and threshold-map of the matching contract.

7 Flavor Transport, Generation Structure, and the Yukawa Dictionary

The gauge branch explains why the realized low-energy sector has three generations, but it does not by itself produce the full flavor dictionary. This paper therefore needs a separate flavor chapter. The imported realized-branch input here is the D9 count package $N_c = 3$ and $N_g = 3$. This chapter is the bridge between that structural result and the matter-family chapters for quarks, charged leptons, and neutrinos.

The flavor derivation is deliberately constructive. It does not claim that the full OPH flavor observable is theorem-level. Instead, it builds the object chain that a closed theorem would have to pass through. The derivation chain has the following main architecture: start from a refinement-indexed family transport kernel, derive a centered generation-bundle branch generator, lift that to same-label transport data, derive the induced overlap-edge transport cocycle, reduce the cocycle to a persistent flavor observable, and then push that observable into common sector-response objects for the downstream quark, charged-lepton, and neutrino derivations.

Mathematically, this derivation is where "why this family exists" becomes a concrete technical question. The realized Standard Model branch gives three generations structurally; the flavor derivation is where one tries to turn that three-generation existence statement into transport, splitting, suppression, phase, and excitation data. The relevant objects are the intermediate transport and spectral structures, not the final fermion masses themselves, that later mass readouts consume.

The active chain, in slightly compressed form, is:

1. normalize a refinement-indexed family transport kernel;
2. derive a centered generation-bundle branch generator on the realized three-generation charged bundle;
3. lift this to same-label edge-line transport and then to an overlap-edge cocycle with explicit defect and gap bookkeeping;
4. reduce those data to projectors, spectral gaps, pair suppressions, and cycle phases;
5. push the resulting family object into sector-response objects and then into suppression/phase tensors for the downstream matter derivations.

Three lane-specific claim boundaries are attached to this region of the derivation. The shared excitation dictionary is the common proof-facing base. Above that base, the charged lane carries exact centered readback, a closed common-shift no-go, the declared same-label q_e readback, a source-side determinant character for any fixed formal exponent vector

$$S_M = \sum_e M_e^{\mathrm{ch}} \log q_e,$$

a determinant-line lift on theorem-grade physical charged data, and an algebraic mass readout from theorem-grade $A_{\mathrm{ch}}(P)$. The theorem lane does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The charged-lepton theorem gap is the determinant trace-lift attachment $3\mu(r) = S_M(r)$ on the charged determinant channel. The charged determinant channel has a corpus-limited no-go boundary: the uncentered trace lift is not emitted. The neutrino lane carries a target-informed weighted-cycle candidate above a template family kernel, with compare-only bridge and absolute-attachment diagnostics. The quark lane carries a theorem-grade source-spread obstruction: its ordered profile shapes leave a free $(\mathbb{R}_{>0})^2$ fiber. Selected-class descent proves representative independence but does not choose either positive modulus. Target-anchored mixed-convention mass coordinates and GeV-valued mass textures remain audit data; they are neither a source-only running sextet nor physical dimensionless Yukawa matrices. A separate compare-only microphysical bridge records edge-statistics transport on a diagnostic surface. Together these boundaries mark where the flavor derivation leaves the common transport backbone and passes to lane-specific closure contracts.

8 Quark Family Derivation

The quark lane carries a source-only non-identifiability theorem on the public physical quark frame class f_P chosen by P. The source equations determine the ordered up- and down-sector profile rays but leave their endpoint spans as two independent positive moduli. Selected-class descent does not remove this freedom, so no numeric mass row is emitted. Same-family and restricted common-refinement artifacts reproduce their chosen targets only after those moduli are obtained by target inversion. Their rows also mix renormalization conventions, while their GeV-valued matrices are mass textures rather than physical dimensionless Yukawa matrices. This theorem does not classify all public quark frame classes.

8.1 What quarks are in this derivation

Quarks are the color-charged elementary constituents of hadronic matter. In nature, the up quark and down quark dominate protons and neutrons, while the strange quark, charm quark, bottom quark, and top quark appear in progressively heavier and more unstable sectors. In the OPH particle derivation, the quark lane has a closed downstream algebraic readout conditional on a source-only physical spread datum. The target-free source equations do not emit that datum: they fix two ordered profile rays and leave their positive endpoint spans independent. The compatible fiber is $(\mathbb{R}_{>0})^2$, so the six numeric rows are not source-only predictions.

The quark route starts from the shared flavor excitation dictionary, internalizes the target-free mass bridge on the D12 mass ray, proves selected-bridge-fiber representative independence for the attached physical spread datum on f_P, and applies the affine mean law once both positive moduli are supplied. The stored target-anchored matrices have GeV-valued singular values drawn from several comparison conventions. They are mass textures, not physical dimensionless Yukawa matrices. The same-label left-handed selector surface closes to the singleton σ_{ref}. That lower object is a negative sheet-selector statement on the selected D12 sheet. It does not break the independent positive-rescaling action on the two sector profiles.

8.2 Emitted quark rows

Running quark masses are coordinates on a renormalization-group trajectory. Finite renormalizations and scale changes alter those coordinates while preserving physical amplitudes, so OPH can emit an RG-covariant trajectory or invariant but cannot derive the human choice of an $\overline{\text{MS}}$ chart. The chart must be declared after source emission. The top row is a pole extraction coordinate rather than a sixth member of one common running-mass packet. All six rows are withheld because the spread pair is non-identifiable from the source corpus. A physical Yukawa construction would additionally transport every running coordinate to one common scale, convert the top coordinate, emit the running Higgs expectation value in the same scheme, and apply $y_q(\mu) = \sqrt{2}\, m_q(\mu)/v(\mu)$.

8.3 Target-free mass bridge and selected-class public theorem

The same-label left-handed solver surface closes to the singleton σ_{ref} (`sigma_ref`). This is a negative same-sheet selector statement: same-sheet rephasing preserves CKM moduli, so it does not move that selected sheet to the physical CKM shell. Same-sheet overlap scans, chirality-swapped basis diagnostics, and other non-sector-attached orbit improvements are compare-only and do not change the selected-class theorem.

A separate target-free mass bridge is internalized on the emitted D12 ray. On the minimal light branch

$$y_u = c_u\,\varepsilon^6, \qquad y_d = c_d\,\varepsilon^6, \qquad \varepsilon = \frac{1}{6},$$

the light-quark overlap-defect theorem emits

$$\Delta_{ud}^{\text{overlap}} = \frac{1}{6}\log\frac{c_d}{c_u}.$$

The emitted same-family D12 mass object is the ray

$$D12_{ud}^{\text{mass}} \equiv \mathcal{R}_{D12}^{ud} \qquad \text{(paper id } \texttt{D12\_ud\_mass\_ray}\text{)},$$

and on that ray the same scalar is equivalently the one-scalar law

$$\Theta_{ud}^{\text{mass}} := \texttt{quark\_same\_family\_value\_law}.$$

The exact scalar identities are

$$\Delta_{ud}^{\text{overlap}} = \frac{t_1}{5}, \qquad \log\frac{c_d}{c_u} = \frac{6}{5}\,t_1, \qquad t_1 = 5\,\Delta_{ud}^{\text{overlap}} = \frac{5}{6}\log\frac{c_d}{c_u}.$$

The D12 mass-side package is therefore functorial:

$$\begin{aligned}
\eta_Q^{\text{centered}} &= -\frac{1-x_2^2}{27}\,t_1,\\
\kappa_Q &= -\frac{t_1}{54},\\
x_2 &= -0.5175863354681689.
\end{aligned}$$

The odd source package is also forced:

$$\beta^{\mathrm{source}}_{u,\mathrm{diag},B} = \frac{t_1}{10},$$
$$\beta^{\mathrm{source}}_{d,\mathrm{diag},B} = -\frac{t_1}{10},$$

$$\texttt{source\_readback\_u\_log\_per\_side} = \left(-\frac{t_1}{10},\, 0,\, +\frac{t_1}{10}\right),$$
$$\texttt{source\_readback\_d\_log\_per\_side} = \left(+\frac{t_1}{10},\, 0,\, -\frac{t_1}{10}\right),$$

For any supplied positive spread pair one obtains

$$\tau_u = \frac{\sigma_d}{10(\sigma_u+\sigma_d)}\,t_1, \qquad \tau_d = \frac{\sigma_u}{10(\sigma_u+\sigma_d)}\,t_1.$$

These identities do not select σ_u or σ_d. The mass bridge is not part of the selected-class theorem boundary. The builder-facing pure-B payload pair is a lower implementation object beneath the exact theorem surface, and `intrinsic_scale_law_D12` is the derived wrapper above $\Theta^{\mathrm{mass}}_{ud}$.

Write the same-label left-handed physical carrier as

$$\Sigma^{\mathrm{phys}}_{ud} := \left\{(\sigma_{\mathrm{id}}, \tau, U_{u,L}, U_{d,L}, V_{\mathrm{CKM}}, I_{\mathrm{CKM}}) : V_{\mathrm{CKM}} = U^\dagger_{u,L} U_{d,L}\right\} / \sim,$$

where

$$(U_{u,L}, U_{d,L}, V) \sim (U_{u,L}D_u,\ U_{d,L}D_d,\ D_u^\dagger V D_d)$$

for diagonal $D_u, D_d \in U(1)^3$. On the selected public quark frame class f_P, represented on the realized lane by the explicit common-refinement transport-frame class $[F_0^\dagger F_1]$, the exact $\Sigma^{\mathrm{phys}}_{ud}$ element and the attached exact sigma datum are independent of the declared representative inside the selected bridge fiber. Therefore the exact sigma readout descends uniquely to the selected audit/support surface on f_P. It is not a public source-only mass prediction because the physical sigma datum remains target-derived.

Sigma descent non-selection. Let $R_{\mathrm{decl}}(f_P)$ be the declared selected bridge fiber over the public quark frame class. If a map

$$\Sigma : R_{\mathrm{decl}}(f_P) \to \mathbb{R}^4$$

is constant on that fiber, then it descends to a well-defined public datum $\overline{\Sigma}(f_P)$. This proves representative independence only. It does not select the value of $\overline{\Sigma}(f_P)$, because every constant vector in $\mathbb{R}^4$ would also descend. Therefore a value obtained from the current-family running-quark target surface remains target-derived after descent.

Common-scale physical rejection of the reciprocal-ray candidate. A physical comparison first converts all six quark eigenvalues to dimensionless Yukawa singular values in one effective theory, scheme, and scale. Using the Standard Model $\overline{\mathrm{MS}}$ values at M_Z tabulated by Antusch, Hinze, and Saad [7], the two ordered log-gap ratios are

$$\rho_u = 1.1108888543, \qquad \rho_d = 0.7519410008, \qquad \rho_u\rho_d = 0.8353228768.$$

The reciprocal-ray law requires the last product to equal one. It is therefore falsified on the properly typed target surface. Even after granting the four endpoint Yukawas (y_u, y_t, y_d, y_b), the minimax reciprocal shape misses the held-out y_c and y_s by 21.556% at M_Z. The result persists under running: the products are 0.836092, 0.837752, 0.845227, and 0.843611 at 10^3, 10^5, 10^{12}, and 10^{16} GeV, with best held-out errors between 19.93% and 21.42%. The stored $\rho = 1.294285$ formula misses the charm coordinate by 56.5% at M_Z, even under the endpoint grant.

The generic physical interface is therefore

$$(\mu_u, \sigma_u, \rho_u, \mu_d, \sigma_d, \rho_d) \in \mathbb{R}^6,$$

with an exact inverse to the six ordered positive Yukawas. The older three-scalar theorem remains exact only inside its imposed reciprocal-ray, affine-mean, fixed-shared-scale subfamily. The two-spread counterfamily below remains a valid restricted non-identifiability lower bound, but it is not a parameterization of the whole physical interface.

A separate selector audit sharpens the missing source object. Generation-blind flavor-singlet scalars cannot emit a fixed nonzero bifundamental Yukawa matrix. Simultaneous S_3 invariance restricts a fixed matrix to the span of I and J, which has a degenerate eigenspace and aligned up/down frames. A physical construction therefore needs a source-derived flavor-orbit selector, such as a spontaneous invariant functional or genuine bifundamental boundary data, together with a quark–Higgs carrier. The calibration simulator contains no quark, Higgs, or Yukawa coupling and its enumerated observables have zero Fisher information for these coordinates. Its completed 64k runs remain at the calibration null: the two fusion-tower readings are 1.495245 and 1.484301, while the dense first/full readings are 1.498275 and 1.497823. Those values are not quark-mass evidence.

Restricted source-spread non-identifiability theorem. Remove every running-mass target, exact-witness, fitted-spread, and compare-only ancestor. Grant the remaining source packet its strongest ordered three-point shape law. For $\rho_{\text{ord}} > 0$, define

$$v_u = \frac{1}{3(1+\rho_{\text{ord}})}\big(-(2\rho_{\text{ord}}+1), \rho_{\text{ord}}-1, \rho_{\text{ord}}+2\big),$$

$$v_d = \frac{1}{3(1+\rho_{\text{ord}})}\big(-(\rho_{\text{ord}}+2), 1-\rho_{\text{ord}}, 2\rho_{\text{ord}}+1\big).$$

Both vectors have zero trace and unit endpoint span. Their adjacent-gap ratios are ρ_{ord} and ρ_{ord}^{-1}, respectively. Conversely, those three conditions determine each profile up to one positive endpoint span. Hence every compatible pair is

$$E_u = \sigma_u v_u, \qquad E_d = \sigma_d v_d, \qquad (\sigma_u, \sigma_d) \in (\mathbb{R}_{>0})^2.$$

The group $(\mathbb{R}_{>0})^2$ acts freely and transitively by independent rescaling of the two spans while fixing the source shape data. The requested four-tuple

$$\left(\frac{\sigma_u+\sigma_d}{2}, \frac{\sigma_u-\sigma_d}{2}, \sigma_u, \sigma_d\right)$$

is not invariant under this action. No unique source-only spread package follows from the stated corpus.

No-extra-axiom MAR non-definability theorem. This ambiguity is not merely a missing implementation. Fix any generic MAR-admissible one-Higgs three-generation package and write

$$Y_q = U_{q,L}\ \mathrm{diag}\Big(e^{\mu_q+\sigma_q v_{q,1}}, e^{\mu_q+\sigma_q v_{q,2}}, e^{\mu_q+\sigma_q v_{q,3}}\Big)\, U_{q,R}^{\dagger}, \qquad q \in \{u, d\},$$

with $\sum_i v_{q,i} = 0$. For every $\lambda_u, \lambda_d > 0$, replace σ_q by $\lambda_q \sigma_q$ while holding the frames fixed. Gauge representations, anomaly cancellation, hypercharges, one-Higgs Yukawa completability, the CKM matrix, intrinsic CP capability, and the weak-sector counting clause are unchanged. Moreover

$$\det \exp(\lambda_q \sigma_q v_q) = \exp\left(\lambda_q \sigma_q \sum_i v_{q,i}\right) = 1,$$

so determinant normalization does not select either modulus.

MAR assigns every member the same complexity vector

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}}, N_{\mathrm{nonab}}, N_c, N_g),$$

which contains no Yukawa eigenvalue. The members are nevertheless physically inequivalent because the declared physical-equivalence relation preserves Yukawa invariants. Axiom 3 also does not remove the family: its gauge-invariant local constraint values and associated multipliers are not numerically specified and no map from P to quark Yukawa multipliers is emitted. Thus Axioms 1–5 plus fixed P admit a free $(\mathbb{R}_{>0})^2$ family of equal-MAR-score quark spectra. There is no smallest positive rescaling; its infimum is zero, which would give the massless or degenerate boundary rather than the observed spectrum. Therefore the stated axioms do not define a unique quark mass spectrum. This conclusion uses no additional axiom and can be overturned only by deriving, from the existing axioms, a source functional that is nonconstant on this explicit counterfamily.

The kernel interface: the exact content of the missing derivation. Two companion theorems make that boundary executable. First, the admissibility battery recorded for the family-transport lane (positive-semidefinite hermitian descendants, open three-cluster gaps at every level, a simple centered spectrum, the conjugacy-Riesz margin, persistent projector labeling, and overlap-edge amplitudes above the floor) places no constraint on the spectrum of the centered compressed branch generator: for every pair $(r, s) \in (\mathbb{R}_{>0})^2$ there is a certificate-passing two-level kernel whose generator has raw gap ratio exactly r and spectral span exactly s. The construction places the target spectrum by unitary conjugation, $T = Q\,\mathrm{diag}(\sqrt{\mu})\,Q^\dagger$, so the descendant spectrum is exact, and shrinks the refinement drift geometrically until the Riesz margin passes. The persistence certificates are therefore a filter, not a generator: a kernel derivation that merely passes them cannot emit the ordered ratio constant, the mean-law coordinate, or the spans.

Second, the previously recorded three-scalar interface is exact only on its imposed reciprocal-ray subfamily. At fixed shared scale g_{ch}, that subfamily factors through

$$(r, \sigma_u, \sigma_d) \in (\mathbb{R}_{>0})^3, \qquad \rho_{\mathrm{ord}} = \frac{3}{2+r}, \qquad x_2 = \frac{r-1}{r+1},$$

with the rays, the affine mean coefficients A_{ud}, B_{ud}, the sector means, and the six coordinates closed-form in the triple, and with the explicit left inverse $\sigma_u = \frac{1}{2}\ln(m_t/m_u)$, $\sigma_d = \frac{1}{2}\ln(m_b/m_d)$, and $r = 3/\rho - 2$ at $\rho = \ln(m_c/m_u)/\ln(m_t/m_c)$; the forward map is injective and the executed round trip closes at machine precision inside that subfamily. It is not the general interface of two ordered three-point spectra.

Indeed, for any $x \neq \pm 1$, let

$$L(x) = \mathrm{ctr}(-1, x, 1), \qquad Q(x) = \mathrm{ctr}(1, x^2, 1).$$

Their Gram determinant is

$$\det \mathrm{Gram}(L, Q) = \frac{4}{3}(1 - x^2)^2.$$

Thus L, Q form a basis of the centered three-vector plane, and every centered sector spectrum has unique coordinates $E_q = a_q L + b_q Q$. Once both a_q and b_q are free, x, and hence r, is a basis choice rather than an additional invariant of the spectrum. The general common-scale eigenvalue interface has six scalar coordinates: two centered coordinates and one mean for each sector. A proposed r plus four centered coordinates plus two means is a redundant seven-coordinate chart. The old ray law removes one centered coordinate per sector and thereby recovers its special three-scalar chart.

By the freedom theorem and the non-definability theorem above, the corpus selects neither the ray triple nor the six-scalar physical interface. The executable acceptance harness remains useful only as a fail-closed score of the legacy ray subfamily (`family_transport_kernel_admissibility_freedom_theor` `quark_kernel_three_scalar_interface_theorem.json`, `quark_kernel_normalization_acceptance_current`

The edge data do not remove the ambiguity. Even after granting source values $S_{13}, S_{23}, \delta_{21} > 0$, every positive pair can be written as

$$\sigma_u = S_{13} + c_u \delta_{21}, \qquad \sigma_d = S_{23} + c_d \delta_{21}$$

for suitable (c_u, c_d). The source equations emit no rule fixing these two coefficients. The specific coefficients used by the numerical candidate are therefore a declared ansatz. Their ancestry begins at a hand-written family-transport template whose own metadata calls for replacement by an OPH-derived kernel.

Define

$$\sigma_{\text{seed}}^{ud} := \frac{\sigma_u + \sigma_d}{2}, \qquad \eta_{ud} := \frac{\sigma_u - \sigma_d}{2},$$

$$A_{ud} := \frac{1}{2(1 + \rho_{\text{ord}} - x_2^2)},$$

$$B_{ud} := \frac{1}{2\Big(1 - x_2^2 - \frac{x_2^2}{1+\rho_{\text{ord}}}\Big)},$$

$$\rho_{\text{ord}} = 1.294284936377706.$$

Then

$$g_u = g_{\text{ch}} \exp\!\big(-(A_{ud}\sigma_{\text{seed}}^{ud} - B_{ud}\eta_{ud})\big), \qquad g_d = g_{\text{ch}} \exp\!\big(-(A_{ud}\sigma_{\text{seed}}^{ud} + B_{ud}\eta_{ud})\big).$$

The Jacobian from (σ_u, σ_d) to $(\log(g_u/g_{\text{ch}}), \log(g_d/g_{\text{ch}}))$ has determinant $-A_{ud}B_{ud} \neq 0$. The two free moduli therefore change the mass readout; they are not a gauge redundancy. Given a source-only physical spread datum, the affine mean law emits (g_u, g_d) algebraically on f_P. The target-derived packet is retained as a mixed-convention audit witness. Its dimensionful matrices are mass textures. The conditional selected-class route can be written as

$$f_P + \Sigma_{ud}^{\text{source}}(P) \Longrightarrow (\sigma_u, \sigma_d, \sigma_{\text{seed}}^{ud}, \eta_{ud})_{\text{phys}} \Longrightarrow (g_u, g_d) \Longrightarrow (m_u, m_d, m_s, m_c, m_b, m_t),$$

followed, only after common-scale RG transport and division by the running Higgs expectation value, by dimensionless Yukawa matrices. The first arrow is obstructed by the free $(\mathbb{R}_{>0})^2$ action on the stated source corpus.

Retrospective $S_3/D12$ two-mode formula-discovery witness. A frozen ansatz supplied after the preceding no-go was established reproduces the six mixed-convention comparison coordinates numerically. Its exact mathematical component starts from the transposition Cayley graph of S_3. The adjacency spectrum is $3, 0, -3$ with multiplicities $1, 4, 1$, so the Laplacian spectrum is $0, 3, 6$ and

$$\frac{e^{-3\tau} - e^{-6\tau}}{1 - e^{-3\tau}} = e^{-3\tau}.$$

This identity is a finite-group theorem. No theorem presently identifies the three distinct isotypic heat values with three generations or supplies the proposed heat time

$$\tau_f = \frac{P}{4} - \frac{\pi\alpha_U}{5}.$$

The ansatz sets $w = \pi\alpha_U$, $r = e^{-3\tau_f}$, $\rho = 3/(2+r)$, and $x = (r-1)/(r+1)$, and then uses

$$\begin{aligned} e_u &= S_{13} + \frac{\rho\delta_{21}}{1+\rho}, & e_d &= S_{23} + \frac{\delta_{21}}{2(1+\rho-x^2)}, \\ \bar{\sigma}_u &= e_u - w\Delta S_{13}, & \bar{\sigma}_d &= e_d - w(1-\Delta S_{13}), \\ a_u &= e_u + \frac{w}{2}, & a_d &= e_d - \frac{w}{5}, \\ b_u &= b_u^{\mathrm{ray}} - \frac{w\rho}{10}, & b_d &= b_d^{\mathrm{ray}} - \frac{w}{4}. \end{aligned}$$

The ray coordinates have the symbolic form

$$b_u^{\mathrm{ray}} = a_u \frac{-\rho x + \rho - x - 1}{(1+\rho)(x^2-1)}, \qquad b_d^{\mathrm{ray}} = a_d \frac{-\rho x - \rho - x + 1}{(1+\rho)(x^2-1)}.$$

The ansatz feeds $\bar{\sigma}_u, \bar{\sigma}_d$ into the candidate affine mean law above and sets $E_q = a_q L + b_q Q$. With the frozen repository numbers, this gives

$$r = 0.3179975211, \qquad \rho = 1.2942205385, \qquad x = -0.5174535369,$$
$$(a_u, b_u) = (5.6430129216, -4.9927828217), \qquad (a_d, b_d) = (3.3952170977, -1.8369623926).$$

Conditional normalized-trace lemma. Let $\ell : M_d(\mathbb{C}) \to \mathbb{C}$ be complex linear, invariant under all unitary conjugations, and normalized by $\ell(I) = 1$. Unitary conjugacy of rank-one projectors and additivity over an orthonormal resolution of I give $\ell(A) = \mathrm{Tr}(A)/d$, so every rank-one channel has weight $1/d$. Likewise, a width operator on $M_d(\mathbb{C})$ invariant under the full $U(d) \times U(d)$ left-right action is scalar by Schur's lemma; total Hilbert–Schmidt width t therefore assigns t/d^2 to each unit slot. Consequently, *if* the physical heat, up-odd, down-odd, up-even, and down-even responses are respectively identified with one isotropic slot in $M_5(\mathbb{C})$ of total width $5w$ and rank-one modules of dimensions $2, 5, 10, 4$, the coefficients $w/5, w/2, w/5, \rho w/10, w/4$ follow. The full invariance and normalization of the response law, those five module assignments, their signs and orientation, and exclusion of competing assignments are hypotheses outside OPH. Thus this lemma closes the denominator arithmetic conditional on the physical channel functor; it does not retroactively remove post-selection from the formula.

The per-particle comparison table is excluded from the public prediction ledger. The machine audit retains it as a retrospective diagnostic: the evaluator emits dimensionless coordinates, inserts an unproved one-GeV unit, and compares them to a table that mixes light-quark $\overline{\mathrm{MS}}$ values at 2 GeV, heavy-quark self-scale values, and a separate top extraction.

The raw diagonal residual sum against the supplied central values and nominal standard deviations is 1.1653, with a maximum relative residual of 0.2946%. It is not a likelihood or goodness-of-fit statistic: discovery used these targets, there is no source-theory covariance, and the rows are not one common-scale spectrum. In the postdeclared 219,615-member denominator grammar, the selected tuple $(5, 2, 5, 10, 4)$ is the unique minimum of that raw residual sum, but eight formulas tie its best maximum error. The grammar freezes the continuous inputs, graph, selected entries, assignments, signs, and functional forms after discovery and therefore is not a global look-elsewhere correction.

Most importantly, this evaluator is runtime-target-separated but not source-only in ancestry. $S_{13}, S_{23}, \delta_{21}, \Delta S_{13}$, and $g_{\rm ch}$ descend from the explicitly hand-written `family_transport_kernel` template. The last quantity is the dimensionless template-eigenvalue mean plus its minimum gap; treating it as GeV supplies the missing unit by hand. The selected $P_{\rm cand}$ branch is itself computed through an internal Stage-5 quark-spectrum continuation. Moreover, the edge laws add log-overlap suppression to a linearly scaling $TT^\dagger$ gap, and ΔS_{13} is a selected basis-dependent matrix entry. Thus the ansatz is archived as `post_hoc_target_informed_repository_template_ansatz`, with promotion disabled. It neither contradicts the non-identifiability theorem nor supplies a physical postdiction.

Sufficient Flavor Source Closure contract. The new proof audit also isolates a sufficient conditional completion, but does not instantiate it. A physical theorem would require all of the following on one acyclic, target-free source DAG:

1. a unique interval-certified source root for $P, \alpha_U(P)$, with no Stage-5 quark continuation in its ancestry;
2. a physical S_3 family-carrier attachment, the heat-time law $\tau_f = P/4 - \pi\alpha_U/5$, the ρ, x dictionary, and the two edge-response laws;
3. a source-labeled simple-spectrum family generator and charged seed with exact common-refinement intertwiners, monomial transport of the label rays, positive common normalization, and selected edge magnitudes bounded away from zero;
4. a physical response-channel functor that proves the normalized-trace hypotheses, module assignments, signs, orientation, competing-channel exclusion, and the affine mean law on that carrier, on a domain where the A, B denominators do not vanish;
5. a common-scale physical readout with $m_q(\mu) = y_q(\mu)v(\mu)/\sqrt{2}$, including field normalization, units, and the running Higgs expectation value in the same scheme;
6. a source-only RG packet with existence and non-blowup across the required intervals, a locally Lipschitz beta system, frozen threshold ordering and matching maps, top conversion, and the declared comparison charts.

Given these hypotheses, composition of the single-valued maps makes the final sextet unique. This is a sufficient proof-obligation contract, not a completed or proved-minimal OPH theorem. The numerical table above is not its corollary: the evaluator's coordinates are dimensionless and no instance of the final two receipts has been supplied or executed. What the counterfamily proves as a necessary condition is narrower and decisive: any successful source functional must be nonconstant on the independent (λ_u, λ_d) centered-spread rescaling orbit.

RSCC as a quarantined retrospective specification. A later Representation-Slot Cumulant Closure (RSCC) proposal makes part of the preceding contract more explicit without closing it. It removes $S_{13}, S_{23}, \delta_{21}, \Delta S_{13}$, and the repository $g_{\rm ch}$ decimal from the leaf evaluator. In their place it declares

$$F = \mathbb{C}^3_{\rm perm} \oplus V_{\rm std} \cong \mathbf{1} \oplus 2V_{\rm std}, \qquad \dim F = 5, \qquad \dim F_0 = 4,$$

and the composite response dimensions

$$(29, 432, 22, 32, 840, 1008, 432, 1584).$$

These integers are arithmetically correct for the declared direct sums and tensor products. If a continuous Gaussian source space, full unitary isotropy, reversible independent sheets, the listed effect ranks, and the listed signs are additionally assumed, the second-cumulant identity gives

$$\log \mathbb{E}[e^X] = \frac{1}{2}\operatorname{Var} X = \frac{r}{d}w^2.$$

This conditional identity does not construct the physical effects or select the modules. In particular, $\dim \operatorname{End}_{S_3}(\mathbf{1} \oplus 2V_{\mathrm{std}}) = 5$, so S_3-invariance does not force one scalar covariance on the two-copy multiplicity space. Pooling heterogeneous sums such as $\operatorname{End}(F) \oplus F_0$ into a single $1/29$ response requires an additional symmetry that mixes physically distinct blocks. The proposed F also has heat spectrum $0^{(1)}, 3^{(4)}$, not the regular representation's additional sign-sector value $6^{(1)}$; the physical regular-heat-to-family attachment is open. Finally, a genuine Gaussian covariance term is nonnegative. The negative w^2 entries require a separate signed response or subtraction law, not orientation reversal alone.

With $w = \pi\alpha_U$, the declared RSCC formulas are

$$\bar{\sigma}_u = 3P + \left(5 + \frac{4}{15}\right)w + \frac{w^2}{29}, \quad \bar{\sigma}_d = 2P + \frac{4}{15}w - \frac{w^2}{432},$$
$$a_u = 3P + \left(5 + \frac{4}{5}\right)w + \frac{w^2}{22}, \quad a_d = 2P + \left(1 + \frac{1}{32}\right)w + \frac{w^2}{420},$$

and

$$\frac{g_{\mathrm{ch}}}{v} = 2\exp\left[-2\pi + \frac{P}{1008} + \frac{w^2}{432} - \frac{w^2}{1584}\right].$$

Conditional on these formulas and the inherited heat-time, ray, even-response, affine A, B, and exponentiation laws, the downstream map is deterministic. Using the D10 v-display gives a maximum residual of 0.2943587% and a raw diagonal nominal-residual sum of 1.1613966 against the same mixed-convention comparison table. Neither number is a likelihood: the ledger was designed after the target-derived effective coordinates were visible, there is no common-scale RG packet or theory covariance, and the supplied RSCC comparison changes the nominal u-row uncertainty relative to the preceding bundle.

The ancestry and ablation checks sharpen the status. The target-inferred denominators for four visible effective coordinates are approximately

$$29.19665, \qquad 428.18579, \qquad 21.90604, \qquad 836.47734,$$

close to the selected $29, 432, 22, 840$, while no prospective grammar of alternative module expressions was frozen. Removing all RSCC w^2 terms and the entire δ_g correction, but retaining the lower-order scaffold and inherited readout, improves the same retrospective maximum residual to 0.2142293% and the raw residual sum to 0.6694007. This does not make the ablation a physical theory; it shows that the detailed covariance ledger is not selected by the numerical agreement. Moreover, the imported P_{cand} remains a source-audit witness with an internal Stage-5 quark-model ancestor. Its optional D10 GeV display uses $P = 1.63094$ and $\alpha_U = 0.0411249804\ldots$, not the RSCC witness pair, and the canonical D10 artifact marks its selector candidate-only and detects mixed sources.

RSCC therefore records an explicit, falsifiable retrospective specification for parts of receipts F2–F5, but discharges none of F1–F6. The claimed rescaling-orbit statement must also be scoped correctly:

$$\|\lambda E_q - E_q\|^2 = (\lambda - 1)^2 \|E_q\|^2$$

has its minimum at one only on the orbit through the declared RSCC vector. For a generic base vector C_q, the minimizer of $\|\lambda C_q - E_q\|^2$ is $\langle C_q, E_q\rangle/\|C_q\|^2$, and no present OPH dynamics requires minimizing this postulated residual. RSCC remains a post-hoc module-ledger ansatz, not a replacement for the Flavor Source Closure contract or a physical quark-mass postdiction.

Further-theorem audit and conditional QFRC rigidity. The subsequent audit corrects the finite-MaxEnt premise. On a finite spectrum, maximizing entropy under mean and covariance constraints gives a discrete exponential-quadratic Gibbs law, not a Gaussian density. Explicitly, on support $\{-2, -1, 0, 1, 2\}$,

$$p_A = (1/12, 1/6, 1/2, 1/6, 1/12), \qquad p_B = (1/16, 1/4, 3/8, 1/4, 1/16)$$

both have mean zero and variance one, while $\kappa_4(p_A) = 0$ and $\kappa_4(p_B) = -1/2$. The actual finite-support MaxEnt law at variance one has a nonzero fourth cumulant (approximately -0.46815). Hence neither finite MaxEnt nor the first two moments entails RSCC's Gaussian two-cumulant truncation. Gaussianization can be recovered only conditionally from an exported triangular array with a Lindeberg or adequate mixing/bounded-dependency condition, normalization, and covariance convergence; no such array is emitted by the source.

The Quark Flavor Register Closure (QFRC) proposal instead states an exact primitive-path certificate. Conditional on QF1–QF9, normalized trace fixes each path weight as rank over the declared register dimension, including

$$-\frac{1}{5}, \frac{4}{15}, \frac{1}{29}, -\frac{1}{432}, \frac{4}{5}, \frac{1}{22}, \frac{1}{32}, \frac{1}{420}, -\frac{1}{10}, -\frac{1}{4}, \frac{1}{1008}, \frac{1}{432}, -\frac{1}{1584}.$$

Exact record projectors also allow only the zero or unit scalar multiplier, and inert tensor refinement preserves normalized trace weights. This is a useful rigidity theorem, but its hypotheses contain the required physical content: the typed register construction, exhaustive primitive-path catalogue, effect ranks, structural multiplicities, signs, winding character, refinement intertwiners, implementation invariance, and positive-gap branch selector. A neutral finite register can be changed while preserving the broad structural signature and changing the normalized response, so the present broad axioms do not select QFRC. That countermodel is scoped to the broad signature; it is not a realization of every stronger carrier condition one might add.

The selector results sharpen the same boundary. An equivariant section requires a stabilizer-fixed candidate in each orbit fiber; an ambiguous transitive fiber need not have one. Conversely, a complete finite invariant candidate class with a unique positive-gap minimizer has a robust selector when refinement error is below half the gap. QFRC supplies neither a physically complete candidate class nor a source-derived cost and gap. The absolute-scale rescaling no-go, finite-renormalization scheme ambiguity, interval-root schema, and piecewise-RG uniqueness theorem are likewise exact obstruction or conditional well-posedness statements. No actual target-free source map with strict interval inclusion, operational clock, or frozen beta/threshold/matching packet accompanies the proposal. Therefore QFRC composes algebraically to the frozen RSCC packet but closes none of F1–F6 and authorizes no numerical quark-mass postdiction.

Targeted POFT carrier-emission assay. The Primitive Oriented Family Transport proposal supplies explicit complex 3×3 matrices T_0, T_1, but the simulator test must not manufacture them through a fitted readout. The frozen direct observable is therefore the mean of the natural three-label permutation matrices carried by the saved oriented S_3 edges. Its singular-value ratios are invariant under family permutations, unitary row/column phases, and an overall scale, so they give a necessary comparison before any entrywise match is considered. POFT requires

$$s(T_0)/s_1 = (1, 0.55147\ldots, 0.25517\ldots), \qquad s(T_1)/s_1 = (1, 0.54625\ldots, 0.27451\ldots).$$

A fresh 4,096-patch BW run, an independent 4,096-patch fusion run, a dense 65,536-patch population run, and a 65,536-patch BW run contain 830,066 edges in total. Their corresponding triples are

$$(1, 0.00522, 0.00134), \quad (1, 0.00552, 0.00361), \quad (1, 0.00179, 0.00002), \quad (1, 0.00071, 0.00028).$$

Source-node block bootstraps retain the same near-rank-one result. Every state lies within the predeclared 0.02 Haar-null tolerance and more than 0.5407 from both POFT targets under the predeclared 0.05 POFT tolerance. Moreover, the saved state schema contains only edge endpoints, integer S_3 labels, and geometry: it exports no complex oriented family amplitude and no paired coarse/fine edge intertwiner. Hence all direct T_0, refined T_1, and joint physical-emission receipts are false. The conclusion is scoped to the natural direct carrier. A source-derived complex lift could define a different carrier. Choosing its phases or paths to reconstruct the proposed matrices would make the simulator test circular.

Maximal theorem-emitted quark package theorem. Let

$$\begin{aligned} P := p_a & \\ &+ (\text{Axioms } 1\text{–}5) \\ &+ \text{Assumption } 6 \\ &+ \text{the emitted OPH nodes D1–D10} \\ &+ \text{the listed public D12 quark objects.} \end{aligned}$$

Then the quark-side package can be written in four layers:

1. the emitted D12 mass ray

$$D12^{\text{mass}}_{ud} = \mathcal{R}^{ud}_{D12} = \left\{ \lambda \left(\frac{1}{5}, -\frac{1 - x_2^2}{27} \right) : \lambda \geq 0 \right\}, \qquad \lambda = \text{ray\_modulus} = t_1;$$

2. the same-label left-handed selector value

$$\sigma_{ud} = \sigma_{\text{ref}},$$

   with canonical token `D12::same_label_left::reference_sheet`, which is a negative sheet-selector statement;

3. the separate target-free mass bridge

$$P \vdash \Delta^{\text{overlap}}_{ud} = \frac{1}{6} \log \frac{c_d}{c_u}, \qquad P \vdash \Theta^{\text{mass}}_{ud} = \texttt{quark\_same\_family\_value\_law};$$

4. the selected-class support wrapper on f_P, which proves representative independence for the attached physical spread datum. It does not select either modulus. Conditional mass readout would also require an RG-covariant trajectory and a declared comparison chart; physical Yukawa matrices require the additional common-scale dimensionless conversion.

Separate same-family and common-refinement artifacts reproduce their chosen target rows exactly. Their sigma datum is obtained by inversion of those rows. The associated GeV-valued matrices have the same status: target-anchored mass-texture audits. A continuation sidecar also backreads a mass-side scalar after identifying coefficient ratios with target mass ratios. None of these surfaces breaks the free source-spread action.

8.4 Quark Theorem Boundary

The conditional downstream closure route is

$$f_P \quad + \quad \Sigma_{ud}^{\mathrm{source}}(P) \Longrightarrow (\sigma_u, \sigma_d, \sigma_{\mathrm{seed}}^{ud}, \eta_{ud})_{\mathrm{phys}} \Longrightarrow (g_u, g_d) \Longrightarrow (m_u, m_d, m_s, m_c, m_b, m_t),$$

The first arrow is not identified by the source theory. The last arrow denotes scheme-labelled mass coordinates only after an RG trajectory and comparison chart are specified. Physical dimensionless Yukawa matrices lie one common-scale normalization beyond it. Selected-fiber descent and global frame classification are logically separate from both obstructions.

8.5 Strong-CP branch

The selected-class quark wrapper carries target-anchored mass textures on the public quark frame class f_P. Strong CP is a separate phase-side invariant. The available corpus does not derive the bare QCD angle θ_{QCD}, does not emit the physical anomaly-invariant combination $\bar{\theta}$, and does not prove that the physical strong-CP phase vanishes.

This phase problem is independent of the source-spread and common-scale Yukawa obstructions. A closure would require a source-emitted quark mass matrix at one declared scale, its physical determinant-line phase contribution, and a theorem fixing the topological-angle contribution on the realized branch. The present GeV mass textures do not supply that input.

9 Charged-Lepton Family Derivation

The charged-lepton derivation differs from the quark derivation in a precise way. It does not emit public charged masses from P. It emits an exact same-family witness, a closed common-shift no-go, the declared same-label q_e readback, a determinant-line lift on theorem-grade physical charged data, and a downstream algebraic mass readout from theorem-grade $A_{\mathrm{ch}}(P)$. For any fixed formal source exponent vector $M_\bullet^{\mathrm{ch}}$, the same-label readback defines a source-side determinant character

$$S_M = \sum_e M_e^{\mathrm{ch}} \log q_e.$$

The theorem lane does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The electron, muon, and tau rows are therefore reported as n/a on the public theorem lane.

The mathematical split is equally precise. The charged-lepton derivation starts from the ordered charged package, proves that the realized support is a one-dimensional linear subray, exposes the canonical quadratic support-extension direction, maps that into the charged excitation gaps, closes a two-scalar support-extension law shell, isolates the smaller eta source-readback primitive on that same carrier, and then builds the log-spectrum and forward shape/scale surface. Those centered objects are common-shift invariant. The determinant line fixes the physical affine scalar once theorem-grade physical charged data are present, and the mass readout is then algebraic. The charged theorem boundary has two distinct pieces. One piece is promotion of the latent charged sector-response candidate to theorem-grade $\widehat{C}_e$. The other is source-to-physical determinant attachment. For a fixed formal source exponent vector $M^{\mathrm{ch}}_{\bullet}$, that attachment is the identity

$$3\mu(r) = \sum_e M^{\mathrm{ch}}_e \log q_e(r),$$

equivalently zero determinant-normalization defect

$$N_{\mathrm{det}}(P) = s_{\mathrm{det}}(P) - \sum_e M^{\mathrm{ch}}_e \log q_e(P),$$

on the charged determinant channel.

Physically, the electron is the light charged lepton that makes atoms and chemistry possible, the muon is its heavier unstable cousin, and the tau lepton is the heaviest charged lepton. A promoted charged-lepton derivation would have to explain how those three rows emerge from the shared flavor dictionary without Koide-assisted fitting. The theorem-grade lane has to derive promotion of the latent charged sector-response candidate $\widehat{C}^{\mathrm{cand}}_e$ to theorem-grade $\widehat{C}_e$ by closing the branch-generator splitting theorem. On theorem-grade physical Y_e, a refinement-stable uncentered lift collapses the determinant-line section and affine anchor to one descended physical affine scalar $\mu_{\mathrm{phys}}(Y_e)$, with

$$\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\mathrm{phys}}(Y_e)\,\mathbf{1}, \qquad s_{\mathrm{det}}(Y_e) = 3\mu_{\mathrm{phys}}(Y_e), \qquad A_{\mathrm{ch}}(Y_e) = \mu_{\mathrm{phys}}(Y_e).$$

Theorem 9.1 (Charged same-carrier source-pair readback)**.** *Fix the ordered charged carrier*

$$(-1, x_2, 1),$$
$$x_2 = -0.5175863354681689.$$

If the charged source pair

$$(\eta_{\mathrm{ext}}, \sigma_{\mathrm{ext}}) = (\eta_{\mathrm{source\_support\_extension\_log\_per\_side}}, \sigma_{\mathrm{source\_support\_extension\_total\_log\_per\_side}})$$

is emitted on that carrier, then the centered charged logs are

$$e_{\mathrm{log,centered}} = -\frac{(3+x_2)\sigma_{\mathrm{ext}} - \eta_{\mathrm{ext}}}{6},$$

$$\mu_{\mathrm{log,centered}} = \frac{x_2\sigma_{\mathrm{ext}} - \eta_{\mathrm{ext}}}{3},$$

$$\tau_{\mathrm{log,centered}} = \frac{(3-x_2)\sigma_{\mathrm{ext}} + \eta_{\mathrm{ext}}}{6},$$

and therefore the charged masses are

$$m_e = g_e\, e^{e_{\mathrm{log,centered}}}, \qquad m_\mu = g_e\, e^{\mu_{\mathrm{log,centered}}}, \qquad m_\tau = g_e\, e^{\tau_{\mathrm{log,centered}}}.$$

This theorem is exact on the same-carrier shell, but the absolute values are not emitted. The visible scalar order is

$$\eta_{\mathrm{ext}} \quad \text{then} \quad \sigma_{\mathrm{ext}},$$

and the charged absolute scale g_e is unresolved. Builder-facing code therefore exposes η_{ext} and then σ_{ext} as the first same-carrier residuals, but the theorem-grade boundary lies above that surface. If the latent candidate $\widehat{C}^{\mathrm{cand}}_e$ is promoted, then η_{ext} and σ_{ext} become charged spectral invariants instead of separate primitive goals, and the absolute-scale burden is pushed to one affine-covariant absolute charged anchor A_{ch}. In the local chain, $\widehat{C}_e$ itself is undeclared: only the centered compressed generation-bundle branch-operator candidate $\widehat{C}_e^{\mathrm{cand}}$ is on disk, and the operator-side gate for that package is the upstream promotion theorem `oph_generation_bundle_branch_generator_splitting`, not a new ad hoc charged operator ansatz. Its exact gate clause is the compression-descendant commutator statement `compression_descendant_commutator_vanishes_or_is_uniformly_quadratic_small_after_central_split`. The centered common-shift quotient is closed negatively, so centered data alone do not emit the affine anchor A_{ch}.

Theorem 9.2 (Charged absolute-scale underdetermination)**.** *Let*

$$E_e^{\mathrm{centered}} = \left(e_{\mathrm{log,centered}}, \mu_{\mathrm{log,centered}}, \tau_{\mathrm{log,centered}}\right)$$

be the centered charged log triple emitted from the charged source pair $(\eta_{\mathrm{ext}}, \sigma_{\mathrm{ext}})$. Then for every $c \in \mathbb{R}$,

$$Y_e(c) := \exp(c)\,\mathrm{diag}\!\left(e^{E_e^{\mathrm{centered}}}\right)$$

has the same charged spectral invariants

$$\eta_{\mathrm{ext}}, \qquad \sigma_{\mathrm{ext}}, \qquad \gamma_{21}, \qquad \gamma_{32},$$

the same centered log vector, and the same ratio data. Only the absolute masses scale:

$$(m_e, m_\mu, m_\tau) \longmapsto e^c\,(m_e, m_\mu, m_\tau).$$

Hence the charged OPH chain determines only the quotient class

$$E_e^{\mathrm{centered}} \in \mathbb{R}^3/\langle(1,1,1)\rangle,$$

not the absolute scalar $g_e = e^c$.

The proof is immediate from the centered sum rule

$$e_{\mathrm{log,centered}} + \mu_{\mathrm{log,centered}} + \tau_{\mathrm{log,centered}} = 0,$$

which implies

$$\det Y_e^{\mathrm{shape}} = 1, \qquad \det Y_e = g_e^3.$$

All emitted charged invariants depend only on differences of log entries, so a common shift in the $(1,1,1)$ direction leaves them unchanged. This is the exact reason the available corpus closes the P-driven charged lane as a no-go instead of as a mass theorem: the theorem surface fixes the centered shape, while the determinant-line landing fixes the absolute normalization.

The charged absolute-scale lane is explicitly typed. The charged scale is a linear quantity

$$g_e = e^{\mu_e^{\mathrm{abs}}},$$

so the log-coordinate μ_e^{abs} must not be mixed directly with centered log gaps. The same-family writeback therefore records the type-consistent shell

$$\begin{aligned}\mu_{e,\mathrm{seed}}^{\mathrm{abs}} &= \log(0.9231656602589082)\\ &= -0.07994658034676537,\end{aligned}$$

$$\mu_{e,\mathrm{cand}}^{\mathrm{abs}} = \mu_{e,\mathrm{seed}}^{\mathrm{abs}} - \gamma_{\mathrm{min}} = -0.38231224060567365,$$

$$g_e^{\mathrm{cand}} = e^{\mu^{\mathrm{abs}}_{e,\mathrm{cand}}} = 0.6822819838027987.$$

This is a representation-consistency shell only, not a charged-mass theorem. It fixes the linear-vs-log coordinate discipline. This surface does not emit a theorem-grade value law for g_e. The audited shortcut

$$\Delta_e^{\mathrm{abs}} = 0.30236566025890826,$$
$$g_e = 0.6822819838027987$$

is not a theorem-grade closure. It merely chooses one representative on the common-shift orbit and does not land on the physical charged masses.

The next theorem object beyond this shell is one theorem-grade affine-covariant section A_{ch} of the quotient map, satisfying

$$A_{\mathrm{ch}}(\ell + c\,\mathbf{1}) = A_{\mathrm{ch}}(\ell) + c.$$

The clean realization of that section is an uncentered charged response lift carrying a determinant line, in which

$$A_{\mathrm{ch}} = \tfrac{1}{3}\log\det(Y_e) = \tfrac{1}{3}\,\mathrm{tr}(\log Y_e).$$

Once such an anchor exists,

$$g_e = e^{A_{\mathrm{ch}}(\ell)}, \qquad \Delta_e^{\mathrm{abs}} = \log g_{\mathrm{ch}}^{\mathrm{shared}} - A_{\mathrm{ch}}(\ell).$$

Compare-only charged continuation bridge. For comparison, there is a sharper charged continuation bridge under three extra continuation assumptions:

1. a uniform $\mathbb{Z}_6$ center-label ensemble, so $\varepsilon = 1/6$;
2. the balanced positive Hermitian circulant branch, so $\rho/a = 1/\sqrt{2}$;
3. the phenomenological charged-family phase choice $\delta = 2/9$.

These are continuation inputs, not consequences of the OPH axioms. For the positive-semidefinite Hermitian square-root-mass carrier

$$C = aI + \rho\left(e^{i\delta}R + e^{-i\delta}R^2\right), \qquad a, \rho \ge 0,$$

where R is the regular cyclic shift, $R^3 = I$. The commutant of the regular C_3 action is spanned by I, R, R^2, which makes C the general Hermitian C_3-equivariant carrier. Fourier diagonalization gives

$$r_k = a + 2\rho\cos\!\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2.$$

In a positive spectral chamber, $m_k = s r_k^2$ and $\sqrt{m_k} = \sqrt{s}\, r_k$. The roots-of-unity identities

$$\sum_{k=0}^{2}\cos\!\left(\delta + \frac{2\pi k}{3}\right) = 0, \qquad \sum_{k=0}^{2}\cos^2\!\left(\delta + \frac{2\pi k}{3}\right) = \frac{3}{2}$$

give the Koide invariant

$$Q = \frac{\sum_k m_k}{(\sum_k \sqrt{m_k})^2} = \frac{1 + 2(\rho/a)^2}{3},$$

so $\rho/a = 1/\sqrt{2}$ is exactly equivalent to $Q = 2/3$. At balance, positivity holds for $|\delta| \le \pi/12$ (mod $2\pi/3$), and throughout that chamber Koide is independent of δ. Outside it, physical square roots are absolute eigenvalues and the physical Koide ratio differs from the signed trace expression. Equivalently, if $E_0 = 3a^2$ and $E_c = 6\rho^2$ are the singlet and charged-plane Hilbert–Schmidt powers, then

$$Q = \frac{1 + E_c/E_0}{3}, \qquad Q = \frac{2}{3} \iff E_c = E_0.$$

The positive square-root-mass vector lies on the 45° Koide cone about the democratic direction. Circulant symmetry leaves the relative norm of the singlet and charged plane free.

The finite OPH response packet sets that norm. Let

$$\mathcal{V}_K = \mathbf{1} \oplus \chi \oplus \bar{\chi}, \qquad \mathcal{H}_{\mathrm{or}} = \mathbb{C}^2, \qquad \mathcal{A}_K = B(\mathcal{V}_K \otimes \mathcal{H}_{\mathrm{or}}) \simeq M_6(\mathbb{C}).$$

A one-state record cannot distinguish orientation reversal, so $\mathbb{C}^2$ is the minimal faithful orientation record. Application of Minimal Admissible Realization (MAR) to this response-local register is an explicit hypothesis. With q_+ one oriented outcome, define

$$E_+ = P_0 \otimes I_{\mathrm{or}} + P_c \otimes q_+, \qquad Z_0 = P_0 \otimes I_{\mathrm{or}}, \qquad Z_c = P_c \otimes q_+.$$

The blocks Z_0, Z_c both have rank two. Born–Lüders conditioning of the unique MaxEnt state $I_6/6$ on E_+ gives

$$p_0 = p_c = \frac{1}{2}.$$

The orientation-blind singlet has event action zero, while the selected charged orientation has action $\ln 2$. Charged multiplicity two times its weight $1/2$ equals the singlet weight. The unique normalized trace on $M_6(\mathbb{C})$ fixes these probabilities without the free trace parameter present on $\mathbb{C} \oplus M_2(\mathbb{C})$.

In $L^2(B(\mathbb{C}^3), \tau_3)$, the operators I, R, R^2 are orthonormal, and

$$\mathcal{J}(e_0) = I, \qquad \mathcal{J}(e_+) = R, \qquad \mathcal{J}(e_-) = R^2$$

is unitary and intertwines the C_3 action, orientation reversal, and the two block projections. The canonical GNS square-root amplitude is

$$\xi = \sqrt{p_0}\, e_0 + \sqrt{\frac{p_c}{2}} \left(e^{i\delta} e_+ + e^{-i\delta} e_- \right),$$

so its response has $a = \sqrt{p_0}$ and $|b| = \sqrt{p_c/2}$. Hence

$$\frac{|b|^2}{a^2} = \frac{p_c}{2p_0} = \frac{1}{2}, \qquad \frac{|b|}{a} = \frac{1}{\sqrt{2}}, \qquad Q = \frac{2}{3}.$$

An orientation action gap ΔS gives the general form

$$\frac{p_c}{p_0} = 2e^{-\Delta S}, \qquad \frac{|b|^2}{a^2} = e^{-\Delta S}, \qquad Q(\Delta S) = \frac{1 + 2e^{-\Delta S}}{3}.$$

The physical OPH Koide theorem is conditional on faithful trace-preserving u.c.p. maps Φ and Ψ from the source process to a physical response process and back, with $\Psi\Phi = \mathrm{id}$. Kadison–Schwarz applied to both maps puts every source element in the multiplicative domain of Φ, so Φ is a $*$-monomorphism and an exact L^2 isometry. If it also intertwines Z_0, Z_c with physical response records and sends the conditioned source state to the normalized positive square-root-mass response $C = (Y_e^\dagger Y_e)^{1/4}$, then it transports $p_0 = p_c$ to $E_0 = E_c$ and proves physical Koide. For an accepted finite chiral checkpoint with three left and three right family modes, positive kinetic metrics, and a committed neutral Higgs direction, the reversible-checkpoint conditions force $\Phi = \mathrm{Ad}_{J_L \oplus J_E}$, preserving chirality and the regular C_3 action. If $M_F = X_F^2$ is the positive source response, then

$$\widehat{Y}_e = \frac{\sqrt{2}}{v} J_L M_F J_E^\dagger, \qquad \mathcal{M}_L = J_L M_F J_L^\dagger.$$

The source and canonical physical square-root responses have equal block powers. This closes the finite source-scale mass response. MAR selection of the accepted chiral sector, its physical carrier and refinement data, positivity, mass-scheme descent, and the all-orders threshold or resonance continuation are work in progress. The relation $\mathrm{Hom}_{C_3}(\mathbf{1}, \chi \oplus \bar{\chi}) = 0$ prevents the neutral source grammar from producing a labeled charged-family vector through symmetry. A charged source tensor, connection, or selected orbit must supply the phase and individual ratios.

The count $(N_c + 1)/(2N_c N_g) = 2/9$ is arithmetic. A physical phase theorem requires a regular-C_3 family bundle, an oriented family connection whose declared loop has holonomy $\exp[i\beta_{\mathrm{EW}}/(2N_c N_g)]$, and an attachment identifying that holonomy with the phase of the charged Fourier coefficient. Hypercharge acts as a common scalar on the generation copies, while the shift R permutes them. Pure finite A_5/C_3 holonomy yields cube-root phases. Construction of the required continuous family connection and its source selector is work in progress. Under the stated connection and attachment hypotheses, $\delta = 2/9$. The phase of one equal link differs by a factor of three from the total triangular holonomy. With the scale-free normalization $a = 1$, the ordered roots

$$r_k = a + 2\rho \cos\left(\delta + \frac{2\pi k}{3}\right)$$

become

$$\begin{aligned} r_e &= 0.040349908219207475, \\ r_\mu &= 0.5802119201475368, \\ r_\tau &= 2.3794381716332555. \end{aligned}$$

Under those assumptions the bridge selects the same-carrier pair

$$\begin{aligned} \eta_{\mathrm{ext}} &= -6.729586682888832, \\ \sigma_{\mathrm{ext}} &= 8.154061112725994, \end{aligned}$$

with ordered gap values

$$\begin{aligned} \gamma_{21} &= 5.33160859254774, \\ \gamma_{32} &= 2.822452520178255, \\ \kappa_{\mathrm{ext}} &= -4.59605680397234, \end{aligned}$$

and centered logs

$$E_{\mathrm{log,centered}} = [-4.495223235091244,\ 0.836385357456495,\ 3.6588378776347503].$$

Against the charged references, this continuation-centered shape has residual norm

$$\left\| E^{\mathrm{cont}}_{\mathrm{log,centered}} - E^{\mathrm{ref}}_{\mathrm{log,centered}} \right\| \approx 2.13 \times 10^{-5},$$

so this compare-only branch is a near-exact centered-shape closure up to one common absolute scale. The theorem lane remains unpromoted because the public affine absolute anchor is external to this near-exact centered charged-shape branch. The compare-only common scale required for exact absolute masses is

$$g_e^\star = 0.04577885783568762.$$

Equivalently, relative to the stored shared-budget seed

$$g_{\rm ch}^{\rm shared} = 0.9231656602589082,$$

the missing affine absolute anchor on the determinant-line route would have to take the target value

$$\Delta_e^{\rm abs,\star} = \log \frac{g_{\rm ch}^{\rm shared}}{g_e^\star} = 3.003986333402356.$$

This identifies the public charged theorem boundary: the compare-only branch nearly solves the centered charged shape, but the theorem-grade derivation lacks promotion of the latent candidate $\widehat{C}_e^{\rm cand}$ to theorem-grade $\widehat{C}_e$, and then one affine-covariant absolute anchor $A_{\rm ch}$ that would turn that centered readback into public charged masses on the theorem lane.

Frozen complete Stage-5 postdiction. The historical Stage-5 continuation also supplies a complete, reference-free-at-runtime numerical postdiction on its frozen legacy branch when its additional exponent and determinant prescriptions are retained. Besides the three continuation assumptions above, set

$$\boldsymbol{n}_e = (7, 4, 3), \qquad \det M_e = \frac{v(P)^3}{2\, 6^{14}},$$

and normalize the squared-root direction by its geometric mean:

$$m_i = \frac{v(P)}{(2\, 6^{14})^{1/3}} \frac{r_i^2}{(r_e^2 r_\mu^2 r_\tau^2)^{1/3}}.$$

At the legacy D10 running-tree value $P = 1.63094$, $v = 246.76711732749683\,\mathrm{GeV}$, the frozen formula emits

$$\begin{aligned} m_e^{\rm S5} &= 0.510882243295\,\mathrm{MeV}, \\ m_\mu^{\rm S5} &= 105.635282871\,\mathrm{MeV}, \\ m_\tau^{\rm S5} &= 1776.579124017\,\mathrm{MeV}. \end{aligned}$$

Against the PDG-2026 comparison packet these differ by approximately -228, -219, and -197 ppm, respectively; the maximum absolute mass error is 0.02284%. Re-evaluating the same formula on the public-pixel probe $P = 1.6309681897$, $v = 246.6174823334856\,\mathrm{GeV}$, gives

$$(m_e, m_\mu, m_\tau) = (0.510572453797,\ 105.571227599,\ 1775.501839453)\,\mathrm{MeV},$$

with maximum absolute error 0.08347%. Thus the sharper 0.02284% number is explicitly legacy-branch-specific. The builder does not read charged reference masses, but this remains a postdiction rather than a theorem-grade prediction because the balanced carrier and 2/9 phase were prior empirical art, the electron exponent was selected inside a historical residual scan that fixes only exponent differences, the factor $2^{1/6}$ is asserted rather than derived, and the corpus does not derive the determinant attachment or the sorted-root-to-family map. Nor is a renormalization coordinate and QED/electroweak map from this historical mass coordinate to pole masses supplied, so the ppm comparison is descriptive rather than scheme-certified. The frozen executable receipt is `code/particles/runs/leptons/charged_stage5_frozen_candidate.json`; any change made to improve agreement must create a new candidate rather than overwrite this one.

Frozen retrospective icosahedral face-incidence candidate. The screen geometry supplies a sharper candidate carrier than a scalar twelve-port moment. The twenty outward-oriented icosahedral faces form A_5/C_3, and the face stabilizer cyclically permutes the three corners. Thus each face has a canonical local regular-C_3 fiber, and an equivariant Hermitian circulant has a face-representative-independent unordered spectrum. This is an exact geometric lemma. It is not yet a physical generation attachment: the sixty face-corner flags form the regular A_5 torsor, not one canonical three-dimensional matter-family space.

A submitted continuation on this carrier adds three finite-incidence equalizers for the common affine coordinate κ, amplitude defect χ_ρ, and phase defect ζ_δ. On its frozen submitted tuple it emits

$$\begin{aligned} m_e^{\text{face}} &= 0.510998950843394\,\text{MeV},\\ m_\mu^{\text{face}} &= 105.658375501481\,\text{MeV},\\ m_\tau^{\text{face}} &= 1776.930000014351\,\text{MeV}. \end{aligned}$$

The 0.000300-ppm headline uses the frozen rounded reference fields. Against the same packet's higher-precision tau central value 1776.93246513409 MeV, the submitted tau residual is -1.387289 ppm. The archive is hash-bound and its builder is reference-separated at runtime. The construction is nevertheless target-informed: the equalizers were found after the masses were known, and their three values agree to approximately one part per million or better with the three coordinates obtained by inverting the comparison triple inside the same ansatz. Moreover, the submitted tuple combines the truncated probe $P = 1.6309681897$ and its v coordinate with α_U from the distinct full public pixel. Evaluated coherently, the same formulas have maximum mass residuals against the rounded fields of approximately 0.424 ppm on the canonical public branch and 84.10 ppm on the source-audit branch. The public pixel is itself a conditional endpoint branch, and no charged running-to-pole theorem is supplied.

A follow-up theorem package explicitly declares the diagonal affine repair map

$$T(\kappa, \chi_\rho, \zeta_\delta) = (s_\kappa - q_\kappa \kappa,\ s_\chi + q_\chi \chi_\rho,\ s_\zeta + q_\zeta \zeta_\delta).$$

For that declared map it proves $\|DT\|_\infty = 0.0015356510519\ldots < 1$, so Banach's theorem closes existence, uniqueness, and iterative stability of its fixed point. This is a real conditional closure, not a derivation of the repair dynamics. The same package gives a source-multiplier family $T_{\boldsymbol{\lambda}}$ with the same displayed symmetry, analytic degree, Jacobian, and contraction but different masses. It therefore proves scoped selector non-identifiability under those properties. It does not prove non-entailment from every OPH axiom. In addition, exact zeroth-order block balance gives $|b|/a = 1/\sqrt{2}$, whereas the endpoint operator uses $|b|/a = e^{-\chi_\rho}/\sqrt{2}$; a source-derived bare-to-endpoint repair bridge must stage those two statements.

Proposition 9.3 (Finite CFQ schema satisfiability)**.** *The stipulated CFQ register-and-path class is nonempty. There is an explicit finite algebraic model with register dimensions*

$$(50, 31, 10, 512, 77, 21, 27, 5),$$

connected transition graphs, normalized tracial states, rank-one event projectors, and the eight declared path weights

$$\left(\frac{1}{50}, -\frac{1}{31}, -\frac{1}{310}, \frac{1}{512}, \frac{1}{77}, \frac{1}{21}, \frac{1}{27}, \frac{1}{135}\right).$$

Each noncentral event in $B(H_r)$ admits an accepted/rejected central record dilation in $B(H_r) \otimes D_2$. The graph family is covariant under the sixty proper icosahedral rotations, and normalized path weights are unchanged under inert ancillary stabilization $X \mapsto X \otimes I_k$.

Proof. For each declared connected graph, the diagonal matrix units and both directed units on every edge generate E_{ij} along graph paths and hence generate the full matrix algebra. The normalized trace of a rank-one event is the reciprocal register dimension, and tensor products multiply the two depth-two weights. For an event P, the pinching map

$$\mathcal{E}_P(X) = PXP + (I-P)X(I-P)$$

is a unital trace-preserving conditional expectation. Its instrument writes $P\rho P$ and $(I-P)\rho(I-P)$ into orthogonal accepted/rejected pointers in the central D_2 factor. Explicit permutation intertwiners preserve graph incidence and ranks for all sixty proper rotations. Finally, $\tau_{dk}(P \otimes I_k) = \tau_d(P)$, and the partial-trace coarse map retracts the inert embedding. These facts construct one model of the declared schema. □

Proposition 9.3 proves formal nonemptiness. It supplies no derivation of the charged source. The submitted builder hard-codes the eight dimensions, path automaton, coupling degrees, grading, signs, clock, and scalar response. Its verifier duplicates that expected schema and exhausts only the authored automaton, rather than the physically admissible path space of OPH dynamics. The dependency DAG omits those design choices and the known retrospective formula history, so runtime absence of mass files does not establish historical no-target ancestry. In adversarial audit, target-corrupted response, weight, phase, and builder fields received a submitted pass; its negative-control harness also accepts any nonzero verifier exit. The construction therefore resolves the abstract event-versus-central-record issue at fixed cutoff and proves internal consistency, while physical response selection, cofinal screen refinement, and charged-sector attachment remain open.

Proposition 9.4 (Conditional charged nature and singularity transport)**.** *Let M_F be the positive face endpoint. Suppose a physical chiral three-family carrier supplies a natural unitary J_L, canonical kinetic metrics, and*

$$\frac{v^2}{2}\widehat{Y}_e\widehat{Y}_e^\dagger = J_L M_F^2 J_L^\dagger. \tag{NI6}$$

Then its positive left charged response is $J_L M_F J_L^\dagger$. Suppose in addition that the exact renormalized charged kernel is supplied, its singular lines satisfy the declared infrared and regularity conditions, and a CFQ–Dyson map has singularity readout $J_L M_F J_L^\dagger$. Regular invertible field changes preserve the zero set, and a regular Nielsen factorization $\partial_\xi K = A_\xi K + K B_\xi$ makes each simple zero gauge independent.

Proof. The first statement follows from uniqueness of the positive square root. Multiplication of K on the left and right by analytic invertible matrices multiplies its determinant by nonvanishing factors. Jacobi's identity gives

$$\partial_\xi \det K = (\operatorname{tr} A_\xi + \operatorname{tr} B_\xi) \det K,$$

so a simple zero cannot move while the Nielsen factors are regular. These are the standard mixed-fermion and gauge-independence implications used in the pole literature [11, 12]; the stable charged-line infrared qualification follows the infraparticle boundary [13, 14]. □

Proposition 9.4 closes the downstream logic after its physical premises are supplied. It does not supply them. NI6 is the desired operator attachment squared, and the submitted RP4 premise sets the Dyson singularity readout equal to the face response. The archive's explicit kernel $K_0(s) = sI - M_F^2$ has zero self-energy; it is a free algebraic existence witness rather than the interacting charged-lepton 1PI kernel. All physical NI1–NI8 and RP1–RP8 receipt fields remain open.

The builder embeds the target-informed hybrid coordinates as defaults without hash-binding their parent receipts. Its verifier reconstructs the fixed point and response shape but never checks $M_F = g_{\mathrm{end}} S_F$: a coherent change of the ordered spectrum by factors $(2, 1/2, 1)$, together with the matching mass matrix, Yukawa matrix, and pole roots, preserves the determinant and satisfies the tested mass relations. The package is therefore a conditional gate theorem, not physical nature identification, an interacting pole theorem, or a promoted mass prediction.

The valid status is therefore `FROZEN_RETROSPECTIVE_TARGET_INFORMED_FACE_CANDIDATE`, with public promotion false. Promotion requires a quotient-visible face-to-charged-family intertwiner, the $\ln 2$ MaxEnt and ensemble-to-Yukawa bridge, a charged connection deriving both the $2/9$ base phase and its correction, a normalized $\mathbb{Z}_6$ determinant character, an OPH dynamical theorem selecting the physical CFQ registers, state, exhaustive path category, grading, clock, and response rather than declaring them, cofinal refinement naturality, one receipt-bound coherent source tuple frozen before comparison, and a mass- scheme map. The exact receipts are `charged_icosahedral_face_carrier_frontier.json` and `charged_face_incidence_conditional_theorem.json`, together with the conditional CFQ weight receipt; the compare-only audits are `charged_icosahedral_completion_v1_review.json` and `charged_face_incidence_conditional_theorem_review.json`, the CFQ rigidity review, and `charged_cfq_carrier_v1_review.json`. This candidate is the leading explicit completion conjecture, but it does not change the theorem mass rows.

9.1 Absolute charged-lepton mass intervals from electromagnetic transport

The missing absolute anchor is one real scale. Fixing A_{ch} is equivalent to fixing the single family rescaling $Y_e \mapsto e^{\kappa} Y_e$, under which every charged mass scales as $m_i \mapsto e^{\kappa} m_i$ while the centered shape and every mass ratio are invariant. The declared charged antecedents are invariant under that rescaling, which is the content of the common-shift no-go: the gauge representations, the anomaly structure, the complexity vector, the determinant algebra, and the same-label readback take the same value on the whole family $\{e^{\kappa} Y_e\}$, so no source object built from those antecedents alone selects κ.

The rescaling is not a symmetry of the declared electromagnetic transport. The charged leptons enter the photon vacuum polarization, and the leptonic packet in the fine-structure endpoint map carries

$$P_{\mathrm{lep}}(\kappa) = \frac{1}{3\pi} \sum_i \left[2 \log \frac{M_Z}{m_i} - \frac{5}{3} \right], \qquad \frac{\mathrm{d}P_{\mathrm{lep}}}{\mathrm{d}\kappa} = -\frac{2}{\pi}.$$

The endpoint pixel fixed point $P = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$ consumes P_{lep}, so the emitted endpoint and the pixel move with κ. The common-shift no-go omits this transport from its antecedent list. Source-only, the system is the stiff curve $(P(\kappa), \kappa)$ with $|\mathrm{d}P/\mathrm{d}\kappa| \approx 6 \times 10^{-5}$, and the absolute masses are work in progress on the source branch.

On the empirical-closure surface the transport identifies κ. Inverting the on-shell decomposition

$$a_0 + g = A^{-1}(0)\,(1 - P_{\mathrm{lep}}(\kappa) - \Delta_{\mathrm{had}} - \Delta_{\mathrm{top}})$$

with the frozen anchor $a_0 = 128.308$, the certified anchor-gap interval $g \in [0.649, 0.855]$, the empirical hadronic payload $\Delta_{\mathrm{had}} = 0.02676 \pm 0.00075$, the frozen continuation mass ratios, and the measured $A^{-1}(0)$ supplied as a compare-only exclusion inside the interval solve path gives

$$\kappa \in [-0.346, +0.337], \qquad \kappa_{\mathrm{central}} = -0.004.$$

The charged masses then carry certified intervals,

$$
\begin{aligned}
m_e &= 0.5089\ [0.362,\ 0.716]\ \mathrm{MeV}, && \text{measured } 0.5110\ \mathrm{MeV},\\
m_\mu &= 105.22\ [74.8,\ 148.0]\ \mathrm{MeV}, && \text{measured } 105.66\ \mathrm{MeV},\\
m_\tau &= 1.7695\ [1.258,\ 2.489]\ \mathrm{GeV}, && \text{measured } 1.7769\ \mathrm{GeV},
\end{aligned}
$$

with central values within about 0.4% of measurement and the physical triple inside every interval. The interval width reduces to the empirical hadronic payload uncertainty and the anchor gap, which are the two open objects of the fine-structure lane. A closed source hadronic spectral measure and a closed anchor bridge tighten the intervals around the measured values.

The continuation bridge of the previous paragraph is a consistency test on this scale. Its internal determinant prescription reproduces the physical leptonic packet at $|\kappa| \approx 0.006$, so that prescription is a tested ansatz on the absolute scale rather than a free parameter. The construction adds no axiom. The source-only charged no-go and its promotion gate keep their status; the interval lane carries the absolute masses on the empirical-closure surface in the interim.

9.2 Conditional completed-response candidate and the symmetry-breaking frontier

A submitted completion package closes the charged response conditionally by stipulating a typed, incidence-complete response architecture on an oriented icosahedral flag: eight finite registers of dimensions $50, 31, 10, 512, 77, 21, 27, 5$, eight primitive path classes with normalized rank-one trace weights, a public-block amplitude, an oriented process phase, and a $\mathbb{Z}_6$ determinant character. Conditional on that architecture, the three-channel response map is a Banach contraction with contraction constant 1.5×10^{-3}; its unique fixed point, the regular-C_3 shape functional, and the determinant character emit one dimensionless charged triple on the strict source-audit branch with zero runtime charged reference input:

$$
\frac{(m_e, m_\mu, m_\tau)}{E_\star} = (4.18511 \times 10^{-23},\ 8.65348 \times 10^{-21},\ 1.45532 \times 10^{-19}),
$$

with mass ratios $m_\mu/m_e = 206.7683$, $m_\tau/m_e = 3477.3655$, $m_\tau/m_\mu = 16.8177$, and unclosed-clock displays $(0.51096, 105.649, 1776.78)$ MeV. A segregated compare-only audit places this coherent branch about 84 ppm below the frozen comparison triple; the comparison file is excluded from the candidate ancestry (`charged_mcpr_completion_conditional.json`).

The symmetry-breaking audit fixes the claim class of that candidate. The register dimensions, path table, signs, amplitude, phase, and determinant exponent are declared model inputs, so the triple is a declared-model coordinate on the completed theory. The Koide subsection derives the finite public-block amplitude $1/\sqrt{2}$ from the connected M_6 response register. The candidate supplies no recoverable attachment between that event and a physical chiral mass channel, which leaves its mass claim conditional. The arithmetic identity $(N_c + 1)/(2N_cN_g) = 2/9$ supplies no oriented phase; a holonomy reading requires link variables, a loop, and a nonlocal readout map. Finally, a product of fourteen normalized rank-one traces is neither a $\mathbb{Z}_6$ character nor a determinant, and the canonical determinant norm carries the kinetic factor $\det \mathcal{M}_L = (v/\sqrt{2})^3\, |\det Y_e|/\sqrt{\det Z_L \det Z_E}$; the historical 6^{-14} weight is a candidate positive path weight.

The same audit reduces the source-side completion to a finite symmetry-breaking program on the icosahedral carrier. The vertex permutation module decomposes as $\mathbb{R}^{12} \cong W_1 \oplus W_3 \oplus W_3' \oplus W_5$; the traceless quadrupole map $Q(w) = \sum_i w_i(p_i p_i^T - I/3)$ satisfies $Q = QP_5$ and $Q^*Q = \frac{8}{5}P_5$, so it sees exactly W_5, and the two adjacent gaps of its spectrum span the centered family plane. A unique invariant MaxEnt state has zero expectation on every nontrivial irreducible module, and the homogeneous twelve-port branch has the unique uniform minimizer, so charged family shape requires a source-selected W_5 orbit. Because $\mathrm{Sym}^2_0(\mathbf{3}) \cong \mathrm{Sym}^2_0(\mathbf{3'}) \cong W_5$, the family attachment is multiplicity-one up to one sign once the chiral family fibers carry a three-dimensional A_5 representation. The open objects are an A_5-invariant effective action on W_5 with a unique simple-spectrum minimizing orbit (the invariant algebra has one quadratic, two cubic, and two quartic coefficients, so the first decisive quantity is the W_5-restricted Hessian scalar h_5 at the homogeneous state), the physical A_5 family lift, the normed determinant-line descent with kinetic factors, the interacting charged kernel packet, the operational scale, and the prospective freeze.

The affine scale has a second admissible closure through electromagnetic transport. With the exact one-loop Ward kernel $I(z) = \int_0^1 x(1-x)\log(1+zx(1-x))\,\mathrm{d}x$ in closed form, the charged response $\mathcal{W}_Q(\mu;\ell) = \frac{2}{\pi}\sum_i I(q^2 e^{-2(\mu+\ell_i)})$ is a smooth, strictly decreasing bijection of the common log-scale μ onto $(0,\infty)$ at fixed centered shape ℓ, with high-energy slope $-2/\pi$. A source-complete spacelike Ward endpoint pair with a source-complete nonleptonic subtraction therefore fixes a unique μ_{ch}, hence the determinant line $|\det(M_e/E_\star)| = e^{3\mu_{\mathrm{ch}}}$ and the electron ratio $m_e c^2/E_\star$ consumed by the cesium clock packet. The common-shift no-go of the preceding subsections holds on the reduct without mass-dependent electromagnetic transport; a Ward endpoint response and a determinant-line basepoint are the two admissible closures of that orbit. The Ward inversion, its interval enclosure, and its fail-closed source-packet schema are encoded in `charged_ward_determinant_line.json`; the endpoint pair, the nonleptonic subtraction, the higher-order monotonicity certificate, and the atomic transport factor are its open parents.

10 Neutrino Family Derivation

The neutrino lane contains one exact no-go and one failed comparison candidate. The intrinsic isotropic branch is excluded by its spectral cap. The weighted-cycle construction descends from two hand-written family-transport matrices and declared label, orientation, cycle, and exponent choices. Its exponent law was promoted after candidate laws had been ranked against the measured splitting ratio. The resulting point therefore has retrospective, target-informed template status. The bridge and absolute attachment use compare-only information and carry no prediction status.

For the isotropic matrix $M = aI + \rho C$ with unit-modulus off-diagonal entries, the general Gershgorin estimate applied to $M^\dagger M$ gives

$$\max_{i,j} |\Delta m^2_{ij}| \le 8a\rho + 4\rho^2.$$

The live bound is $1.52304 \times 10^{-6}\,\mathrm{eV}^2$, about three orders of magnitude below the atmospheric scale. The previously quoted $8a\rho$ expression omitted the $4\rho^2$ term and was not a valid general bound.

The frozen weighted-cycle candidate gives

$$\begin{aligned}\theta_{12} &= 34.225904631810025^\circ,\\ \theta_{23} &= 49.72282845058266^\circ,\\ \theta_{13} &= 8.686355527700156^\circ,\end{aligned}$$

$$\begin{aligned}\delta_{\mathrm{PMNS}} &= 305.58061231449796^\circ,\\ J &= -0.02753115613565372,\end{aligned}$$

and the dimensionless hierarchy invariant

$$\frac{\Delta m^2_{21}}{\Delta m^2_{32}} = 0.030721110097966534.$$

The official NuFIT 6.1 normal-ordering profile gives

$$\Delta\chi^2_{23,\delta} = 20.11955 \quad \text{with the tabulated atmospheric likelihood}, \qquad \Delta\chi^2_{23,\delta} = 18.43528 \quad \text{without it},$$

at $(\sin^2\theta_{23}, \delta_{\mathrm{CP}}) = (0.582056, -54.419^\circ)$ [10]. Both values exceed the two-parameter 3σ contour value 11.83. Separate marginal interval checks had hidden this correlation. The published profiles overlap and are not summed; their maximum is a lower bound on the unavailable full fixed-candidate $\Delta\chi^2$.

The historical shared-basis construction set

$$M_{\mathrm{shared}} = U^*_{e,\mathrm{left}}\, M_{\mathrm{wc}}\, U^\dagger_{e,\mathrm{left}}, \qquad U_{\nu,\mathrm{shared}} = U_{e,\mathrm{left}}\, U_{\mathrm{wc}},$$

the identity

$$U^\dagger_{e,\mathrm{left}} U_{\nu,\mathrm{shared}} = U_{\mathrm{wc}}.$$

This cancellation is a tautology because $U_{\nu,\mathrm{shared}}$ was defined as $U_{e,\mathrm{left}}U_{\mathrm{wc}}$. It does not independently identify the physical charged-lepton basis or validate the PMNS point. No theorem in the theorem stack places the f-labelled weighted-cycle operator in the charged-lepton mass basis.

The stored charged-basis artifact also fails its own promotion gate. Its source has `closure_state: open`, and its three normalized shape singular values differ only at about the 10^{-12} relative level, so its left singular vectors do not define a stable physical family basis. The charged-lepton continuation is support-extension gated and does not supply a replacement labelled basis. At source level, the physical PMNS matrix is therefore unformed.

Taking the stored basis declarations literally gives a sharply different diagnostic. Reordering M_{wc} from $[f3, f1, f2]$ to the independently declared shared basis $[f1, f2, f3]$, computing its Takagi matrix $U_\nu^{(f)}$, and then forming $U^\dagger_{e,\mathrm{left}} U_\nu^{(f)}$ yields

$$\theta_{12} = 38.6812^\circ, \qquad \theta_{23} = 76.2759^\circ, \qquad \theta_{13} = 44.8283^\circ,$$

with $\delta = 324.2140^\circ$ and $J = -0.0233164$. In particular, $\sin^2\theta_{13} = 0.4970$ and $\sin^2\theta_{23} = 0.9437$, both outside the corresponding NuFIT 6.1 profile grids. This is a diagnostic conditional on the stored matrices and label order. It is not a source-closed prediction because the charged basis and family kernel also descend from template-level inputs.

The correct transported Takagi identity is

$$U^T_{\nu,\mathrm{shared}} M_{\mathrm{shared}} U_{\nu,\mathrm{shared}} = \mathrm{diag}(m_i) \in \mathbb{R}_{>0}.$$

Here U_{wc} denotes the canonical phase-fixed Takagi unitary, not the raw arbitrary-phase eigensystem of $M^\dagger_{\mathrm{wc}} M_{\mathrm{wc}}$. Equivalently, $U^T_{\mathrm{wc}} M_{\mathrm{wc}} U_{\mathrm{wc}}$ is positive diagonal in the basis where M_{wc} was declared. The earlier expression with $U^T_{\mathrm{PMNS}} M_{\mathrm{shared}} U_{\mathrm{PMNS}}$ was algebraically false. Canonical Takagi readout under the declared charged-basis assumption and the electron-row gauge $U_{e1} \in \mathbb{R}_{>0}$ gives the comparison-only Majorana pair

$$\alpha_{21}^{(\mathrm{Maj})} = 153.6185177794357^\circ, \qquad \alpha_{31}^{(\mathrm{Maj})} = 257.0032408220805^\circ.$$

The declared exponent law uses the positive transport-load segment between $\chi = 1+\epsilon$ and $1+\gamma_{1/2}$: on a one-dimensional affine segment, the balanced selector and the least-distortion selector for any positive translation-invariant quadratic form coincide at the midpoint, so

$$D_\nu = \frac{\chi + (1+\gamma_{1/2})}{2}, \qquad p = 1 + \gamma + \frac{\epsilon}{D_\nu}.$$

The midpoint fact does not derive the segment endpoints, the exponent law, the cycle topology, or their application to neutrino transport. Git history records target-ranked model selection before this law was promoted. The formula is therefore part of the frozen candidate rather than a source-side consequence of OPH. On the declared positive selector segment there is a stronger compare-only two-parameter adapter: under the declared normal-ordering hypothesis, solving τ_ν against the representative central ratio and then solving λ_ν against Δm^2_{32} gives

$$(m_1, m_2, m_3) = (0.01745663295,\ 0.01948419960,\ 0.05308139066)\ \text{eV},$$

$$\begin{aligned}\Delta m^2_{21} &= 7.49 \times 10^{-5}\ \text{eV}^2,\\ \Delta m^2_{31} &= 2.5129 \times 10^{-3}\ \text{eV}^2,\\ \Delta m^2_{32} &= 2.438 \times 10^{-3}\ \text{eV}^2.\end{aligned}$$

These exact central numbers are compare-only. The bridge audit also found target feedback in the reduced invariant, so the following values are retained only as diagnostic coordinates:

$$\begin{aligned}C_\nu &= 0.9994295999075177,\\ P_\nu &= 6.699825740519345,\\ B_\nu &= P_\nu C_\nu = 6.696004159297337,\end{aligned}$$

and therefore

$$\lambda_\nu = \frac{m_{\star,\text{eV}}}{q_{\text{mean}}^{p_\nu}} P_\nu C_\nu = 1.7237014208357415,$$

$$m_i = \lambda_\nu \widehat{m}_i, \qquad \Delta m^2_{ij} = \lambda_\nu^2 \widehat{\Delta}_{ij}.$$

The normalized same-label overlap-defect calculation is exact after the template and declared selectors are supplied. The positive-segment adapter, bridge corridor, and bridge-coordinate sidecars remain diagnostic surfaces. In particular, the following quantity is only a diagnostic coordinate:

$$C_\nu := \frac{B_\nu}{I_\nu^{1/2}\,\widehat{\text{ratio}}^{1/2}\,\text{sum\_defect}^{-1}}$$

It was defined above the declared proxy and normalizer. The attached stack also factors exactly through $q_e = q_{\text{mean}} qbar_e$, so a $qbar_e$-only collapse law for the bridge factor does not follow from the conditional algebraic surface stated here. The best algebraically complete conditional local object beneath that bridge is the defect-weighted same-label edge family $q_e = \sqrt{g_e d_e}$ together with the induced μ_e family, but that family sits below C_ν and the induced paper-facing amplitude B_ν instead of replacing them. No source-only PMNS, hierarchy, Majorana-phase, or absolute-mass prediction survives these gates.

Cosmological-neutrino pressure surface. The compare-only absolute attachment displays

$$\sum_i m_{\nu_i} = 0.09001192964464505\ \text{eV}.$$

Under the declared normal-ordering hypothesis, standard relic-neutrino inheritance, and absent an explicitly declared extra OPH relativistic coherence channel, the diagnostic cosmological-neutrino-background branch uses

$$N_{\text{eff}}^{\text{OPH}} = 3.044,$$

with the usual broad, low-amplitude normal-ordering free-streaming imprint. This is a diagnostic cosmological pressure surface for the rejected candidate. The displayed mass sum sits below the Planck+BAO 0.12 eV bound [8] but is exposed to strict DESI DR2 ΛCDM-style bounds near 0.0642 eV under their model assumptions [9]. A cosmological upper bound below the displayed compare-only value would exclude the displayed absolute-mass continuation under a model class that also respects oscillation lower limits and the OPH background assumptions.

On the same-label neutrino-only branch, the centered eta-class is exactly S_3-isotropic, so the same-label data are edge-constant and the solar $1-2$ split cannot open there by itself. The first solar mover therefore has to come from realized flavor-side same-label gap/defect readback, not from another neutrino-only selector tweak.

10.1 Intrinsic neutrino eta-chain

The proof-facing input is smaller than the raw same-label matrix payload. It compresses to the same-label scalar certificate

$$(g_e,\ \omega_e)_{e\in\{12,23,31\}},\qquad \omega_e = \text{same-label overlap}_e^2,\qquad d_e = 1-\omega_e,$$

modulo one common scale. From that certificate one forms

$$q_e = \sqrt{g_e d_e},\qquad \eta_e = \log q_e - \frac{1}{3}\sum_f \log q_f,\qquad e\in\{12,23,31\},$$

the intrinsic selector depends only on the centered class

$$[\eta_e] \in \mathbb{R}^3/\langle(1,1,1)\rangle.$$

Equivalently one may work with any positive normalized family

$$\mu_e = \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

Common scaling cancels identically, so the intrinsic selector is determined by the centered eta-class alone.

This factorization result is conditional on the scalar inputs; it is not a source-closure result. The scalar values are numerically complete, but their gap data inherit the template family-transport kernel and their overlap data inherit a candidate-only line lift. The code propagates those upstream statuses, so the intrinsic spectrum remains a conditional diagnostic until both inputs close at source level.

On the isotropic neutrino-only branch one has

$$(\eta_{12},\eta_{23},\eta_{31}) = (0,0,0),$$

which gives the conditional intrinsic singular values

$$(s_0,s_1,s_2) = (2.3986448447627196,\ 2.3986448447627196,\ 2.590074050773907)\times 10^{-12}\ \text{GeV},$$

with ascending squared-mass gaps

$$\begin{aligned} s_1^2 - s_0^2 &= 0,\\ s_2^2 - s_0^2 &= 9.549864971855843\times 10^{-25}\ \text{GeV}^2. \end{aligned}$$

This is the exact neutrino-only isotropy obstruction behind the solar-splitting boundary.

Theorem 10.1 (Fixed isotropic reference plus same-label scalars suffice for the intrinsic deformation)**.** *Fix the isotropic reference M_0, equivalently its parameters (a, ρ, Ω). Assume the same-label gap and overlap scalars are supplied on all realized arrows. Then the full intrinsic neutrino mass-eigenstate bundle factors through the scalar certificate*

$$(g_e, \omega_e)_{e \in \{12,23,31\}}$$

or equivalently through the centered eta-class $[\eta_e]$. No additional raw matrix payload is needed to form the intrinsic selector, the depressed-cubic spectrum, the ascending gaps, or the intrinsic spectral subspaces. When the singular spectrum is simple, canonical Takagi congruence fixes an ordered column basis up to column signs. At a degeneracy, including the isotropic $s_0 = s_1$ point, only the degenerate spectral subspace and its projector are fixed. Ascending singular-value sorting does not select the physical normal or inverted mass-eigenstate labels.

Theorem 10.2 (Exact principal-branch selector from the centered eta-class)**.** *Fix the cycle sum*

$$\Omega = \psi_{12} + \psi_{23} + \psi_{31},$$

and a positive weight family μ_e. Define

$$\mu_{\min} = \min_e \mu_e, \qquad F_{\max} = \sum_e \arcsin\left(\frac{\mu_{\min}}{\mu_e}\right).$$

If $|\Omega| < F_{\max}$, then on the principal branch $\psi_e \in (-\pi/2, \pi/2)$, the affine energy

$$A(\psi) = \sum_e \mu_e (1 - \cos\psi_e)$$

has a unique minimizer. It is given by

$$\psi_e^\star = \arcsin(\lambda/\mu_e),$$

where λ is the unique solution of

$$\sum_e \arcsin(\lambda/\mu_e) = \Omega, \qquad \lambda \in (-\min_e \mu_e, \min_e \mu_e).$$

Proof. The Euler–Lagrange equations are $\mu_e \sin\psi_e = \lambda$. On the principal branch the scalar function

$$F(\lambda) = \sum_e \arcsin(\lambda/\mu_e)$$

is strictly increasing because

$$F'(\lambda) = \sum_e \frac{1}{\sqrt{\mu_e^2 - \lambda^2}} > 0.$$

Its range on $(-\mu_{\min}, \mu_{\min})$ is $(-F_{\max}, F_{\max})$, so $F(\lambda) = \Omega$ has a unique solution under the stated condition. Strict convexity follows because the Hessian of A is

$$\nabla^2 A = \operatorname{diag}(\mu_e \cos\psi_e),$$

which is positive definite on the principal branch and is positive definite after restriction to the affine plane $\sum_e \psi_e = \Omega$. $\square$

For the live normalized family

$$(\mu_{12}, \mu_{23}, \mu_{31}) = (1.4497003,\ 0.9968526,\ 0.5534471),$$

one has $F_{\max} = 2.5511001$, $|\Omega| = 1.7561443$, and positive domain margin 0.7949558. The corrected theorem domain therefore leaves the stored numerical selector unchanged.

Let

$$a = m_\star = \frac{v^2}{\mu_u}, \qquad \rho = |(M_0)_{12}|.$$

Then the intrinsic Majorana matrix is

$$M(\eta) = \begin{pmatrix} a & \rho e^{i\psi_{12}} & \rho e^{i\psi_{31}} \\ \rho e^{i\psi_{12}} & a & \rho e^{i\psi_{23}} \\ \rho e^{i\psi_{31}} & \rho e^{i\psi_{23}} & a \end{pmatrix}.$$

Theorem 10.3 (Exact intrinsic spectral cubic)**.** *Let $H = M^\dagger M$. Then*

$$H = dI + T, \qquad d = a^2 + 2\rho^2,$$

where T has zero diagonal and off-diagonals

$$x_{12} = 2a\rho \cos\psi_{12} + \rho^2 e^{i(\psi_{23}-\psi_{31})},$$

$$x_{23} = 2a\rho \cos\psi_{23} + \rho^2 e^{i(\psi_{31}-\psi_{12})},$$

$$x_{13} = 2a\rho \cos\psi_{31} + \rho^2 e^{i(\psi_{23}-\psi_{12})}.$$

Define

$$P = |x_{12}|^2 + |x_{23}|^2 + |x_{13}|^2, \qquad Q = \Re(x_{12}x_{23}\overline{x_{13}}).$$

Then the eigenvalues λ_k of T are exactly the three real roots of

$$\lambda^3 - P\lambda - 2Q = 0,$$

and the intrinsic squared masses are

$$m_k^2 = d + \lambda_k.$$

Equivalently,

$$\lambda_k = 2\sqrt{\frac{P}{3}} \cos\left(\frac{1}{3}\arccos\left(\frac{3\sqrt{3}\,Q}{P^{3/2}}\right) - \frac{2\pi k}{3}\right).$$

Corollary 10.4 (Intrinsic eta-chain spectral closure)**.** *Once the centered eta-class is emitted at the flavor boundary, the intrinsic neutrino branch emits*

$$s_0 \le s_1 \le s_2, \qquad s_1^2 - s_0^2,\ s_2^2 - s_0^2,\ s_2^2 - s_1^2,$$

and the corresponding intrinsic spectral subspaces with no PMNS import and no flavor-label leakage. A columnwise Takagi basis is emitted only on the simple-spectrum locus. The physical normal-ordering assignment is $(\nu_1, \nu_2, \nu_3) = (s_0, s_1, s_2)$, while the inverted-ordering assignment is $(\nu_3, \nu_1, \nu_2) = (s_0, s_1, s_2)$. The source does not choose between them, so physical ordering remains open.

The matrix used in this corollary is Majorana, so its physical column matrix is the Takagi matrix U_ν defined by

$$U_\nu^T M U_\nu = \operatorname{diag}(s_0, s_1, s_2) \in \mathbb{R}_{\geq 0}.$$

Earlier artifacts exported the left SVD matrix, which diagonalizes $MM^\dagger$, instead of the Takagi matrix, whose columns diagonalize $M^\dagger M$. The corrected congruence has maximum off-diagonal residual 5.4×10^{-28} GeV on the live anisotropic matrix. Conditional on the stale stored charged matrix, the corrected intrinsic combination would give

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta) = (45.0137^\circ,\ 2.50489^\circ,\ 2.53597^\circ,\ 210.6251^\circ).$$

That diagnostic is strongly incompatible with the NuFIT angle surface, but it is not an OPH prediction because the charged basis is open and template-derived.

10.2 Perturbative laws around the isotropic point

Write

$$\psi_e = \varphi + \delta_e, \qquad \delta_{12} + \delta_{23} + \delta_{31} = 0.$$

At first order in the centered eta-class,

$$\delta_e = -\tan\varphi\,\eta_e + O(\eta^2).$$

Define

$$\sigma^2 = \frac{2}{3}\sum_e \delta_e^2.$$

Assume $2a\cos\varphi + \rho \neq 0$ and that the collective state is separated from the isotropic doublet. Then the two ascending doublet eigenvalues satisfy

$$s_{0,1}^2 = m_d^2 \mp 2a\rho|\sin\varphi|\,\sigma + O(\delta^2),$$

so

$$s_1^2 - s_0^2 = 4a\rho|\sin\varphi|\,\sigma + O(\delta^2) = 4a\rho\frac{\sin^2\varphi}{|\cos\varphi|}\sqrt{\frac{2}{3}\sum_e \eta_e^2} + O(\eta^2).$$

Under the declared normal-ordering hypothesis, the ascending atmospheric gaps obey

$$\Delta m_{31}^2 = \Delta_{\mathrm{atm,iso}} + \frac{1}{2}\Delta m_{21}^2 + O(\delta^2), \qquad \Delta m_{32}^2 = \Delta_{\mathrm{atm,iso}} - \frac{1}{2}\Delta m_{21}^2 + O(\delta^2),$$

so the first-order invariant atmospheric object is the collective-to-doublet centroid gap

$$\Delta_{\mathrm{cent}} = s_2^2 - \frac{1}{2}(s_0^2 + s_1^2) = \Delta_{\mathrm{atm,iso}} + O(\delta^2).$$

Its quadratic shift is

$$\delta\Delta_{\mathrm{cent}} = -a\rho\,\sigma^2\frac{a(4\cos^2\varphi - 1) + 6\rho\cos\varphi}{2a\cos\varphi + \rho} + O(\delta^3).$$

The projective largest-mass right-singular direction, in the phase gauge that approaches the real democratic vector at the isotropic point, obeys the first-order deformation law

$$u_3 = u + \kappa\begin{pmatrix}\delta_{23}\\ \delta_{31}\\ \delta_{12}\end{pmatrix} + O(\delta^2), \qquad u = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad \kappa = \frac{\sqrt{3}\,(2a\sin\varphi + 3i\rho)}{9(2a\cos\varphi + \rho)}.$$

This equation fixes only the complex line. If v_3 denotes its normalized right-hand side, the canonical Takagi representative used by the executable pipeline is

$$u_3^{\mathrm{Tak}} = e^{-\frac{i}{2}\arg(v_3^T M_\nu v_3)} v_3, \qquad (u_3^{\mathrm{Tak}})^T M_\nu u_3^{\mathrm{Tak}} = s_3 > 0,$$

up to the unavoidable column sign. In particular, at the isotropic point the real democratic vector is generally outside positive Takagi gauge. Equivalently,

$$u_3 = u - \tan\varphi\, \kappa \begin{pmatrix} \eta_{23} \\ \eta_{31} \\ \eta_{12} \end{pmatrix} + O(\eta^2).$$

One exact demonstration uses centered eta-class

$$(\eta_{12}, \eta_{23}, \eta_{31}) = \begin{array}{c}(0.13631072512014578, -0.18362987264261327, \\ \qquad\qquad 0.0473191475224675),\end{array}$$

and returns the ascending intrinsic singular values

$$(s_0, s_1, s_2) = (2.3929601069646055, 2.4048200109774875, 2.589606227283229) \times 10^{-12}\ \mathrm{GeV},$$

with

$$\begin{aligned} s_1^2 - s_0^2 &= 5.690121167370743 \times 10^{-26}\ \mathrm{GeV}^2, \\ s_2^2 - s_0^2 &= 9.798023388600230 \times 10^{-25}\ \mathrm{GeV}^2. \end{aligned}$$

These are exact outputs of the intrinsic eta-chain once the fixed isotropic reference and centered eta-class are supplied. They are not flavor-labeled OPH rows, and ascending sorting does not pick a physical ordering. Under the declared normal-ordering hypothesis $(\nu_1, \nu_2, \nu_3) = (s_0, s_1, s_2)$; under the inverted-ordering hypothesis $(\nu_3, \nu_1, \nu_2) = (s_0, s_1, s_2)$. The source-side label rule is absent.

10.3 Weighted-cycle bridge rigidity and absolute attachment

The weighted-cycle route supplies a frozen PMNS/hierarchy candidate. The reduced invariant C_ν was selected on a compare-only correction surface, with $B_\nu = P_\nu C_\nu$ retained as a diagnostic amplitude parameterization:

$$\begin{aligned} C_\nu &= G^2 Q S^{-1/2} = 0.9994295999075177, \\ P_\nu &= 6.699825740519345, \\ B_\nu &= P_\nu C_\nu = 6.696004159297337. \end{aligned}$$

The blocked absolute attachment displays

$$\begin{aligned} \lambda_\nu &= \frac{m_{\star,\mathrm{eV}}}{q_{\mathrm{mean}}^{p_\nu}}\, P_\nu C_\nu = 1.7237014208357415, \\ m_i &= \lambda_\nu \hat{m}_i, \\ \Delta m_{ij}^2 &= \lambda_\nu^2 \widehat{\Delta m_{ij}^2}. \end{aligned}$$

The basis permutation, holonomy orientation, and CP sign lack source-side selection. Exhaustive enumeration of both stored orientations and all row and mass-column relabelings of the stored candidate matrix finds no normal-ordering-consistent relabeling that passes both NuFIT 6.1 correlated 3σ gates. This excludes a convention-only rescue of that matrix; it neither derives nor exhausts possible source-side physical charged-basis placements. The one-parameter atmospheric-anchor slice and the two-parameter positive-segment adapter remain compare-only continuations.

11 Hadrons, QCD, and the Emergence of Ordinary Matter

The hadron derivation is the most operationally demanding part of the particle program. The source-only artifact is not a fitted hadronic residual and not a bare scalar ρ_{had}. It is an OPH-QCD hadronic precision backend: a QCD quotient ensemble, source QCD parameter map, Euclidean slab/vacuum-transfer construction, hadronic Hilbert quotient, Ward-normalized electromagnetic current ledger, positive two-current spectral export $d\rho_Q^{(2)}$, higher-point and transition spectral exports, same-scheme remainder Ξ_Q, and systematics ledger. A source-only hadron mass row requires a working OPH hadron backend, such as GLORB/Echosahedron, that can emit these objects with manifest provenance and production systematics. Local surrogates, ordinary Chrome/Oracle workers, and bookkeeping scripts cannot promote this sector. The two-current spectral measure is the marginal needed for running-α and HVP transport; it is insufficient for HLbL and rare-decay long-distance amplitudes without the four-current and transition sectors. The source-backend boundary has an empirical closure policy. The empirical closure row class uses a separate $e^+e^- \to$ hadrons payload class for the hadronic spectral contribution and stays separate from source-only OPH rows. The source-only derivation nevertheless keeps the mathematical execution bridge, deterministic runtime receipt, writeback/evaluation path, and surrogate validation machinery as non-promoting scaffolding for the backend lane.

11.1 Seeded $2+1$ family and unquenched measure

Let

$$m_l := \frac{m_u + m_d}{2}, \qquad \rho_l := \frac{m_u + m_d}{2\,\Lambda^{(3)}_{\overline{\text{MS}}}}, \qquad \rho_s := \frac{m_s}{\Lambda^{(3)}_{\overline{\text{MS}}}}.$$

The seeded fixed-physics family is

$$a\Lambda_n = a\Lambda_{\text{seed}}\, 2^{-n}, \qquad a\Lambda_{\text{seed}} = \sqrt{\rho_l \rho_s},$$

$$am_l^{(n)} = a\Lambda_n \rho_l, \qquad am_s^{(n)} = a\Lambda_n \rho_s,$$

$$\beta_n = 6 + \frac{9}{2\pi^2} \log \frac{1}{a\Lambda_n}, \qquad L_n = \left\lceil \frac{\lambda_L^{\text{target}}}{a\Lambda_n} \right\rceil, \qquad T_n = \left\lceil \frac{\lambda_T^{\text{target}}}{a\Lambda_n} \right\rceil.$$

The unquenched ensemble measure is

$$d\mu_n(U) = Z_n^{-1} \exp[-S_g(U;\beta_n)] \det D_l(U; am_l^{(n)})^2 \det D_s(U; am_s^{(n)})\, dU.$$

On this branch, $N_f = 2+1$ and QED is off.

11.2 Runtime receipt and deterministic cfg/source contract

The emitted execution law is

$$U_{n,c} = K_n^{N_{\text{therm}} + (\text{cfg\_index})\, N_{\text{sep}}}(U_{\text{cold}}; \text{seed}_{n,c}),$$

with stop-time formula

$$t_{\text{stop}}(n,c) = N_{\text{therm}} + (\text{cfg\_index})\, N_{\text{sep}}.$$

The deterministic cfg seed law is

$$\text{seed}_{n,c} = \text{bytes.fromhex}(\text{cfg\_seed\_hash}_{n,c}),$$

with

$$\text{cfg\_seed\_hash} = \text{SHA256}\Big(\text{Serialize}\big(\text{ensemble\_id}, \beta, L, T, am_l, am_s, \text{cfg\_index}\big)\Big).$$

The source set is fixed as

$$S_{n,c} = \{[0,0,0,0],\ [L_n//2, L_n//2, L_n//2, T_n//2]\}.$$

The non-null external runtime receipt used in the surrogate execution is

$$N_{\text{therm}} = 2048, \qquad N_{\text{sep}} = 512.$$

These values are surrogate execution inputs only; they are not claimed as theorem outputs.

The production geometry summary makes the runtime boundary concrete. On the emitted seeded $2+1$ family there are three ensembles and six configurations total. If one stores all four links at every site as full double-complex 3×3 matrices, the naive raw gauge storage estimate is

$$576 \text{ bytes/site},$$

which gives total naive raw gauge storage

$$2.80071464105088 \times 10^{14} \text{ bytes}$$

over the production schedule. The normalized backend correlator dump needed by the analysis layer is tiny:

$$195264 \text{ bytes}$$

for the full π_{iso}, $N_{\text{iso,dir}}$, and $N_{\text{iso,ex}}$ payload on the production schedule. The hadron requirement is the backend export bundle that would feed the normalized dump from real production execution.

11.3 Operational reading of the required computation

It is useful to separate the lightweight analysis layer from the missing backend layer. Informally, the repository-side control plane is explicit: the seeded ensemble family, the deterministic cfg/source contract, the export manifest, the jackknife evaluator, the forward-window selector, and the publication-budget schema are all explicit. The missing step is the production export of Ward-projected hadronic spectral data from a real OPH hadron backend.

Technically, the required production computation is the standard lattice-QCD stable-channel workflow on the emitted family: run unquenched RHMC/HMC for the three seeded ensembles; for each realized cfg and fixed source solve the clover-improved Wilson light/strange systems; construct the zero-momentum π_{iso}, $N_{\text{iso,dir}}$, and $N_{\text{iso,ex}}$ two-point sequences; write the backend bundle; convert that bundle into the repo-side production dump; then let the existing jackknife and forward-window machinery emit $am_{X,\text{ground}}$, R_X, $m_X[\text{GeV}]$, and the published σ_{stat}, δ_{cont}, δ_{vol}, and δ_χ fields. This gives a complete source-only execution contract and a separate empirical display policy.

The engineering burden is therefore asymmetric. Local execution is sufficient for manifest generation, surrogate validation, and downstream readout tests, because those stages operate on a tiny correlator payload. The physical branch is different: it needs backend hardware and execution semantics that are absent from the repository environment. This paper therefore treats GLORB/Echosahedron-class OPH hardware, or an equivalent working OPH hadron backend, as the source-only backend requirement. Production hadron masses enter the particle pipeline only with a backend-emitted Ward-projected spectral measure and its uncertainty budget. This is a source-backend scope boundary.

11.4 Stable-channel correlators, effective masses, and forward windows

The pion stable channel is

$$p_\pi^{(n,c,s)}(t) = \sum_x \Re\,\mathrm{tr}_{c,\mathrm{spin}}\left[\gamma_5 S_l(x;s)\gamma_5 S_l(s;x)\right].$$

The nucleon stable channel is

$$p_{N,\mathrm{dir}}^{(n,c,s)}(t) = \sum_x G_d, \qquad p_{N,\mathrm{ex}}^{(n,c,s)}(t) = \sum_x G_x,$$

$$p_N^{(n,c,s)}(t) = p_{N,\mathrm{dir}}^{(n,c,s)}(t) - p_{N,\mathrm{ex}}^{(n,c,s)}(t).$$

Cfg/source and ensemble averaging are

$$\bar{p}_X^{(n,c)}(t) = \frac{1}{|S_{n,c}|}\sum_{s\in S_{n,c}} p_X^{(n,c,s)}(t),$$

$$C_X^{(n)}(t) = \frac{1}{|C_n|}\sum_{c\in C_n} \bar{p}_X^{(n,c)}(t).$$

The effective-mass laws are

$$am_{\mathrm{eff},\pi}(t) = \log\frac{C_\pi(t)}{C_\pi(t+1)}, \qquad am_{\mathrm{eff},N}(t) = \log\frac{|C_N(t)|}{|C_N(t+1)|}.$$

The forward window is

$$W_n = \{t : 1 \le t+1 < \lfloor T_n/2 \rfloor\}.$$

The evaluator monitors log-convexity,

$$R_{\log\mathrm{conv},\pi}(t) = C_\pi(t)^2 - C_\pi(t-1)C_\pi(t+1),$$

$$R_{\log\mathrm{conv},N}(t) = |C_N(t)|^2 - |C_N(t-1)|\,|C_N(t+1)|,$$

tail-drop,

$$D_X(t) = am_{\mathrm{eff},X}(t) - am_{\mathrm{eff},X}(t+1),$$

and mirror suppression,

$$M_X(t) = \exp[-am_{\mathrm{eff},X}(t)(T_n - 2t)].$$

The selected forward window is the longest contiguous run satisfying finite effective masses, nonnegative log-convexity up to tolerance, nonnegative tail-drop up to tolerance, mirror suppression below threshold, and local plateau flatness. The candidate ground-state mass is then the weighted window average

$$am_{X,\mathrm{ground}} = \frac{\sum_{t\in W_X^{\mathrm{sel}}} w_t\, am_{\mathrm{eff},X}(t)}{\sum_{t\in W_X^{\mathrm{sel}}} w_t}, \qquad w_t = \frac{1}{\max(\sigma_t^2, \varepsilon)}.$$

11.5 Statistics, systematics, and dimensional readout

Delete-1 jackknife is performed over the cfg axis after source averaging inside each cfg. With n_{cfg} configurations and integrated autocorrelation time $\tau_{\text{int,cfg}}$, the effective cfg count is

$$n_{\text{eff,cfg}} = \frac{n_{\text{cfg}}}{2\,\tau_{\text{int,cfg}}}.$$

The published statistical error is

$$\sigma_{\text{stat},X} = \text{JKstderr}(am_{X,\text{ground}}).$$

The machine-readable systematics field uses

$$\sigma_{\text{sys},X} = \sqrt{\delta^2_{\text{cont},X} + \delta^2_{\text{vol},X} + \delta^2_{\chi,X}}.$$

Continuum, volume, and chiral proxies are encoded as

$$R_X^{(n)} = \frac{am_X^{(n)}}{a\Lambda_n}, \qquad R_X^{(n)} \approx R_X(0) + c_X(a\Lambda_n)^2,$$

$$\delta^{(n)}_{\text{cont},X} = a\Lambda_n \left| R_X^{(n)} - R_X(0) \right|,$$

$$\delta^{(n)}_{\text{vol},\pi} = am_\pi^{(n)} \frac{e^{-am_\pi^{(n)} L_n}}{\max(am_\pi^{(n)} L_n, 1)},$$

$$\delta^{(n)}_{\text{vol},N} = am_N^{(n)} \frac{e^{-am_\pi^{(n)} L_n}}{\max(am_\pi^{(n)} L_n, 1)},$$

$$\delta^{(n)}_{\chi,\pi} = a\Lambda_n \left| R_\pi^{(n)} - \langle R_\pi \rangle_n \right|,$$

$$Q_N^{(n)} = \frac{R_N^{(n)}}{R_\pi^{(n)}}, \qquad \delta^{(n)}_{\chi,N} = a\Lambda_n \langle R_\pi \rangle_n \left| Q_N^{(n)} - \langle Q_N \rangle_n \right|.$$

These are surrogate publication proxies, not production physical systematics. Given a ground-state candidate,

$$R_X = \frac{am_{X,\text{ground}}}{a\Lambda^{(3)}_{\overline{\text{MS}}}}, \qquad m_X[\text{GeV}] = R_X\,\Lambda^{(3)}_{\overline{\text{MS}}}[\text{GeV}],$$

with

$$\Lambda^{(3)}_{\overline{\text{MS}}} = 0.3344017073 \text{ GeV}.$$

11.6 Surrogate execution bundle and frontier

The surrogate execution evolves a latent state $z \in \mathbb{R}^d$ with Hamiltonian

$$H(z,p) = S(z) + \frac{1}{2}\sum_i p_i^2,$$

where

$$S(z) = \frac{1}{2}\sum_i \omega_i^2 z_i^2 + \lambda_4 \sum_i z_i^4 + \kappa \sum_i (z_{i+1} - z_i)^2 + 2\alpha_l \sum_i \log(\mu_l^2 + z_i^2) + \alpha_s \sum_i \log(\mu_s^2 + \tfrac{1}{2} z_i^2).$$

Leapfrog integration plus Metropolis accept/reject gives the surrogate HMC update

$$(z, p) \mapsto (z', p'), \qquad P_{\mathrm{acc}} = \min(1, e^{-\Delta H}).$$

This kernel is a deterministic executable surrogate honoring the emitted receipt and seed law, with no physical lattice-QCD RHMC/HMC claim.

For validation of the execution bridge, the surrogate locks the ground-state masses to the hadron audit proxy values

$$m_{\pi,\mathrm{proxy}} = 0.13497682776768472 \text{ GeV},$$
$$m_{N,\mathrm{iso,proxy}} = \frac{m_p + m_n}{2} = 0.93891875434 \text{ GeV}.$$

Thus

$$am^{(n)}_{\pi,\mathrm{proxy}} = \frac{m_{\pi,\mathrm{proxy}}}{\Lambda^{(3)}_{\overline{\mathrm{MS}}}[\mathrm{GeV}]}\, a\Lambda_n,$$
$$am^{(n)}_{N,\mathrm{iso,proxy}} = \frac{m_{N,\mathrm{iso,proxy}}}{\Lambda^{(3)}_{\overline{\mathrm{MS}}}[\mathrm{GeV}]}\, a\Lambda_n.$$

The surrogate correlator is

$$C^{\mathrm{sur}}_X(t) = A_{X,0}e^{-am_X t} + A_{X,1}e^{-am_{X,\mathrm{ex}} t} + A_{X,\mathrm{mir}}e^{-am_X (T_n - t)}$$

up to a multiplicative correlated noise factor. Direct and exchange nucleon pieces are written as

$$C^{\mathrm{sur}}_{N,\mathrm{dir}}(t) = f_{\mathrm{dir}}\, C^{\mathrm{sur}}_N(t), \qquad C^{\mathrm{sur}}_{N,\mathrm{ex}}(t) = (f_{\mathrm{dir}} - 1)\, C^{\mathrm{sur}}_N(t),$$

so that

$$C^{\mathrm{sur}}_N(t) = C^{\mathrm{sur}}_{N,\mathrm{dir}}(t) - C^{\mathrm{sur}}_{N,\mathrm{ex}}(t).$$

On the finest surrogate ensemble, the resulting diagnostic candidates are

$$m^{\mathrm{sur}}_{\pi,\mathrm{iso}} = 0.135039383836 \text{ GeV}, \qquad m^{\mathrm{sur}}_{N,\mathrm{iso}} = 0.938960210578 \text{ GeV},$$

with worst absolute error approximately 6.26×10^{-5} GeV across the surrogate stable-channel family. These numbers validate the emitted execution bridge. They are not promotable production hadron predictions.

So the hadron derivation closes the full

$$\text{receipt} \to \text{execution} \to \text{writeback} \to \text{evaluation} \to \text{budgets} \to \text{forward-window summary}$$

path on executed surrogate data, while the physical closure requires production unquenched RHMC/HMC, real Dirac solves and baryon contractions, production autocorrelation studies, and production continuum / finite-volume / chiral systematics.

12 Observer-Centric Particle Ontology and Measurement

This paper cannot stop at masses and couplings. OPH is observer-centric at the level of its basic ontology, so a particle chapter also has to explain what a particle *is* in this language and how a measurement turns an excitation surface into an actual observed state. Fortunately, this is one of the places where the OPH theorem surface contains real mathematical content instead of only interpretation.

12.1 Observer patches, overlap consistency, and particle data

Ref. [3] formulates the basic kinematic picture in patch-net language: each observer patch carries a local state, neighboring patches compare a shared interface alphabet, and a global state is physically admissible exactly when neighboring projections agree on the overlap. The algebraic OPH version of the same statement is Axioms 4.1 and 4.2: local physical data are carried by patch algebras $\mathcal{A}(P)$, and those local states must agree on shared subalgebras whenever patches overlap.

The particle interpretation starts from patch-local algebras, overlap-visible observables, edge sectors, and transport data rather than a single absolute global particle basis. A particle row in the reported derivation is a readout claim about a stable or candidate-stable excitation structure visible to a family of observer patches and consistent on their overlaps.

In the observer formulation, an observer is represented by

$$O = (P, \mathcal{A}(P), \rho, R),$$

where P is the observer patch, $\mathcal{A}(P)$ its local algebra, ρ the local state, and R the record algebra. For the particle derivation, R belongs to the quantum observer surface itself. On the exact fixed-cutoff measurement surface it is generated by central record projectors, and practical readout may use approximately commuting projectors that are close to that central reference algebra. That is what makes the observer-facing record surface shareable across overlapping observer descriptions without violating the usual no-cloning constraints on generic quantum states.

12.2 Record algebras and definite outcomes

The integrated measurement appendices and Ref. [4] make the measurement interface precise at fixed cutoff. On the declared operational surface, the completed write/verify slice carries a finite commutative central record algebra. Practical readout may instead use projectors Q_a on the same declared slots with commuting central reference projectors $\widehat{Q}_a$, where

$$\delta_{\mathrm{rec}} := \max_a \|Q_a - \widehat{Q}_a\|.$$

Then

$$\|[Q_a, Q_b]\| \leq 4\,\delta_{\mathrm{rec}},$$

and, whenever $\|\widetilde{\rho} - \rho\|_1 \leq \varepsilon$,

$$\Big|\mathrm{Tr}(\widetilde{\rho} Q_a) - \mathrm{Tr}(\rho \widehat{Q}_a)\Big| \leq \varepsilon + \delta_{\mathrm{rec}}.$$

On the explicit fixed-cutoff screen architecture, the exact record algebra after a completed write/verify cycle is

$$\mathcal{Z}_{\mathrm{rec}}(t) = \mathrm{Alg}\Big(\{P_{m_I^{(J)}(t)}\}_{I \neq J}, \{P_{r_I^{\mathrm{bulk}}(t)}\}_I, \{\Pi_\alpha^{(IJ)}(t)\}_{I<J,\alpha}\Big).$$

Ref. [4] proves that, on the declared operational measurement surface, $\mathcal{Z}_{\mathrm{rec}}(t)$ is commutative and central for the readout instrument being used.

This fixed-cutoff centrality result is what supports definite outcomes without adding a second ontology. The integrated measurement ledger phrases the same idea through edge-center decomposition: on a fixed collar with exact Markov structure aligned with the edge split (the compact paper's Markov-split alignment hypothesis), the state splits into superselection blocks,

$$\rho_{ABC} = \bigoplus_j q_j\, \rho^{(j)}_{Ab_L^{(j)}} \otimes \rho^{(j)}_{b_R^{(j)} C},$$

and the label j is classical center data. The supplement then goes one step further: in the physical algebra there are no interference observables between different sector blocks. So once a particle detection event has been recorded in the observer-accessible center/record algebra, the accessible physics is organized as a classical mixture over those blocks instead of as a superposed measurement record.

Accordingly, a measured particle family or excitation channel should be thought of as an observer-accessible sector or record value carried by the central measurement surface, not as a mysterious absolute collapse of the full universe-state into a preferred basis chosen by hand.

12.3 Born probabilities and post-measurement states

The fixed-cutoff measurement package is also explicit about probabilities and updates. For every event E in the sigma algebra generated by $\mathcal{Z}_{\mathrm{rec}}(t)$, the measurement probability is

$$\mathbb{P}_t(E) = \mathrm{Tr}(\rho_t P_E),$$

where $P_E \in \mathcal{Z}_{\mathrm{rec}}(t)$ is the projector for that event. Conditioning on the event then produces the post-measurement state

$$\rho_t|_E = \frac{P_E \rho_t P_E}{\mathrm{Tr}(\rho_t P_E)}.$$

In the supplement, the same statement appears as the Lüders update on the record algebra. In the synthesis paper *Observers Are All You Need* [2], this fixed-cutoff Born/Lüders package and its Bell/CHSH extension are imported as theorem-bearing, not non-theorem commentary.

This directly answers the particle-state question in the present framework. Measurements give particles actual states by conditioning the observer-accessible state on the central record algebra. A detector click, a stable overlap-sector readout, or a pointer value does more than announce a pre-existing classical label; it defines the conditioned state for subsequent observer-accessible physics. If the same event is re-read without any accepted repair move touching its support, the result in Ref. [4] shows that the re-read probability is 1. So the measurement interface is well defined and operationally auditable.

12.4 Particle identity as transport-stable structure

Once one asks "which event happened?" and then "which particle family is this?", the flavor derivation becomes essential. The active flavor derivation does not start with the names electron, muon, tau lepton, or up quark built into the fundamental observable. It starts with transport kernels, generation-bundle data, same-label eigenline transport, overlap-edge cocycles, and a persistent flavor observable carrying intrinsic labels $f1, f2, f3$. This is an important discipline condition in the paper. At the deepest flavor surface used here, family identity is transport-stable intrinsic structure first and named experimental family assignment second.

The shared Yukawa/excitation dictionary carries substantial weight even though it is not itself the leading missing object. The flavor pipeline exports an invariant base: projectors, spectral gaps, pair suppressions, cycle phases, and common sector-response objects. The nonpromoting component is the map from that intrinsic base to the named low-energy fermion families when the required sector bridge is not emitted. The quark derivation emits named rows because the forward branch is far enough along to support direct numerical comparison there; the charged-lepton derivation does not do so at theorem grade. The neutrino weighted-cycle construction remains a target-informed template candidate and does not close on a physical flavor-labeled surface.

Accordingly, particle identity is a refinement-stable observer-accessible transport pattern whose named interpretation is inherited from the available dictionary and comparison surfaces. Where that dictionary is not emitted by the theorem surface, the paper states that boundary explicitly.

13 H3 Record Charts and Cross-Boundary Token Stitching

The previous subsection concerns sector and family identity: which transport-stable excitation type is being read. A different question is whether two localized record tokens, seen near a chart or partition boundary, are the same continuing observer-visible token. This section closes that second question at the finite-certificate level. It does not add a particle species, mass, charge, or scattering theorem. It says when a bounded patch system is allowed to stitch local H^3 proto-records into nonbranching record-worldlines, and when it must report ambiguity.

13.1 H3 chart and clock atlas

The existence, Lorentz naturality, and exact dimension of this atlas are imported from the compact cap-normal H^3 reconstruction theorem. That theorem identifies sky directions with $q(\Omega) = (1, \Omega)$, represents oriented round caps by $n_C = (\cot\alpha, \csc\alpha\,\mathbf{c})$, and maps each cap to a geodesic half-space in H^3. The present section begins one gate later: it supplies record supports, observer clocks, chart transitions, gauge transport, and worldline stitching inside that chart. The cap-normal chart, cap-response localization, overlap descent, and worldline stitching are separate certificates.

Fix a curvature radius $R_H > 0$ and write

$$H^3_{R_H} = \{X \in \mathbb{R}^{1,3} : \langle X, X\rangle_L = -R_H^2,\ X^0 > 0\}, \qquad \langle X, Y\rangle_L = -X^0Y^0 + X^1Y^1 + X^2Y^2 + X^3Y^3.$$

The spatial distance used by the stitch certificate is

$$d_H(X, Y) = R_H \operatorname{arcosh}\left(-\frac{\langle X, Y\rangle_L}{R_H^2}\right).$$

An H^3 record atlas is a tuple

$$\mathcal{A}_H = (R_H, \{U_i, \chi_i, u_i, J_i, \vartheta_i\}, \{G_{ji}, F_{ji}\}, E, \nabla, \mathsf{Prov}_{R_H}, \mathsf{Prov}_u)\,.$$

Here χ_i are $H^3_{R_H}$ charts, $u_i \in H^3_{R_H}$ is the selected local observer-frame anchor when such an anchor is claimed, J_i are chart domains for record support, and $\vartheta_i : J_i \to T$ are observer-clock maps into a common comparison-time line T. Prov_{R_H} records whether R_H is a unit convention, an imported physical scale, or an independently derived OPH scale; #309 supplies only the unit chart convention by itself. Prov_u records which observer, tetrad, or clock object selected the anchors u_i. The transition $G_{ji} \in \mathrm{SO}^+(1,3)$ is the H^3 chart transition, while F_{ji} is the sector or gauge transport functor on the record payload. These two holonomies are deliberately separate: atlas holonomy tests geometry, and gauge holonomy tests transported sector data. When two charts describe the same transported observer frame, the atlas additionally requires

$$G_{ji}u_i = u_j.$$

Theorem 13.1 (H3 geometric naturality)**.** *For every $A \in \mathrm{SO}^+(1,3)$, the map $X \mapsto AX$ preserves $H^3_{R_H}$, d_H, geodesics, exponential and logarithm maps, parallel transport, Hausdorff distances between compact record supports, and Fréchet means whenever the mean is unique.*

Proof. The defining equation of $H^3_{R_H}$ and the formula for d_H depend only on the Lorentz inner product and the future sheet. Elements of $\mathrm{SO}^+(1,3)$ preserve both. The Levi-Civita connection, geodesic exponential, logarithm, and parallel transport are functorial for isometries, so the listed constructions commute with A. Hausdorff distance and unique Fréchet means are metric constructions, hence are preserved by any isometry. □

13.2 Cap-response localization balls

The compact paper supplies the conditional inverse theorem that turns calibrated modular cap responses into localized record supports. For every descended record token i and clock slice t, this particle/worldline layer consumes a certificate

$$R_i(C,t,O) = \omega_{i,O}\Big(\sigma_t^{C,O}(M_{C,0,O})\Big), \qquad y_{i,j} = F_j(X_i(t)) + e_{i,j},$$

on a compact source domain $\Omega \subset H^3_{R_H}$, with quantitative observability

$$\alpha \bar{d}(X,Y) \le |F(X) - F(Y)|_W \le L\bar{d}(X,Y), \qquad |e_i|_W \le \sigma_i(t).$$

Given an ε-net and residual tolerance $\tau_i(t)$, the imported theorem emits

$$S_i(t) = B_H(\widehat{X}_i(t), r_i(t)), \qquad r_i(t) = R_H\left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma_i(t) + \frac{1}{\alpha}\tau_i(t)\right].$$

A point estimate $\widehat{X}_i(t)$ is a unique finite localization output only when $\Delta_{\mathrm{loc},i}(t) > 0$. If this gap fails, the input to the stitch theorem is an ambiguity set or support ball, not an arbitrary token-ID tie-break. Distinct record sources at a shared clock slice require the certified separation

$$d_H(\widehat{X}_i, \widehat{X}_k) > r_i + r_k + m_{\mathrm{sep}}$$

for a declared positive margin m_{sep}.

13.3 Proto-record extraction and descent

A raw local proto-record is a connected component of an invariant detector scalar above a frozen threshold, together with its certified localization ball or support set, local clock interval, sector payload, and uncertainty collar. The detector scalar is built from record-algebra data, not from implementation object IDs. Extraction is chart-natural when transporting the raw field by G_{ji} and F_{ji} carries extracted components in chart i to the extracted components in chart j, up to the declared support collar.

Before temporal stitching, overlapping charts are descended. The overlap-descent relation joins two proto-records only when their transported supports lie in the same connected component on a real chart overlap and the same-component margin dominates the support collar. If distinct component separation does not dominate the collar, the descent emits an ambiguity certificate instead of choosing a label. On component-connected covers, this descent produces canonical global descended records by the usual sheaf gluing argument: cocycle-compatible local components with positive separation margins have a unique global component, and the construction is natural under atlas restriction.

13.4 Boundary-crossing germs and transported residuals

A cross-boundary candidate edge is admissible only when the two descended records meet a real patch interface in adjacent observer-clock intervals and share a common comparison chart. The interface contact must be oriented and transverse: the signed-distance interval brackets the interface, and the normal velocity has a positive lower bound. Tangencies, grazing contacts, and file or shard-name boundaries without a physical interface are rejected. If the local slab contains an interaction marker, the record is routed to the interaction solver instead of stitched by the free-propagation rule.

For an admissible pair, the certificate transports the left record into the right comparison chart and evaluates frozen residuals

$$r_t, \quad r_\partial, \quad r_x, \quad r_v, \quad r_{\text{int}},$$

for clock adjacency, boundary contact, H^3 position, transported velocity or finite difference, and interaction absence. The cost function is declared before seeing the proposed matching. Record IDs, global IDs, shard-local IDs, and stitch keys are forbidden in admissibility, costs, and tie-breaking. All H^3 position and velocity residuals are evaluated as certified interval or uncertainty-ball distances. A point-estimate residual may be printed as a diagnostic, but it cannot replace the localization radius in an admissibility or matching certificate.

Lemma 13.2 (Candidate-edge naturality)**.** *The admissible candidate-edge set and its residual intervals are invariant under common* $\mathrm{SO}^+(1,3)$ *chart changes and under gauge-bundle trivialization changes.*

Proof. Every geometric term is computed from d_H, oriented interface data, and the common clock line, all of which are preserved by Theorem 13.1 and clock-map compatibility. Sector and gauge residuals compare transported payloads after applying the declared connector, so changing a local trivialization conjugates both sides of the comparison. The residual intervals and the resulting admissibility predicate are therefore unchanged. □

13.5 Assignment gap, nonbranching paths, and ambiguity

Let L and R be the descended record sets on two adjacent time slabs. A stitch matching is a partial one-to-one assignment between L and R, augmented by declared appearance and disappearance interval costs. Its certified gap is

$$\Delta_{\text{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*),$$

where M_* is the proposed matching, $\overline{C}(M_*)$ is its upper cost bound, and $\underline{C}(M)$ is the lower cost bound for every competing matching. The localization gap Δ_{loc} and stitch-assignment gap Δ_{cert} are distinct receipts: the first certifies a unique spatial support inside one clock slice, while the second certifies a unique temporal assignment across slices.

Theorem 13.3 (Robust ID-independent stitch assignment)**.** *If the candidate-edge set is natural, the costs are frozen and ID-independent, the proposed matching is one-to-one, and* $\Delta_{\text{cert}} > 0$*, then* M_* *is the unique certified minimum-cost stitch. The certified matching is natural under chart changes, gauge trivializations, and relabeling of implementation record IDs.*

Proof. The strict gap says every competing matching has lower cost bound above the proposed matching's upper cost bound, so no competitor can tie or beat it under any realization inside the certified intervals. Lemma 13.2 makes the candidate graph and cost intervals invariant under chart and gauge presentation changes. Because IDs are excluded from admissibility and costs, relabeling them changes only bookkeeping fields, not the optimum. ☐

Theorem 13.4 (Unavoidable ambiguity)**.** *If a nontrivial permutation of the visible descended records preserves the atlas, clocks, interfaces, sector/gauge payloads, candidate graph, and cost intervals, then no natural ID-independent finite procedure can certify a unique stitch across that slab.*

Proof. A natural procedure must commute with the permutation. A unique certified output would therefore be fixed by the permutation. But the permutation sends one equally admissible stitch to a distinct equally admissible stitch. Choosing between them would introduce a label or presentation dependence. The only natural finite output is an ambiguity label. ☐

The certified stitch graph is then the graph obtained by chaining the unique matchings across adjacent slabs. Since each matching is partial one-to-one, every connected component is a nonbranching path, ray, or isolated record. These are the H^3 record-worldlines of this section.

13.6 Refinement naturality and optional Lorentzian lift

For a refinement $s \to r$, let Q_{sr} contract fine descended records to coarse records and let η_{sr} bound the cost distortion between the fine and coarse stitch problems. If the coarse certified gap obeys $\Delta_r > 2\eta_{sr}$, then the contracted fine optimum is isomorphic to the coarse optimum. This is the refinement-naturality gate: a cross-boundary stitch is not paper-grade until the coarse/fine contraction agrees within the declared margin.

The H^3 certificate is spatial and observer-record-level. A Lorentzian worldline lift is a separate optional criterion requiring lapse/shift data, a timelike speed margin, and compatibility with the emergent four-dimensional chart. Without that lift, the certificate proves record-token continuation in the H^3 observer chart, not a full spacetime geodesic theorem.

Theorem 13.5 (Certified H3 record-worldline stitching)**.** *Given a valid observer-facing H^3 chart atlas, a common operational clock atlas, and a passing cap-response localization certificate for every descended record token and clock slice, chart-natural proto-record extraction, overlap descent with positive localization-ball separation margins, real transverse interface-crossing germs, sector and gauge transport continuity, a frozen ID-independent one-to-one assignment with $\Delta_{\text{cert}} > 0$, no interaction marker in the free-propagation slab, and a passing coarse/fine contraction gate, the emitted stitch graph is a unique natural nonbranching H^3 record-worldline graph. If any positive localization gap, separation margin, or assignment gap is absent, the natural output is* AMBIGUOUS*; if an interaction marker is present, the output is* INTERACTION REQUIRED*; if atlas, interface, localization, or transport gates fail, the stitch is rejected.*

Proof. The atlas, localization, and extraction gates make local proto-records presentation-independent. The positive Δ_{loc} gate licenses point outputs; otherwise the procedure carries support balls or ambiguity sets forward. Descent turns overlap-compatible local components into descended records before any temporal decision is made. The interface and transport gates define a natural candidate-edge graph by Lemma 13.2. The strict assignment gap gives a unique ID-independent matching by Theorem 13.3. Chaining partial one-to-one matchings gives nonbranching components, and the refinement gate makes the result stable under the declared coarse/fine contraction. The alternative statuses are exactly the negations of these gates, with Theorem 13.4 forcing ambiguity when the visible data do not separate a unique continuation. ☐

13.7 Scope boundary of the measurement chapter

This whole chapter has to preserve one final distinction. The fixed-cutoff measurement interface is theorem-bearing on the microphysics surface: central record algebra, Born probabilities, Lüders conditioning, repeated-read stability, and the stated two-wing Bell/CHSH theorem stack are written there as fixed-cutoff statements. What is *not* proved at the same level is the stronger global observer-continuation or strange-loop proposal sometimes discussed in broader OPH notes. This paper should therefore use the fixed-cutoff measurement package directly, while leaving the larger metaphysical closure proposal out of the main technical chain.

14 Discussion: Matter, Antimatter, Supersymmetry, and Continuation Questions

This section is intentionally not written in the same voice as the structural and quantitative sections above. The OPH theorem surface distinguishes sharply between recovered-core results, quantitative-closure branches, secondary quantitative branches, and continuations. Matter versus antimatter, supersymmetry, and several adjacent questions lie on that boundary. They are too important to omit from a particle-spectrum paper, but they are not part of the closed prediction surface. The quark section above is likewise an obstruction and target-audit surface, not a set of public numeric rows. Its $S_3/D12$ numerical table is a quarantined formula-discovery witness, not an exception to that rule.

14.1 Matter and antimatter

The Standard Model structural branch fixes the chiral gauge architecture within which matter and antimatter are defined. Once a chiral fermion representation is present, its conjugate representation gives the corresponding antiparticle content. In that limited sense, OPH explains why matter and antimatter both exist: they are paired by the realized gauge and chiral structure of the low-energy branch.

What is *not* structurally closed is the cosmological asymmetry between them. The integrated continuation notes record a baryogenesis continuation tied to the realized $\mathbb{Z}_6$ quotient and the electroweak topological channel. In that continuation, the entropy deficit of the $\mathbb{Z}_6$ quotient gives a suppression factor

$$\varepsilon = e^{-\ln 6} = \frac{1}{6},$$

and the same notes compare powers of that factor with the 12-fermion electroweak baryon-violating channel. But that comparison is only a suppression-counting heuristic. It is not a dynamical derivation of the observed baryon asymmetry.

But the observer-side claim notes are explicit about the scope of that estimate: the baryon asymmetry scale is a Phase-III continuation, and suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism. The declared derivation also does not expose a baryogenesis output row. So this paper can report the available OPH continuation idea, but it should not present a sharp matter/antimatter abundance prediction as if it were a closed theorem or an emitted output.

The required ingredients are specific: a concrete out-of-equilibrium epoch, justified defect/sphaleron production dynamics, derived CP-odd source terms, transport and washout control, and a freeze-out computation of the final asymmetry. The available corpus does not have a baryogenesis theorem package. The same continuation also notes that CP-odd overlap data are generically available because central overlap phases are sent to their inverses under CP, but that observation is not a baryogenesis theorem.

14.2 Why the declared branch does not require supersymmetry

The D10 appendix below contains the dedicated gauge-coupling quantitative-closure discussion unification. Its main point is not that supersymmetry has been mathematically ruled out in every possible extension of the program. Its point is narrower and more relevant to this paper: the declared branch does not *need* a supersymmetric partner spectrum in order to explain why unification-like running might appear.

At one loop, the integrated D10 appendix records the familiar fact that Standard Model beta-function coefficients do not produce successful naive unification, whereas MSSM-like coefficients do. The OPH continuation idea is then that edge-sector heat-kernel weights can shift the effective beta-function coefficients in an MSSM-like direction through geometric or entropic sector multiplicity instead of through an actual low-energy superpartner spectrum. In that reading, "unification-like behavior" and "a supersymmetric particle zoo" come apart.

That continuation is narrower than a derived unification theorem. The appendix records one calibration branch in which Peter–Weyl multiplicities, the printed one-loop running frame, the threshold conventions, and an additional fermionic-grading restriction together produce an MSSM-like benchmark shift. It does not prove that those effective loop multiplicities, statistics restrictions, and decoupling conventions are uniquely forced by OPH edge sectors alone.

For this paper, the correct conclusion is therefore scope-limited and modest:

1. the derivation reconstructs the realized Standard Model branch without having to introduce a supersymmetric partner for every known field;
2. the integrated D10 appendix contains a continuation-level argument that some unification-style running features could arise from edge-sector structure instead of MSSM particle content;
3. the beta-shift package itself sits on explicit D10 calibration assumptions instead of a closed edge-sector theorem;
4. none of this is a universal theorem that supersymmetry is impossible.

So if the paper asks "why is there no supersymmetry in the derived spectrum?", the best technical answer is: because the realized branch stops at the Standard Model content, and the existing unification discussion is trying to reproduce the relevant running behavior through a separate D10 calibration package, without extending the particle content to a full supersymmetric multiplet structure.

14.3 Three Generations And Flavor Closure

The structural Standard Model branch fixes $N_g = 3$ on the realized MAR-admissible branch. This result is separate from the harder flavor closure problem. Three generations are structurally recovered; the full flavor dictionary, excitation map, and CKM closure are not emitted on this theorem surface, while the charged source landing from P to physical charged data is a corpus-limited no-go and the neutrino lane has no source-closed flavor prediction. Its weighted-cycle candidate fails the NuFIT 6.1 correlated profile [10]. The flavor chapter and the later family chapters therefore spend substantial space on constructive object boundaries as well as final numbers.

14.4 Ordinary matter and the hadron backend boundary

A second distinction belongs in discussion form instead of in the results sections: the difference between "elementary rows exist" and "ordinary matter is fully derived." Ordinary visible matter is dominated by hadrons, especially protons and neutrons. The OPH derivation has a hadron pipeline, with source-only hadron masses gated by a production backend export. The paper emits the eligible elementary rows and retains the weighted-cycle coordinates only as a rejected comparison record, while the full observed matter spectrum is not emitted from a source-only OPH backend.

14.5 How to read these continuation questions

The reader should therefore interpret this whole discussion section with the same rule used by *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [5]. Quark source-spread selection, charged-lepton source landing, fuller flavor-labeled neutrino, and source-only hadron mass theorems sit beyond the closed structural core; failure in those continuation branches would not undo the recovered-core gauge and gravity chain or erase the independently supported bosonic rows.

15 Conclusion

The particle-spectrum derivation is nontrivial and much more concrete than a qualitative aspiration. The structural theorem surface fixes the realized Standard Model branch and its carrier roles. The conditional quadratic-action theorem supplies classical Maxwell, perturbative pure-Yang–Mills, and pure-Einstein modes, while their quantum-particle gates remain open. The completion pipeline records those fail-closed carrier gates, the hierarchy/naturality bridge on the selected P/N branch, the Higgs/top row on the declared D10/D11 surface, and the two-modulus quark source-spread obstruction on f_P. The new quark audit adds exact S_3 heat-spectrum and L, Q-basis lemmas plus a target-informed template ansatz whose mixed-chart residual is 0.2946%. The later RSCC candidate removes five direct template decimals but replaces them with an unproved target-informed module/effect/sign/cumulant ledger; its lower-order ablation fits the same table better and every F1–F6 receipt remains open. Neither diagnostic adds a dimensionful source mass, a common-scale Yukawa packet, or a prediction row. The neutrino lane records an isotropic no-go, a rejected target-informed weighted-cycle candidate, and open source-basis, mass-label, and operator gates. The ledger also marks four separate row classes: the candidate P-closure source audit, the electroweak W/Z validation row, the charged-lepton absolute-mass landing, and the hadron backend plus empirical closure split. The W/Z row is a frozen compare-only adapter with its own public validation records. The Ward-projected Thomson lane records the source/root audit, the mixed-provenance root-plus-comparison-pixel diagnostic, and the measured endpoint as separate rows on the declared P-closure surface. In the charged lane the theorem surface carries an exact same-family witness, a determinant-line lift, and an algebraic mass readout from theorem-grade $A_{\mathrm{ch}}(P)$, while the source landing from the D10 quantitative descendants of P to physical charged data is closed as a corpus-limited no-go. Source-only hadron masses require a working OPH hadron backend. The declared empirical route would use a separate $e^+e^- \to$ hadrons payload class; no integrated endpoint payload is emitted here.

That accounting belongs on the page because it is part of the derivation. The paper puts the structural results, the fixed-point fine-structure value, the emitted quantitative rows, and the claim-boundary ledger onto one common surface. The bosonic P-driven trunk is explicit. The local extension surface identifies the shared edge entropy with the D10 nonabelian sum on the lifted product presentation of the realized quotient branch, and the same surface fixes that scalar to $P/4$. The local/global hierarchy bridge adds the exact capacity value $N_{\mathrm{CRC}}^{\mathrm{EW}}$, the zero bridge residual, the 12-port screen sieve, the 24-slot oriented repair register, and $\epsilon_H = 0$. The particle spectrum is therefore organized by one local screen scale instead of by a list of unrelated particle inputs. The claim ledger records scope boundaries for the weak validation pair, charged-lepton witnesses, direct-top auxiliary codomain, strong-CP phase boundary, neutrino comparison tension, source-only hadron backend rows, and empirical hadron closure rows.

A D10 Heat-Kernel Calibration Supplement

This appendix carries the D10 compact-group heat-kernel items needed by the particle surface. The theorem-bearing leaf for the same-overlap thermal/Casimir law sits on the screen-microphysics paper surface at fixed cutoff. What is recorded here is the compact-group / Peter–Weyl lift and beta-shift bookkeeping used by the particle-side calibration branch. The public D10 claim tier is explicitly Phase II rather than recovered-core Phase I. In particular, this appendix does not prove that OPH derives the physical one-loop beta coefficients from edge sectors alone.

A.1 $\mathcal{R}_U$ Unified-Coupling Certificate v0.2

The $\mathcal{R}_U$ certificate is the D10 gate needed by the hierarchy discussion in the synthesis paper. It certifies the unified diffusion coupling used in the electroweak transmutation factor

$$\frac{v}{E_\star} = P^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P)}\right].$$

It does not certify the full clock branch

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}} + \mathcal{R}_{\mathrm{atom}}^{133\mathrm{Cs}}.$$

The full $\mathcal{R}_\gamma$ stack requires source-only electromagnetic endpoint, charged-lepton absolute-scale, cesium nuclear, and atomic spectral enclosures. The v0.2 package is a numerical witness for $\mathcal{R}_U$, with formal promotion gated by outward-rounded interval arithmetic.

Frozen source packet. The public-pixel branch uses

$$P_C = \begin{array}{l}\texttt{1.6309682094039593248792798477826489413359828516279250606661}\\ \texttt{507533907793398933432}\end{array}.$$

The source-audit branch uses

$$P_{\mathrm{cand}} = \texttt{1.63097209569432901817967892561191884270169}.$$

The finite heat-kernel cutoffs are

$$N_2 = 128, \qquad N_3 = 64.$$

The one-loop running packet is

$$(b_1, b_2, b_3) = \left(\frac{33}{5}, 1, -3\right), \qquad N_c = 3, \qquad \beta_{\mathrm{EW}} = N_c + 1 = 4.$$

This packet is frozen in the certificate. It is not silently derived in this appendix from the recovered OPH core.

Definition A.1 (No-hidden-free-variable rule for $\mathcal{R}_U$)**.** *The following objects must be fixed before comparison with electroweak or gravity data:*

$$(b_1, b_2, b_3), \quad N_2, \quad N_3, \quad M_U(P), \quad E_{\text{cell}}(P), \quad \beta_{\text{EW}}, \quad \textit{Casimir normalization}, \quad \textit{matching scheme}.$$

Changing any of them after comparison with M_W, M_Z, $\alpha_s(M_Z)$, low-energy electroweak couplings, or G_{exp} demotes the row to calibration.

Dimensionless source map. Set $E_\star = 1$ for the dimensionless calculation. For a trial coupling a, define

$$M_U(P) = e^{-2\pi} P^{1/6}, \qquad E_{\text{cell}}(P) = P^{-1/2},$$
$$v(P, a) = E_{\text{cell}}(P) \exp\left[-\frac{2\pi}{4a}\right].$$

For $i = 1, 2, 3$,

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \qquad \alpha_Y(\mu; P, a) = \frac{3}{5}\alpha_1(\mu; P, a).$$

The source Z-scale is the positive solution of

$$\mu = \frac{v(P, a)}{2} \sqrt{4\pi\alpha_2(\mu; P, a) + 4\pi\alpha_Y(\mu; P, a)}.$$

For SU(2), with $j = n/2$ and $0 \leq n \leq N_2$,

$$d_j = 2j + 1, \qquad C_2(j) = j(j + 1).$$

For SU(3), with $0 \leq p, q \leq N_3$,

$$d_{p,q} = \frac{(p + 1)(q + 1)(p + q + 2)}{2}, \qquad C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}.$$

The group sums are

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \qquad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

Define

$$t_2(P, a) = 4\pi^2\alpha_2(\mu_Z(P, a); P, a), \qquad t_3(P, a) = 4\pi^2\alpha_3(\mu_Z(P, a); P, a),$$

and

$$\Phi_U(P, a) = \bar{\ell}_{\text{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\text{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The certificate value is the zero $\Phi_U(P, \alpha_U(P)) = 0$.

Numerical witness. For $P = P_C$, the v0.2 checker emits

$$\alpha_U(P_C) = 0.041124336195630495,$$
$$\alpha_U(P_C)^{-1} = 24.316501918546496.$$

The corresponding dimensionless hierarchy data are

$$\frac{M_U(P_C)}{E_\star} = 0.0020260720012100037,$$
$$\frac{v(P_C)}{E_\star} = 2.0199803239725553 \times 10^{-17},$$
$$\frac{\mu_Z(P_C)}{E_\star} = 7.502403238986022 \times 10^{-18}.$$

The emitted couplings at μ_Z are

$$\alpha_1 = 0.016885708038988166,$$
$$\alpha_Y = 0.010131424823392899,$$
$$\alpha_2 = 0.033777889448908055,$$
$$\alpha_3 = 0.11833602747445048.$$

The heat-kernel parameters and entropy readouts are

$$t_2 = 1.3334976254578108,$$
$$t_3 = 4.6717191102770705,$$

$$\bar{\ell}_{\mathrm{SU}(2)} = 0.39488144681077636,$$
$$\bar{\ell}_{\mathrm{SU}(3)} = 0.012860605540213493.$$

Thus

$$\bar{\ell}_{\mathrm{SU}(2)} + \bar{\ell}_{\mathrm{SU}(3)} = 0.40774205235098987,$$
$$P_C/4 = 0.4077420523509898,$$

and

$$\Phi_U(P_C, 0.041124336195630495) = 5.55 \times 10^{-17}.$$

A centered derivative check gives

$$\partial_a \Phi_U(P_C, a) \simeq -10.990524683118785.$$

With $\delta = 10^{-6}$, the bracket signs are

$$\Phi_U(P_C, \alpha_U - \delta) = 1.0990748697592423 \times 10^{-5},$$

$$\Phi_U(P_C, \alpha_U + \delta) = -1.0990300680135956 \times 10^{-5}.$$

For the source-audit pixel,

$$\alpha_U(P_{\mathrm{cand}}) = 0.04112424744557487,$$
$$\alpha_U(P_{\mathrm{cand}})^{-1} = 24.316554395881205,$$

$$\frac{v(P_{\mathrm{cand}})}{E_\star} = 2.0198114150099223 \times 10^{-17},$$
$$\frac{\mu_Z(P_{\mathrm{cand}})}{E_\star} = 7.501767385088398 \times 10^{-18}.$$

This branch is the one to cite when avoiding the measured Thomson endpoint in the upstream pixel solve.

Theorem A.2 (Electroweak unified-coupling numerical witness)**.** *Assume the frozen source map, running packet, heat-kernel cutoff policy, and no-hidden-free-variable rule above. Let*

$$I_U = [0.041123336195630494,\ 0.041125336195630496].$$

If outward-rounded interval arithmetic verifies

$$0 \in \Phi_U(P_C, I_U), \qquad 0 \notin \partial_a \Phi_U(P_C, I_U),$$

then there is a unique $\alpha_U(P_C) \in I_U$ satisfying

$$\Phi_U(P_C, \alpha_U(P_C)) = 0.$$

Moreover, the dependency graph of this zero contains no measured G, no Planck unit formed using measured G, no measured Λ, no measured M_Z, no measured M_W, and no measured low-energy gauge coupling.

Proof. Continuity follows because the running couplings, self-consistent Z-scale solution, and finite heat-kernel sums are continuous on the declared positive interval. The endpoint sign change gives existence by the intermediate value theorem. The derivative exclusion gives uniqueness. The dependency statement follows by inspection of the frozen source map: the residual is built only from P_C, the frozen running packet, the self-consistent Z-scale equation, and finite compact-group representation sums. □

Precision policy. The exponential map makes downstream gravity displays sensitive to α_U:

$$\frac{\partial \log G}{\partial \alpha_U} \simeq \frac{\pi}{\alpha_U^2} \approx 1.86 \times 10^3.$$

Therefore the number of digits printed for any downstream G or $\varepsilon_{\mathrm{Cs}}$ row may not exceed the interval precision certified by

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}} + \mathcal{R}_{\mathrm{atom}}^{133\mathrm{Cs}}.$$

The v0.2 witness supports an electroweak hierarchy certificate. It does not support a 52-digit source-only gravity prediction.

A.2 One-Loop Calibration Frame

At one loop, if couplings unify at (M_U, α_U),

$$\alpha_i^{-1}(M_Z) = \alpha_U^{-1} + \frac{b_i}{2\pi} \ln \frac{M_U}{M_Z}.$$

On the D10 branch these are the branch values $(M_U(P), \alpha_U(P))$ emitted by the forward D10 solve; they are not inferred by inverse readback from measured low-energy couplings. Writing $A_i := \alpha_i^{-1}(M_Z)$ and $L := \ln(M_U/M_Z)$, one finds

$$L = \frac{2\pi}{b_1 - b_2}(A_1 - A_2),$$

and the corresponding consistency relation for the D10 lane is

$$A_3^{\mathrm{pred}} = \frac{b_3 - b_2}{b_1 - b_2} A_1 + \frac{b_1 - b_3}{b_1 - b_2} A_2.$$

This appendix keeps that algebra on the page as a calibration relation, not as a theorem that OPH derives a supersymmetric UV spectrum.

A.3 Benchmark coefficient comparison

For the standard one-loop benchmark coefficients,

Model	(b_1, b_2, b_3)	$\alpha_s(M_Z)^{\text{pred}}$
SM	$(41/10, -19/6, -7)$	≈ 0.071
MSSM	$(33/5, 1, -3)$	≈ 0.116
Observed		0.1179 ± 0.0010

At one loop, the printed MSSM-style coefficients reproduce the observed closure far better than the plain SM coefficients. On this D10 branch this is a benchmark matched by the D10 calibration branch, not a theorem that OPH derives a supersymmetric UV spectrum.

A.4 Heat-Kernel Sector Law

Read this subsection as the compact-group merge boundary above the fixed-cutoff microphysics theorem. MaxEnt plus the bi-invariant compact-group package yield the familiar $d_R e^{-tC_2(R)}$ law once the Peter–Weyl lift is supplied; they are not a second, independent proof of the finite-cutoff same-overlap Casimir branch.

Theorem A.3 (Heat-kernel edge-sector weights)**.** *Under MaxEnt with bi-invariant constraints on a compact Lie group G, the sector probabilities take heat-kernel form:*

$$p_R(t) \propto d_R \, e^{-tC_2(R)},$$

where $d_R = \dim R$ and $C_2(R)$ is the quadratic Casimir.

Lemma A.4 (Bi-invariant operators)**.** *If $G = \prod_i G_i$ is a compact semisimple Lie group written as a product of compact simple factors, then any bi-invariant second-order differential operator on G has the form*

$$D = c_0 \mathbf{1} - \sum_i c_i \Delta_{G_i},$$

where Δ_{G_i} is the Laplace–Beltrami operator on the factor G_i.

Remark A.5. *Bi-invariance is equivalent to $D \in Z(U(\mathfrak{g}))$. Linear terms vanish because there are no invariant vectors in the adjoint representation, and the quadratic part is proportional to the Killing form on each simple factor, giving one independent coefficient per factor. The Casimir element then acts as $-\Delta_G$ in the regular representation.*

A.5 Peter–Weyl Multiplicity and Beta Shifts

By Peter–Weyl decomposition, the effective refinement-limit edge representation space is

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*.$$

Entanglement traces over one factor give multiplicity d_R in p_R, but loops see both factors, so the effective multiplicity entering the calibration branch is

$$N_{\text{eff}}(R) = d_R \, p_R.$$

The one-loop beta shift from edge sectors is then

$$\Delta b_a = \sum_R p_R \, d_R \, T_a(R),$$

where $T_a(R)$ is the Dynkin index for gauge factor a. This is the exact mathematical point on the D10 lane where edge-sector heat-kernel weights feed the calibration branch. Matching to MSSM-like one-loop running is a declared Phase II benchmark, not a promoted theorem-level MSSM spectrum claim.

The branch boundary for theorem-level unification closure is explicit. One requires a derived refinement-limit identification of which transportable sectors actually contribute to the running carrier, a derived statistics/grading rule rather than an imposed fermionic-grading restriction, controlled threshold and decoupling conventions on that same carrier, a derived representation content/truncation rule for the sectors retained in the comparison, and a proof that the effective loop multiplicities entering Δb_a are exactly the ones used in this calibration package rather than nearby alternatives. The declared runtime surface does not emit the full RG/matching/threshold/scheme packet: scheme lock, threshold map, beta provenance, and interval composition are separate certificate requirements.

At the branch unification diffusion parameter $t_U(P) \approx 1.64$ selected by the forward D10 solve, the benchmark shift is

$$\Delta b \approx (2.49, 4.38, 3.97) \qquad \text{vs MSSM} \qquad (2.50, 4.17, 4.00).$$

With the additional D10 calibration assumption of a fermionic-grading restriction to half-integer SU(2) sectors, this sharpens to

$$\Delta b \approx (2.50, 4.17, 3.97),$$

which matches the MSSM-style one-loop benchmark at the percent level under that declared calibration package. Equivalently, the benchmark ratio

$$\frac{\Delta b_3}{\Delta b_2} \approx 0.91$$

sits close to the MSSM comparison value 0.96.

B H3 Worldline-Stitch Certificate

This appendix records the finite evidence-facing certificate for Theorem 13.5. It is separate from the D10 heat-kernel calibration package above. A passing certificate is a record-continuation statement in a declared H^3 observer chart, not a particle-species, mass, charge, or scattering-amplitude theorem.

The certificate payload has the following required blocks.

1. **Atlas.** It declares the hyperboloid model $H^3_{R_H} \subset \mathbb{R}^{1,3}$, the Lorentz signature $(-+++)$, the curvature radius R_H, chart domains, and chart transitions $G_{ji} \in \mathrm{SO}^+(1,3)$. Transition residuals must certify Lorentz inner-product preservation, orientation, and future-sheet preservation.

2. **Clock map.** Each local record interval is mapped to a common comparison-time line. The observer-time adjacency margin must dominate the declared clock uncertainty.

3. **Chart-natural extraction.** Raw local records are connected components of a frozen detector scalar on the H^3 chart. The detector scalar, thresholds, support collars, and chart-naturality residuals are included. Implementation record IDs are not inputs.

4. **Overlap descent.** Records are first descended across genuine chart overlaps. The payload records same-component join margins, distinct-component separation margins, support errors, and triple-overlap cocycle residuals.

5. **Interface crossing.** A candidate stitch must cross a real interface in adjacent clock intervals. It includes oriented signed-distance margins and a positive lower bound on normal velocity. Tangential, grazing, and file-boundary-only contacts are rejected.

6. **Transport.** The candidate edge is evaluated in a common chart after sector and gauge transport. The atlas holonomy comparison is separate from the gauge-holonomy comparison.

7. **Assignment.** The matching is one-to-one with appearance/disappearance interval costs. The proposed winner has an upper cost bound, every competitor has a lower cost bound, and

$$\Delta_{\text{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*) > 0.$$

   Record IDs, global IDs, stitch keys, and shard-local IDs are forbidden in admissibility, costs, and tie-breaking.

8. **Refinement.** A coarse/fine pair supplies Q_{sr}, a distortion bound η_{sr}, and a contracted-graph isomorphism check. The coarse gap must satisfy $\Delta_r > 2\eta_{sr}$.

9. **Interaction firewall.** A free-propagation slab is required. If an interaction marker is present, the terminal label is H3_INTERACTION_REQUIRED and the event is handed to the interaction solver.

The allowed terminal statuses are

H3_STITCH_CERTIFIED, H3_STITCH_AMBIGUOUS, H3_STITCH_REJECTED, H3_INTERACTION_REQUIRED, H3_ATLAS_INVALID, H3_CERTIFICATE_INCOMPLETE.

H3_STITCH_CERTIFIED is emitted only when every block above passes and the robust assignment gap is positive. Equal-cost matchings, preserved nontrivial permutations of the visible data, or coarse/fine gaps below margin force H3_STITCH_AMBIGUOUS. Missing fields force H3_CERTIFICATE_INCOMPLETE. Failed atlas, interface, metric, or transport gates force rejection or atlas invalidity. In particular, Euclidean distances on `h3SpatialPoint` coordinates are not certificate distances; the certificate distance is the hyperboloid geodesic d_H from Section 13.

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Chain	OPH output	Role
P-closure and fine structure	$\alpha^{-1}_{\mathrm{root}} = 136.9948351646\ldots$; $A^{\mathrm{fp}}_{\alpha_U} = 137.0359595008\ldots$; empirical endpoint $136.2636589431 \in [136.1583832834, 136.3690975240]$; measured 137.035999177(21)	four distinct rows: source/root witness, mixed diagnostic, empirical hadron closure with anchor gap $[0.6485541111, 0.8547920666]$, and measurement
Classical carrier modes	two transverse Maxwell modes; $2\dim G$ perturbative pure-Yang–Mills modes; two pure-Einstein TT modes	conditional quadratic-action results; quantum particle gates open
Electroweak W/Z	strict source-audit branch (zero-selector law) $(m_W, m_Z) = (80.330,\ 91.119)$ GeV, within 0.05–0.08 percent of measurement with no target input; archived value law (80.3770000154, 91.1879780779) GeV as an audit coordinate; no source-only physical mass emitted	the strict branch carries one discrete two-law choice (9.3 MeV on M_W, 5.1 MeV on M_Z); the value law is an exact implication of the QT1–QT5 quotient-transport premises, whose carrier certificate and source-law selection principle are unemitted; the base-running chart and the target-anchored freeze-once adapter are separate audit objects; no promoted pole pair
Electroweak hierarchy/naturality	$v/E_\star$ on each named pixel branch $N^{\mathrm{EW}}_{\mathrm{CRC}} \simeq 3.53235 \times 10^{122}$ $\mathcal{B}_{\mathrm{EW}} = 0,\ \epsilon_H = 0$	selected dimensionless local/global bridge; physical $E_\star$ remains a separate gate
Higgs/top relation	double-criticality branch $(m_H, m_t) = (125.77,\ 172.63)$ GeV at the frozen boundary-scale candidate $E_\star e^{-\pi} P^{-1/6}$ (two loops), with $m_H = 125.72$ GeV on the fit-free curve at the measured top; target-anchored declared-surface fit (125.1995304097, 172.3523553288) GeV kept separate; no source-only physical mass emitted	the criticality family has zero continuous parameters over the boundary scale; the boundary-scale selection is a theorem modulo two finite carrier facts CF1/CF2 (the variational midpoint principle is proved and its quadratic-cost and placement premises reduce to the axioms; CF1/CF2 are the D11 carrier census the W/Z law also needs); the declared-surface fit is target-anchored and never predicts; the source-root, scale, rigidity, provenance, uncertainty, and complex-pole gates are open
Charged leptons	target-anchored witness values withheld; finite eight-path digital carrier and empirical transport intervals retained as diagnostics	the model proves schema satisfiability and a central record dilation, not physical source selection; the intervals use target-anchored ratios, measured α, and empirical transport
Selected-frame quarks	public numeric rows withheld; reciprocal-ray product 0.835323 and held-out error 21.56% at M_Z	common-scale data reject the reciprocal-ray candidate; the generic interface has six scalars and requires a source-derived flavor-orbit selector
Neutrino absolute attachment	compare-only absolute masses withheld; scale-free and mixing comparison rows retained	weighted-cycle comparison branch
Hadrons	first-principles strong-binding descent plus empirical $e^+e^- \to$ hadrons closure surface	OPH hadron construction for first-principles masses; measured hadron data for empirical closure rows
H3 record localization and worldlines	certified cap-response localization balls followed by cross-boundary record-token continuation, or AMBIGUOUS	conditional observer-record theorem; not a species, mass, or coupling input

Sector	Support status	Derivation stage	Public output(s) / exact sidecar(s)	Caveat / theorem boundary
Carrier roles and modes	structural group/content theorem plus conditional action theorem	explicit transverse/TT quadratic kernels with no promoted particle mass	action, background, and phase are hypotheses; quantization, positive-residue pole, and asymptotic/deconfinement gates remain separate	
Electroweak bosons	strict source-audit branch plus exact chart, conditional quotient-transport theorem, and comparison adapter	source-audit zero-selector law; selected carrier → two-coordinate mass chart; QT1–QT5 ⇒ unique value law	no nonzero mass in the prediction ledger; strict source-audit branch $(m_W, m_Z) =$ (80.330, 91.119) GeV within 0.05–0.08 percent of measurement; the target-free value-law coordinates (80.3770000154, 91.1879780779) GeV, base-running, and freeze-once adapter charts stay in the audit discussion	QT1–QT5 are not derived from the finite carrier; the discrete two-law choice awaits a source-law selection principle; no complex-pole receipt is emitted
Electroweak hierarchy/naturality	selected-branch dimensionless theorem	local $P \to \alpha_U \to v/E_\star$ hierarchy lane → 12-port screen sieve → 24-slot oriented repair register → exact capacity bridge and RG/Higgs square	$N_{\mathrm{CRC}}^{\mathrm{EW}} = 3.5323546226929906511\ldots \times 10^{122}$, $\mathcal{B}_{\mathrm{EW}} = 0$, $\epsilon_H = 0$	Supports the dimensionless hierarchy and naturality identities on the named branch. An independent physical $E_\star$, the public Thomson endpoint, theorem-level W/Z, and the other listed mass rows remain separate surfaces.
Higgs/top stage	double-criticality family from the gauge sector + declared-surface fit sidecar	criticality law $\lambda = 0$, $\beta_\lambda = 0$ at one source scale → zero-parameter (m_H, m_t) family → frozen boundary-scale candidate	no nonzero mass in the prediction ledger; double-criticality branch $(m_H, m_t) =$ (125.77, 172.63) GeV at the frozen boundary-scale candidate, with $m_H = 125.72$ GeV on the fit-free curve at the measured top; the target-anchored declared-surface fit (125.1995304097, 172.3523553288) GeV stays in technical audit prose	Strict promotion inherits the two boundary-scale carrier facts CF1/CF2 (the midpoint selection theorem is proved and its premises reduce to the axioms modulo these) plus the source-root, physical-scale, QT1–QT5, RG/scheme, rigidity, provenance, and complex-pole gates. The companion top coordinate is not a separate public top-mass prediction row.
Quark family	common-scale rejection plus restricted source non-identifiability theorem	one-scheme dimensionless Yukawas → reciprocal-ray test → six-scalar generic interface and flavor-selector boundary	no public numeric quark row; at M_Z, $\rho_u\rho_d = 0.835323$ and the endpoint-granted held-out error is 21.56%	The reciprocal-ray candidate is physically rejected across the tested running scales. The old $(\mathbb{R}_{>0})^2$ result remains a restricted lower-bound obstruction. Mixed-convention template and RSCC residuals are retrospective diagnostics. A flavor-orbit selector, quark–Higgs carrier, and common-scale source transport are absent
Charged leptons	continuation + exact sidecar witness	shared excitation dictionary → ordered charged carrier → exact centered readback → determinant-line lift on theorem-level physical charged data	exact centered readback; same-family target-anchored charged triple withheld from public prediction tables	public charged masses are not emitted from P. The theorem lane does not emit a theorem-level sector-isolated charged determinant exponent vector. It does not attach a source-side determinant character to the physical charged determinant line. The determinant-line lift and algebraic mass readout apply only on theorem-level physical charged data
Neutrinos	target-informed continuation candidate + auxiliary checks	template family transport → same-label scalar certificate → weighted-cycle candidate → compare-only bridge and absolute attachment → shared-basis identity	frozen scale-free candidate and PMNS/Majorana comparison rows; absolute mass attachments withheld from public prediction tables	exact linear algebra conditional on the declared template and selectors. The candidate fails the NuFIT 6.1 correlated $(\sin^2\theta_{23}, \delta_{\mathrm{CP}})$ profile, and neither the scale-free point nor the bridge invariant has prediction or theorem status [10]
Hadrons	strong-binding construction absent; empirical closure policy emitted	OPH-QCD backend contract, including quotient ensemble, source QCD parameter map, Ward current ledger, spectral exports, and empirical $e^+e^- \to$ hadrons payload schema	no first-principles hadron masses; empirical hadron closure rows are separate from first-principles OPH rows	First-principles hadron prediction requires a working OPH hadron construction, such as GLORB/Echosahedron, with the Ward-projected two-current spectral measure, higher-point/transition spectral sectors, same-scheme remainder, and systematics. The empirical closure surface uses separate $e^+e^- \to$ hadrons input and cannot upgrade the first-principles theorem
H3 record-worldline stitching	conditional certificate theorem	declared H_R^3 chart atlas and observer clock atlas → cap-response localization certificate for each descended token and clock slice → localization balls with positive Δ_{loc} for point outputs → overlap descent → real transverse interface-crossing germs → common-chart sector/gauge transport → ID-independent one-to-one assignment gap → coarse/fine contraction check	nonbranching localized record-token paths, rays, or isolated records when both localization and stitch certificates pass; otherwise AMBIGUOUS, REJECTED, or INTERACTION REQUIRED	This is a continuation theorem for observer-visible records in H^3, not a derivation of particle species, masses, gauge charges, or scattering amplitudes. Repeated implementation IDs, nearest-neighbor fits, pre-existing coordinates without response inversion, and file-boundary coincidences are not admissible evidence.

Lane	Exact output(s)	Exact chain on the paper surface	Caveat
Carrier roles and classical modes	no photon/gluon/graviton particle-mass output	axioms $\to$ realized electromagnetic/color/dynamical-metric roles; additional action/background/phase receipt $\to$ transverse/TT quadratic modes	quantum-particle receipt not supplied; confined color has no free asymptotic-gluon claim
Electroweak chart and sidecar	no nonzero source-only mass output	source tuple $\to$ exact two-coordinate chart; QT1–QT5 $\Rightarrow$ candidate value law; inverse adapter separate	all numerical coordinates are confined to the audit discussion; QT1–QT5 remain a finite-carrier certificate obligation and no pole pair is promoted
Hierarchy/naturality bridge	exact dimensionless bridge residual and zero Higgs naturality defect	axioms $\to$ P-fixed source branch $\to$ $N_{\mathrm{CRC}}^{\mathrm{EW}}$ bridge fixed point $\to$ screen sieve and 24-slot repair register $\to$ source-to-Higgs settled-form square	selected-branch theorem for $v/E_\star$ and the naturality identities; it does not supply a physical $E_\star$ or promote the mass rows
Higgs/top exact sidecar	no nonzero source-only mass output	gauge core $\to$ declared Jacobian $\to$ exact inverse slice	benchmark inverse slice only; the conditional coordinates remain outside the prediction ledger and inherit every open full-chain gate
Charged exact witness	exact same-family charged triple withheld	axioms $\to$ shared excitation dictionary $\to$ ordered charged carrier $\to$ exact centered readback $\to$ closed quadratic readout theorem $\to$ same-family exact witness	same-family target-anchored audit witness only; theorem lane carries exact centered readback plus the closed common-shift no-go. The available derivation has a no-go boundary: a theorem-level $\widehat{C}_e$ lift emitting the physical scalar $\mu_{\mathrm{phys}}(Y_e)$ is not part of the emitted theorem surface
Quark physical boundary	numeric prediction rows withheld; common-scale reciprocal-ray failure leads the status; historic formula residuals retained only for audit	six dimensionless Yukawas in one scheme and scale $\to$ reciprocal-ray rejection $\to$ six-scalar interface and selector no-go	at M_Z, $\rho_u = 1.110889$, $\rho_d = 0.751941$, product 0.835323; even with four endpoints granted, the held-out miss is 21.56%. The old two-spread theorem remains a restricted non-identifiability result
Neutrino candidate	frozen scale-free comparison point; absolute mass attachments withheld	template family transport $\to$ same-label scalar certificate $\to$ target-informed weighted-cycle law $\to$ compare-only bridge and absolute attachment $\to$ shared-basis identity	rejected by the NuFIT 6.1 correlated profile; source closure, historical blindness, basis orientation, and absolute normalization all fail promotion gates [10]

Carrier role	Claim tier	Action-level output	Physical degrees of freedom	Quantum-particle boundary
Electromagnetic connection	conditional Maxwell mode	transverse $k^2 = 0$ kernel	two transverse classical modes	photon requires quantization and a positive-residue physical pole
Color connection	conditional perturbative pure-Yang–Mills mode	one transverse $k^2 = 0$ kernel per generator	$2 \dim G$ perturbative modes	deconfined BRST/LSZ receipt required; no free-gluon claim in confined QCD
Metric perturbation	conditional pure-Einstein mode	$D^{\mathrm{TT}} \propto i\Pi^{\mathrm{TT}}/(k^2 + i0)$	two classical TT modes	graviton requires metric quantization, physical Hilbert space, and pole receipt

Particle	Stage	Claim tier	Reported value
W boson	D10 electroweak branch	no source-only physical mass; audit coordinates quarantined	withheld
Z boson	D10 electroweak branch	no source-only physical mass; audit coordinates quarantined	withheld
Higgs boson	Higgs/top critical stage	no source-only physical mass; conditional coordinate not promoted	withheld
Top quark	Higgs/top critical stage	companion coordinate on declared surface; direct-top auxiliary codomain no-go	not a separate public top-mass prediction row

Quark	Claim tier	Public value	Derivation stage
Up quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Down quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Strange quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Charm quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = m_c(\mu)$, comparison chart
Bottom quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = m_b(\mu)$, comparison chart
Top quark	separate extraction coordinate	withheld	cross-section pole-mass chart